Proof reductions by squashing (reductions)

Statements digest, version 0.22.0

Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

Mathematics

1 Squashing relations and their statistics

Definition 1.1 (The system setting of the reductions)**.** The *system setting* of this contract is the following:

(a) the Hilbert space $\mathcal{H}(B)$ of the system B that Bob's physical receiver acts on is separable and may be infinite-dimensional;

(b) every target space $\mathcal{H}(Q)$ is finite-dimensional, with declared dimensions;

(c) every side register other than the adversary's registers, that is, Alice's retained system and the classical registers, is finite-dimensional, with declared dimensions.

Lemma 1.2 (Separation and tensor adjoints in the infinite-dimensional case)**.** *Setting.* B is the system Bob's physical receiver acts on and Q the system it is squashed to. $\mathcal{H}(B)$ is separable and may be infinite-dimensional, while $\mathcal{H}(Q)$ and every side register are finite-dimensional (Definition 1.1, and, for the adversary's registers, the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model), every register of which is finite-dimensional). $\mathcal{T}(B)$ denotes the trace-class operators on $\mathcal{H}(B)$; a *channel* $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ is a completely positive trace-preserving map, and its *adjoint* $\Lambda^\dagger\colon \mathcal{L}(Q) \to \mathcal{L}(B)$ is the unital completely positive map fixed by

$$\operatorname{tr}\left[X\,\Lambda(\rho)\right] = \operatorname{tr}\left[\Lambda^\dagger(X)\,\rho\right] \qquad \text{for all } X \in \mathcal{L}(Q),\ \rho \in \mathcal{T}(B). \tag{1.1}$$

More generally, if $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(B')$, for a Hilbert space $\mathcal{H}(B')$ of any dimension, is only completely positive and trace non-increasing ($\operatorname{tr}\Lambda(\rho) \le \operatorname{tr}\rho$ for $\rho \succeq 0$), then (1.1) for all $X \in \mathcal{L}(B')$ and $\rho \in \mathcal{T}(B)$ fixes a completely positive map $\Lambda^\dagger\colon \mathcal{L}(B') \to \mathcal{L}(B)$, again called its *adjoint*, and $\Lambda^\dagger$ is subunital: $\mathbb{1}_B - \Lambda^\dagger(\mathbb{1}_{B'}) \succeq 0$. Where such a Λ is moreover trace preserving, $\Lambda^\dagger$ is unital, $\Lambda^\dagger(\mathbb{1}_{B'}) = \mathbb{1}_B$, whatever the dimension of $\mathcal{H}(B')$.

Let $\mathcal{H}(B_1), \dots, \mathcal{H}(B_n)$ be separable and $\mathcal{H}(Q_1), \dots, \mathcal{H}(Q_n)$ finite-dimensional.

(i) *(Separation.)* If $X \in \mathcal{L}(B_1)$ satisfies $\operatorname{tr}[X\rho] = 0$ for every $\rho \in S_=(B_1)$, then $X = 0$.

(ii) *(Tensor channels and their adjoints.)* If $\Lambda_i \colon \mathcal{T}(B_i) \to \mathcal{T}(Q_i)$ are channels, then exactly one trace-norm continuous linear map $\Lambda_1 \otimes \cdots \otimes \Lambda_n$ from $\mathcal{T}(B_1 \cdots B_n)$, the trace class of the completed tensor product of the $\mathcal{H}(B_i)$, to $\mathcal{T}(Q_1 \cdots Q_n)$ takes every $\rho_1 \otimes \cdots \otimes \rho_n$ with $\rho_i \in \mathcal{T}(B_i)$ to $\Lambda_1(\rho_1) \otimes \cdots \otimes \Lambda_n(\rho_n)$; it is a channel, and $(\Lambda_1 \otimes \cdots \otimes \Lambda_n)^\dagger = \Lambda_1^\dagger \otimes \cdots \otimes \Lambda_n^\dagger$ on all of $\mathcal{L}(Q_1 \cdots Q_n)$. The same holds with any of the Λ_i replaced by id_{R_i} for a finite-dimensional R_i.

(iii) *(Tensor channels from finite inputs.)* Let $\mathcal{H}(R_1), \ldots, \mathcal{H}(R_n)$ be finite-dimensional, $\mathcal{H}(B_1'), \ldots, \mathcal{H}(B_n')$ separable, of any dimension, and $\Phi_i \colon \mathcal{L}(R_i) \to \mathcal{T}(B_i')$ completely positive and trace preserving. Then exactly one linear map $\Phi_1 \otimes \cdots \otimes \Phi_n$ from $\mathcal{L}(R_1 \cdots R_n)$ to $\mathcal{T}(B_1' \cdots B_n')$, the trace class of the completed tensor product of the $\mathcal{H}(B_i')$, takes every $X_1 \otimes \cdots \otimes X_n$ with $X_i \in \mathcal{L}(R_i)$ to $\Phi_1(X_1) \otimes \cdots \otimes \Phi_n(X_n)$, and it is completely positive and trace preserving. The identity map id_R of a finite-dimensional R is such a Φ_i.

Definition 1.3 (Squashing relation)**.** Let $M = \{M_v\}_{v \in \mathcal{V}}$ be a POVM on B and $F = \{F_v\}_{v \in \mathcal{V}}$ a POVM on Q over the same finite alphabet $\mathcal{V}$, and let $\hat{O} = (\hat{O}_k)_{k=1}^m$ and $O = (O_k)_{k=1}^m$ be tuples of Hermitian operators $\hat{O}_k \in \mathcal{L}(Q)$, $O_k \in \mathcal{L}(B)$. For a channel $\Lambda \colon \mathcal{T}(B) \to \mathcal{T}(Q)$, with adjoint $\Lambda^\dagger$ as in Lemma 1.2, write

$$(M; O) \preceq_\Lambda (F; \hat{O}) \quad :\Longleftrightarrow \quad \Lambda^\dagger(F_v) = M_v \ \forall v \in \mathcal{V} \text{ and } \Lambda^\dagger(\hat{O}_k) = O_k \ \forall k \in \{1, \ldots, m\}, \tag{1.2}$$

and call (1.2) a *squashing relation*: M *squashes to* the *target POVM* F via the *squashing channel* Λ, which *transports* O to $\hat{O}$. The operators $\hat{O}_k$ are the relation's *carried operators*. The empty case $m = 0$ is written $M \preceq_\Lambda F$, and $\preceq$ replaces $\preceq_\Lambda$ when only the existence of a squashing channel is asserted. We call $\Lambda^\dagger(\hat{O}_k)$ the *pull-back* of $\hat{O}_k$.

Lemma 1.4 (Squashing as statistics equivalence)**.** Let $M, F, O, \hat{O}$ be as in Definition 1.3 and let $\Lambda \colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel. The following are equivalent.

(i) $(M; O) \preceq_\Lambda (F; \hat{O})$.

(ii) For every $\rho \in S_=(B)$, every v and every k, $\operatorname{tr}[M_v\rho] = \operatorname{tr}[F_v\Lambda(\rho)]$ and $\operatorname{tr}[O_k\rho] = \operatorname{tr}\left[\hat{O}_k\Lambda(\rho)\right]$.

(iii) For every finite-dimensional side system R, every $\rho \in S_=(BR)$, every $X \in \mathcal{L}(R)$, every v and every k, writing $\sigma \coloneqq (\Lambda \otimes \mathrm{id}_R)(\rho)$,

$$\operatorname{tr}\left[(M_v \otimes X)\rho\right] = \operatorname{tr}\left[(F_v \otimes X)\sigma\right] \qquad \text{and} \qquad \operatorname{tr}\left[(O_k \otimes X)\rho\right] = \operatorname{tr}\left[(\hat{O}_k \otimes X)\sigma\right].$$

Moreover, two channels Λ, Λ' that both realize the relation satisfy $\Lambda^\dagger = \Lambda'^\dagger$ on span $\big(\{F_v\}_{v\in\mathcal{V}} \cup \{\hat{O}_k\}_{k=1}^m\big)$.

Corollary 1.7 (Pull-back of operator bounds)**.** Let $(M;O) \preceq_\Lambda (F;\hat{O})$, let R be a finite-dimensional side system, let $\rho \in S_=(BR)$ and let $\sigma = (\Lambda\otimes\mathrm{id}_R)(\rho)$. Then for every k,

$$\mathrm{tr}_Q\left[(\hat{O}_k \otimes \mathbb{1}_R)\,\sigma\right] = \mathrm{tr}_B\left[(O_k \otimes \mathbb{1}_R)\,\rho\right] \tag{1.3}$$

as operators on R. Consequently, on every such σ:

(i) *Exact values.* If $O_k = c\,\mathbb{1}_B$ then $\mathrm{tr}_Q[(\hat{O}_k\otimes\mathbb{1}_R)\sigma] = c\,\sigma_R$; in particular $\mathrm{tr}\Big[\hat{O}_k\sigma\Big] = c$, and at $c=0$, $\mathrm{tr}\Big[(\hat{O}_k\otimes X)\sigma\Big] = 0$ for every $X \in \mathcal{L}(R)$.

(ii) *Positivity.* If $O_k \succeq 0$ then $\mathrm{tr}_Q[(\hat{O}_k \otimes \mathbb{1}_R)\sigma] \succeq 0$; in particular $\mathrm{tr}\Big[\hat{O}_k\sigma\Big] \geq 0$.

(iii) *Spectral bounds.* If $a\,\mathbb{1}_B \preceq O_k \preceq b\,\mathbb{1}_B$ then $a\,\sigma_R \preceq \mathrm{tr}_Q[(\hat{O}_k\otimes\mathbb{1}_R)\sigma] \preceq b\,\sigma_R$; in particular $a \leq \mathrm{tr}\Big[\hat{O}_k\sigma\Big] \leq b$.

Definition 1.8 (Certificates of transported constraints)**.** The *certificate* of a transported constraint is the hypothesis that makes it hold: for a constraint from Corollary 1.7, the hypothesis of the case that yields it, an operator identity in its case (i) and an operator inequality in its cases (ii)–(iii), both facts about the device and the channel, valid whatever the input; and, for an assumed constraint of Corollary 3.13 (ii), an assumption on the input states.

Definition 1.10 (Measurement channel)**.** For a POVM $M = \{M_v\}_{v\in\mathcal{V}}$ on B and a classical register Y with alphabet $\mathcal{V}$, the *measurement channel* of M is

$$\mathcal{N}_M : \mathcal{T}(B) \to \mathcal{T}(Y), \qquad \mathcal{N}_M(\rho) \coloneqq \sum_{v\in\mathcal{V}} \mathrm{tr}\left[M_v\rho\right]\,|v\rangle\langle v|_Y\,. \tag{1.4}$$

Lemma 1.11 (Measurement-channel factorization)**.** Let M, F be POVMs on B, Q over $\mathcal{V}$ and let $\Lambda : \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel. Then

$$M \preceq_\Lambda F \qquad\Longleftrightarrow\qquad \mathcal{N}_M = \mathcal{N}_F \circ \Lambda. \tag{1.5}$$

2 Protocols, security criteria and exact operational reductions

Definition 2.1 (Measurement-factorizing n-round protocol)**.** An *n-round protocol* with receiver POVM M is a channel

$$\mathcal{E}_M \;=\; \mathsf{K}\circ\mathsf{G}_n\circ\cdots\circ\mathsf{G}_1, \qquad \mathsf{G}_j \;=\; \bar{\mathsf{G}}_j\circ\big(\mathrm{id}\otimes\mathcal{N}_M\big), \tag{2.1}$$

in which the measurement channel $\mathcal{N}_M$ of Definition 1.10 acts on the round-j *measured system* B_j, the quantum system that the *measuring parties*, the parties the declaration names as measuring, receive in round j, and no other system, and the identity acts on everything else. $\bar{\mathsf{G}}_j$ is the round's remaining processing (Alice's measurement on A_j, the round's announcements appended to the public record, and the retained per-round data), and K is the end-of-protocol processing: sifting, parameter estimation and the accept/abort decision, error correction, verification, privacy amplification and the key maps. The *squashed protocol* $\mathcal{E}_F$ is (2.1) with B_j replaced by Q_j, M by F, and $\bar{\mathsf{G}}_j$, K unchanged. One POVM and one channel serve all n rounds: the same M, and in the squashed protocol the same F and the same squashing channel, stands in every G_j, with no dependence on j. That the measuring devices measure by the same M in every round, with no memory between rounds, is Assumption 13.2; F and the squashing channel are objects of the analysis, chosen once for all rounds. That is what makes the composite of the per-round channels a tensor power, such as $\Lambda^{\otimes n}$ for the squashing channel, rather than a product of n different maps.

One measuring party, or several. M is one POVM on the measured system, over the alphabet of the measuring parties' joint outcome. Where one party measures, which is the one-party case, B_j is that party's system, Bob's in a prepare-and-measure declaration. Where several parties measure, $B_j = \bigotimes_f B_{f,j}$ is the product of their party-scoped parts, each factor $B_{f,j}$ belonging to one of them, and (2.1) places *all* of their measurements at the one step $\mathcal{N}_M$, before the round's remaining processing $\bar{\mathsf{G}}_j$ and so before every public message of the round. The declaration says on which side Alice's measurement sits, and each reading is covered by exactly one clause of the first paragraph: the measurement on A_j listed in $\bar{\mathsf{G}}_j$ is that of a party the declaration does *not* name as measuring (Alice's, in a prepare-and-measure declaration), while where she *is* named as measuring, her system is a factor $B_{f,j}$ of the measured system and her measurement is a factor of $\mathcal{N}_M$, not a step of $\bar{\mathsf{G}}_j$. Either way $\bar{\mathsf{G}}_j$ and K touch no B_j, which is all the theorems of this contract read. A declaration realizes the second reading by an embedding: it declares Alice's system A one-dimensional, $d_A = 1$, folds her measured system into the sectors of Q and her outcome components into v, and names the components she holds in its party partition (Definition "Protocol declaration" (protocol-compiler/def:declaration)). That the physical

protocol has that order, every public message of round j sent after every measurement of round j whichever party sends it and whichever party measured, is Assumption A-SEQ of the transmitter (Assumption "Sequential round structure" (transmitter/ass:seq)). The definition says nothing about which party sends which message or about a message's dependence on earlier messages of its own round, and a quantum operation of round j that depended on a message of round j is outside it, as it is outside A-SEQ.

Definition 2.2 (Security criterion)**.** A *security criterion* is a family $f = (f_E)_E$ of functions, one for every admissible adversary system E, defined on states of the protocol's output registers together with E; an *admissible adversary system* is a finite-dimensional system, the form the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) gives the adversary's registers. Its value on the protocol $\mathcal{E}_M$ at an input ρ, a state of the systems the protocol acts on together with E, is

$$\mathsf{sec}(\mathcal{E}_M, \rho) := f_E\big((\mathcal{E}_M \otimes \mathrm{id}_E)(\rho)\big). \tag{2.2}$$

Definition 2.3 (Exact operational reduction)**.** A *protocol presentation* is a triple $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$: a set $\mathbb{A}$, each element of which is called an *attack strategy*, a tuple W of *output registers*, and a map $\mathsf{out}\colon \mathbb{A} \to S_{\leq}(W)$ assigning to each strategy the *full output* the protocol produces under it. The full output is the state on *every* register a security criterion may read (Definition 2.2): both parties' key registers, the public transcript with the acceptance and abort data, and the adversary's final register, and no marginal of them.

Let $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$ and $\mathsf{P}' = (\mathbb{A}', W, \mathsf{out}')$ be presentations *over the same output registers* W. An *exact operational reduction* from P to P' is a map $r\colon \mathbb{A} \to \mathbb{A}'$ with

$$\mathsf{out}(a) \;=\; \mathsf{out}'\big(r(a)\big) \qquad \text{for every } a \in \mathbb{A}. \tag{2.3}$$

We call $\mathbb{A}$ the *physical strategy set* and $\mathbb{A}'$ the *analysed strategy set*, equivalently the *admissible* strategy set, and $r(\mathbb{A}) \subseteq \mathbb{A}'$ the reduction's *image*.

The conditions, in full.

(R1) *(one output tuple)* the two presentations carry literally the same output registers, and which register of P' is which register of P is part of the data rather than inferred. Where a model protocol's adversary register is a *different* system, an identification of it with the physical one has to be declared and the criterion has to be invariant under that identification; this contract declares none and needs none, the adversary's registers being literally the same system in both of its instances;

(R2) *(exact equality)* (2.3) is equality of the two states on the nose: not of a marginal, not up to a distance, and not up to any map on W;

(R3) *(totality and typing)* r is defined at every strategy of $\mathbb{A}$ and takes its values in $\mathbb{A}'$. That its values lie in $\mathbb{A}'$ is a condition on r and on $\mathbb{A}'$ together, and it is the only place where a restriction of the analysed set has to be earned.

What is not required, each item being used by an instance in this contract. r need not be injective, need not be surjective onto $\mathbb{A}'$, and need not preserve any structure of $\mathbb{A}$; the analysed set $\mathbb{A}'$ may be, and in this contract typically is, strictly larger than the image, because what a numerical analysis can optimize over is a set cut out by finitely many conditions and not an image (Definition 6.3); no metric, topology, order, linear or convex structure on $\mathbb{A}$, on $\mathbb{A}'$ or on the output states is used; out and out' need not be linear, affine, continuous or given by channels, and a strategy need not be a channel or a tuple of channels (in Theorem 3.11 a strategy is a state); and nothing is asserted in the other direction, a bound over $\mathbb{A}$ transferring to $\mathbb{A}'$ under no hypothesis of this definition.

Two presentations, one per adversary system. A security criterion in the sense of Definition 2.2 is a *family* $(f_E)_E$, one function per admissible adversary system. Fixing E fixes W and turns each f_E into one function on $S_{\leq}(W)$, so the quantification over E in that definition is a quantification over presentations: a statement of this contract that holds "for every admissible E" exhibits one presentation pair and one reduction per E, and the transfer of Lemma 2.4 is applied to each.

Lemma 2.4 (Transfer and composition of exact operational reductions)**.** Let r be an exact operational reduction from $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$ to $\mathsf{P}' = (\mathbb{A}', W, \mathsf{out}')$ in the sense of Definition 2.3. Write $\mathcal{F}(W)$ for the set of all functions $S_{\leq}(W) \to [-\infty, +\infty]$; a security criterion in the sense of Definition 2.2, at a fixed adversary system, is one such function.

(i) *(transfer, with the same ε and for every criterion at once)* for every $f \in \mathcal{F}(W)$ and every $a \in \mathbb{A}$, $f(\mathsf{out}(a)) = f(\mathsf{out}'(r(a)))$, and consequently

$$\sup_{a \in \mathbb{A}} f\big(\mathsf{out}(a)\big) \;=\; \sup_{b \in r(\mathbb{A})} f\big(\mathsf{out}'(b)\big) \;\leq\; \sup_{b \in \mathbb{A}'} f\big(\mathsf{out}'(b)\big). \tag{2.4}$$

In particular, if $f(\mathsf{out}'(b)) \leq \varepsilon$ for every $b \in \mathbb{A}'$, then $f(\mathsf{out}(a)) \leq \varepsilon$ for every $a \in \mathbb{A}$, with the *same* ε, and simultaneously for every $f \in \mathcal{F}(W)$, no property of f being used;

(ii) *(enlargement of the analysed set)* if $\mathbb{A}' \subseteq \mathbb{A}''$ and $\mathsf{out}''|_{\mathbb{A}'} = \mathsf{out}'$, then the same r is an exact operational reduction from P to $\mathsf{P}'' =$

$(\mathbb{A}'', W, \mathsf{out}'')$. So it is enough that the image $r(\mathbb{A})$ lie in the analysed set, and the last inequality of (2.4) is where that slack is spent;

(iii) *(restriction of the physical set)* if $\mathbb{A}_0 \subseteq \mathbb{A}$, then $r|_{\mathbb{A}_0}$ is an exact operational reduction from $(\mathbb{A}_0, W, \mathsf{out}|_{\mathbb{A}_0})$ to P';

(iv) *(one map, many analysed sets)* let $(\mathbb{A}'_k)_{k\in K}$ be a family of subsets of $\mathbb{A}'$ such that *one and the same* r satisfies $r(\mathbb{A}) \subseteq \mathbb{A}'_k$ for every $k \in K$. Then r is an exact operational reduction from P to $\big(\bigcap_{k\in K} \mathbb{A}'_k, W, \mathsf{out}'\big)$, and by (i) a bound over that intersection transfers. The hypothesis cannot be weakened to the existence, for each k, of *some* exact operational reduction into $\mathbb{A}'_k$: let W be one classical bit, $\mathbb{A} = \{a\}$, $\mathbb{A}' = \{b_1, b_2\}$ with $b_1 \neq b_2$ and $\mathsf{out}(a) = \mathsf{out}'(b_1) = \mathsf{out}'(b_2) = |0\rangle\langle 0|$, and $\mathbb{A}'_k = \{b_k\}$. Each $r_k \colon a \mapsto b_k$ is an exact operational reduction into $\mathbb{A}'_k$, the intersection is empty, so "$f(\mathsf{out}'(b)) \leq 0$ for every b in the intersection" holds vacuously; yet $f(\mathsf{out}(a)) = 1$ for $f(\rho) \coloneqq \mathrm{tr}[|0\rangle\langle 0|\,\rho]$, so the conclusion of (i) is false for the intersection;

(v) *(composition, and how the analysed sets compose)* let $\mathbb{A}'_0 \subseteq \mathbb{A}'$ contain $r(\mathbb{A})$ and let r' be an exact operational reduction from $\big(\mathbb{A}'_0, W, \mathsf{out}'|_{\mathbb{A}'_0}\big)$ to $\mathsf{P}'' = (\mathbb{A}'', W, \mathsf{out}'')$. Then $r' \circ r$ is an exact operational reduction from P to P'', with image $r'(r(\mathbb{A}))$. The analysed set the composite needs is the second reduction's, or any subset of $\mathbb{A}''$ containing $r'(r(\mathbb{A}))$; the first reduction's analysed set enters only as a domain the second must cover, which is what $r(\mathbb{A}) \subseteq \mathbb{A}'_0$ says. Composition is associative and the identity map on $\mathbb{A}$ is an exact operational reduction from P to itself;

(vi) *(the direction, and what the slack costs)* nothing above bounds $\sup_{\mathbb{A}'} f \circ \mathsf{out}'$ by $\sup_{\mathbb{A}} f \circ \mathsf{out}$, and the first can be strictly larger: take $\mathbb{A} = \{a\}$, $\mathbb{A}' = \{r(a), b\}$ with $\mathsf{out}'(b) \neq \mathsf{out}(a)$ and an f separating the two states. An analysed set larger than the image therefore costs *tightness* and never soundness.

3 Output-state equality and security transfer

Theorem 3.1 (Equality of the physical and squashed output states)**.** Let $M \preceq_\Lambda F$ over $\mathcal{V}$ and let $\mathcal{E}_M, \mathcal{E}_F$ be as in Definition 2.1. Then

$$\mathcal{E}_M \;=\; \mathcal{E}_F \circ \big(\mathrm{id} \otimes \Lambda^{\otimes n}\big), \tag{3.1}$$

the identity acting on every register other than $B_1 \dots B_n$; and consequently, for every finite-dimensional adversary system E and every input ρ,

$$(\mathcal{E}_M \otimes \mathrm{id}_E)(\rho) = (\mathcal{E}_F \otimes \mathrm{id}_E)\big(\sigma\big), \qquad \sigma \coloneqq \big(\mathrm{id} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)(\rho). \tag{3.2}$$

In particular $\mathsf{sec}(\mathcal{E}_M, \rho) = \mathsf{sec}(\mathcal{E}_F, \sigma)$ for every security criterion in the sense of Definition 2.2.

Definition 3.2 (Round model and attack sequences, typed by source mode)**.** The round model of the protocol (2.1) is that of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) with per-round system B, Alice's system A and the declared source mode. An attack sequence for the protocol is thus a family $(\mathcal{A}_j)_{j=1}^{n}$ of channels whose output carries the systems the adversary supplies in the declared source mode, the display being that definition's at $S = B$:

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}\big(E_{j-1},\ B_j\, E'_j\big), & \text{trusted-source mode,} \\ \mathrm{CPTP}\big(E_{j-1},\ A_j\, B_j\, E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{3.3}$$

where E_0 is Eve's initial register, E'_j her outgoing memory, and E_j collects E'_j together with whatever public data of round j she retains. In trusted-source mode A_j is *not* an output of $\mathcal{A}_j$: it is supplied by Alice and carried into the round as a separate input. The protocol's output under the sequence is obtained from (2.1) by supplying B_j — and, in untrusted-source mode, A_j as well — as the output of $\mathcal{A}_j$ in round j. The corresponding *squashed sequence* is

$$\mathcal{A}_j^{\mathrm{sq}} \;:=\; \begin{cases} \big(\Lambda \otimes \mathrm{id}_{E'_j}\big) \circ \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1}, Q_j E'_j\big), & \text{trusted-source mode,} \\ \big(\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_{E'_j}\big) \circ \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1}, A_j Q_j E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{3.4}$$

the squashing channel acting on the B_j factor in both cases and on nothing else.

Several measuring parties. Where the measured system of a round is the product $B_j = \bigotimes_f B_{f,j}$ of party-scoped parts (Definition 2.1), the squashing channel is the product $\Lambda = \bigotimes_f \Lambda_f$ of one channel per part, $Q_j = \bigotimes_f Q_{f,j}$, and the alphabet $\mathcal{V}$ of M and F is the product of the parts' alphabets. Neither line of (3.3) changes: B_j is one output system of $\mathcal{A}_j$ whatever its factorisation, so the adversary supplies all of its parts in one channel and is given the same freedom as with one measuring party; in particular the parts may be arbitrarily correlated, and the adversary is not required to act on them independently. Nor does (3.4) change: the product channel still acts on the B_j factor alone, leaving A_j and E'_j untouched. That the product channel realizes a squashing relation for the product family is Proposition 4.8, which this definition does not assert. With one measuring party the product has one factor.

Corollary 3.3 (Output-state equality against adaptive attack sequences)**.** Let $M \preceq_\Lambda F$ and fix a source mode. For every attack sequence $(\mathcal{A}_j)_{j=1}^{n}$

of the corresponding type (3.3), the output state of the physical protocol $\mathcal{E}_M$ under $(\mathcal{A}_j)_j$ equals the output state of the squashed protocol $\mathcal{E}_F$ under $(\mathcal{A}_j^{\mathrm{sq}})_j$ of the same branch of (3.4). Consequently every security criterion takes the same value on the two, and if $\mathcal{E}_F$ is ε-secure against every sequence drawn from families $\mathfrak{A}_j^{\mathrm{sq}}$ with

$$\mathfrak{A}_j^{\mathrm{sq}} \supseteq \begin{cases} (\Lambda \otimes \mathrm{id}) \circ \mathrm{CPTP}\big(E_{j-1}, B_j E'_j\big), & \text{trusted-source mode,} \\ (\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}) \circ \mathrm{CPTP}\big(E_{j-1}, A_j B_j E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{3.5}$$

then $\mathcal{E}_M$ is ε-secure against *all* attack sequences of that type.

Proposition 3.4 (Measurement factorization of the platform's protocol channel)**.** Let $\mathfrak{D}$ be a protocol declaration in the sense of Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration) that satisfies the conditions of Definition "Admissible declaration" (meat-core/def:admissible-declaration), in the format in which A carries one sector, one system is measured, Alice's measurement is declared as a POVM $\{M_x\}_{x\in\mathcal{X}}$, the public discussion is two messages neither of which reads the other, and no entry is scoped to a party. Write a for its joint announcement map, ϕ for its acceptance-statistic map and r for its raw-key map, and let $\mathcal{M}^{\mathrm{sq}}$ be the single-round protocol channel it generates, on AQ with the declared POVM $F = \{F_v\}_{v\in\mathcal{V}}$ on the measured system Q, the system the declaration's measuring parties hold (Definition 2.1). Let $M = \{M_v\}_{v\in\mathcal{V}}$ be a POVM on a separable B, and let $\mathcal{M}^{\mathrm{ph}}$ be the map that the same components define with (B, M) in place of (Q, F),

$$\mathcal{M}^{\mathrm{ph}}(\omega) := \sum_{x,v} |r(x,v)\rangle\langle r(x,v)|_Z \otimes |\phi(a(x,v))\rangle\langle\phi(a(x,v))|_{\widehat{C}} \otimes |a(x,v)\rangle\langle a(x,v)|_I \ \mathrm{tr}\left[(M_x \otimes M_v)\,\omega\right], \qquad \omega \tag{3.6}$$

Then:

(i) both maps satisfy the hypothesis of Definition 2.1 with one remaining processing $\bar{\mathsf{G}}$,

$$\mathcal{M}^{\mathrm{sq}} = \bar{\mathsf{G}} \circ \big(\mathrm{id}_A \otimes \mathcal{N}_F\big), \qquad \mathcal{M}^{\mathrm{ph}} = \bar{\mathsf{G}} \circ \big(\mathrm{id}_A \otimes \mathcal{N}_M\big),$$

where $\bar{\mathsf{G}}$ measures A with Alice's POVM $\{M_x\}$ and applies to the pair (x, v) the classical map $|x, v\rangle\langle x, v| \mapsto |r(x,v)\rangle\langle r(x,v)|_Z \otimes |\phi(a(x,v))\rangle\langle\phi(a(x,v))|_{\widehat{C}} \otimes |a(x,v)\rangle\langle a(x,v)|_I$. Alice's measurement is thus a step of $\bar{\mathsf{G}}$, where Definition 2.1 places the measurement of a party the declaration does not name as measuring, and not of the measurement channel;

(ii) if $M \preceq_\Lambda F$, then for every finite-dimensional E and every $\omega \in S_\leq(ABE)$

$$\big(\mathcal{M}^{\mathrm{ph}} \otimes \mathrm{id}_E\big)(\omega) = \big(\mathcal{M}^{\mathrm{sq}} \otimes \mathrm{id}_E\big)\big((\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_E)(\omega)\big). \tag{3.7}$$

Definition 3.10 (Reachable states and sound constraints)**.** Let $M \preceq_\Lambda F$ and let $S^{\mathrm{ph}} = (S_E^{\mathrm{ph}})_E$ be a family of *physical inputs*, one set $S_E^{\mathrm{ph}} \subseteq S_=(A^n B^n E)$ per admissible adversary system E. A state $\sigma \in S_=(A^n Q^n E)$ is *reachable* if

$$\sigma = \big(\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)(\rho) \qquad \text{for some } \rho \in S_E^{\mathrm{ph}}, \tag{3.11}$$

and S_E^{rch} denotes the set of reachable states. A family $C = (C_E)_E$ of subsets $C_E \subseteq S_=(A^n Q^n E)$, and any constraint whose solution set it is, is *sound* if $S_E^{\mathrm{rch}} \subseteq C_E$ for every admissible E. Both notions depend on Λ and on S^{ph}, which are suppressed in the notation and fixed throughout any one analysis.

Theorem 3.11 (Security transfer from a constrained squashed model). Let $M \preceq_\Lambda F$ and let $\mathcal{E}_M, \mathcal{E}_F$ be as in Definition 2.1. For every admissible adversary system E let $S_E^{\mathrm{ph}} \subseteq S_=(A^n B^n E)$ and $S_E^{\mathrm{sq}} \subseteq S_=(A^n Q^n E)$ satisfy

$$\big(\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)\big(S_E^{\mathrm{ph}}\big) \;\subseteq\; S_E^{\mathrm{sq}}. \tag{3.12}$$

If $\mathsf{sec}(\mathcal{E}_F, \sigma) \le \varepsilon$ for all $\sigma \in S_E^{\mathrm{sq}}$ and all E, then $\mathsf{sec}(\mathcal{E}_M, \rho) \le \varepsilon$ for all $\rho \in S_E^{\mathrm{ph}}$ and all E — with the *same* ε, and simultaneously for every security criterion in the sense of Definition 2.2.

Corollary 3.12 (Imposing and intersecting sound constraints)**.** Let $M \preceq_\Lambda F$ and fix a family S^{ph} of physical inputs.

(i) A family S^{sq} satisfies (3.12) if and only if it is sound.

(ii) An arbitrary intersection of sound families is sound.

(iii) Consequently, if $(C^{(i)})_{i \in I}$ is any collection of sound families and $\mathsf{sec}(\mathcal{E}_F, \sigma) \le \varepsilon$ for every admissible E and every $\sigma \in \bigcap_{i \in I} C_E^{(i)}$, then $\mathsf{sec}(\mathcal{E}_M, \rho) \le \varepsilon$ for every admissible E and every $\rho \in S_E^{\mathrm{ph}}$, with the same ε and for every criterion of the form (2.2).

(iv) The tightest sound family is S^{rch} itself and the loosest is $S_E^{\mathrm{sq}} = S_=(A^n Q^n E)$; enlarging a sound family costs tightness and never soundness.

Corollary 3.13 (Sound constraints from transported operators)**.** Let $(M; O) \preceq_\Lambda (F; \hat{O})$, fix a family S^{ph} of physical inputs and write $R := A^n E$. Let $\hat{O}^{(n)} \in \mathcal{L}(Q^n)$ be Hermitian and let $O^{(n)} := (\Lambda^\dagger)^{\otimes n}(\hat{O}^{(n)})$ be its pull-back. Then

$$\mathrm{tr}_{Q^n}\left[(\hat{O}^{(n)} \otimes \mathbb{1}_R)\,\sigma\right] = \mathrm{tr}_{B^n}\left[(O^{(n)} \otimes \mathbb{1}_R)\,\rho\right] \tag{3.13}$$

for every $\rho \in S_E^{\mathrm{ph}}$ and its image σ, and each of the following families of squashed states is sound:

(i) *(certified)* for each certificate case of Corollary 1.7 — identity, positivity, spectral bounds — read with $B^n, Q^n, \Lambda^{\otimes n}, R$ in place of B, Q, Λ, R, the family of squashed states satisfying its conclusion;

(ii) *(assumed)* $\{\sigma : \mathrm{tr}\big[(\hat{O}^{(n)} \otimes X)\sigma\big] \bowtie c\}$ for a fixed $\bowtie \in \{\leq, =, \geq\}$, a fixed $X \in \mathcal{L}(R)$, Hermitian unless $\bowtie$ is $=$, and a fixed c, provided $\mathrm{tr}\big[(O^{(n)} \otimes X)\rho\big] \bowtie c$ holds for every $\rho \in S_E^{\mathrm{ph}}$.

Operators with a prescribed pull-back are supplied round by round: if $\hat{O}^{(n)}$ is $\hat{O}_k$ on Q_j and $\mathbb{1}_Q$ on the remaining rounds, then $O^{(n)}$ is O_k on B_j and $\mathbb{1}_B$ elsewhere; and a product $\hat{O}_{k_1} \otimes \cdots \otimes \hat{O}_{k_n}$ pulls back to $O_{k_1} \otimes \cdots \otimes O_{k_n}$.

4 Operations on squashing relations

Proposition 4.1 (The squashing relation as a pre-order)**.** $M \preceq_{\mathrm{id}} M$, and $M \preceq_{\Lambda_1} F_1$ together with $F_1 \preceq_{\Lambda_2} F_2$ imply $M \preceq_{\Lambda_2 \circ \Lambda_1} F_2$. The same holds for the relation with carried operators, the operator tuples composing the same way: $(M; O) \preceq_{\Lambda_1} (F_1; \hat{O}^1)$ and $(F_1; \hat{O}^1) \preceq_{\Lambda_2} (F_2; \hat{O}^2)$ give $(M; O) \preceq_{\Lambda_2 \circ \Lambda_1} (F_2; \hat{O}^2)$.

Corollary 4.2 (Monotonicity of the worst-case criterion along the pre-order)**.** For a POVM G and a family S of inputs write $\varepsilon^*(G, S) := \sup_E \sup_{\sigma \in S_E} \mathsf{sec}(\mathcal{E}_G, \sigma)$. Let $F_2 \preceq_{\Lambda_{12}} F_1$ and let both squashed protocols use the same processing $\bar{\mathsf{G}}_j$ and K. Then for every family S,

$$\varepsilon^*(F_2, S) = \varepsilon^*\big(F_1, (\mathrm{id} \otimes \Lambda_{12}^{\otimes n} \otimes \mathrm{id}_E)(S)\big) \;\leq\; \varepsilon^*(F_1, S_1) \quad \text{for every } S_1 \supseteq (\mathrm{id} \otimes \Lambda_{12}^{\otimes n} \otimes \mathrm{id}_E)(S).$$

Lemma 4.3 (Linearity: post-processing, private settings and mixtures)**.** Let $\Lambda \colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel.

(i) *(post-processing)* If $M \preceq_\Lambda F$ and $P(z|v)$ is a stochastic matrix from $\mathcal{V}$ to a finite alphabet $\mathcal{U}$, then the post-processed families $(PM)_z := \sum_v P(z|v) M_v$ and $(PF)_z := \sum_v P(z|v) F_v$ are POVMs and $PM \preceq_\Lambda PF$.

(ii) *(a private classical variable)* For each $t \in \mathcal{T}_{\mathrm{set}}$ let $\{M_{y|t}\}_{y \in \mathcal{Y}}$ be a *conditional* POVM on B — complete for each fixed t, $\sum_y M_{y|t} = \mathbb{1}_B$ — and let $\{F_{y|t}\}_{y \in \mathcal{Y}}$ be conditional POVMs on the same Q with $M_{|t} \preceq_\Lambda F_{|t}$ for *one common* Λ. Let $(q_t)_{t \in \mathcal{T}_{\mathrm{set}}}$ be a probability distribution. Then, over the *joint* alphabet $\mathcal{V} = \mathcal{Y} \times \mathcal{T}_{\mathrm{set}}$,

$$\big\{q_t M_{y|t}\big\}_{(y,t)} \;\preceq_\Lambda\; \big\{q_t F_{y|t}\big\}_{(y,t)}, \tag{4.1}$$

which in the comma notation reads $\{M_{y,t}\} \preceq_\Lambda \{F_{y,t}\}$ with $M_{y,t} = q_t M_{y|t}$ and $F_{y,t} = q_t F_{y|t}$.

(iii) *(mixtures)* In the setting of (ii), $\{\sum_t q_t M_{y|t}\}_y \preceq_\Lambda \{\sum_t q_t F_{y|t}\}_y$.

Definition 4.4 (Joint alphabet, classes and the joint family of a private setting)**.** Let $\mathcal{Y}$ be a finite non-empty ordered *outcome alphabet*, let $\mathcal{T}_{\text{set}}$ be a finite non-empty ordered set of *settings*, and let $(q_t)_{t\in\mathcal{T}_{\text{set}}}$ be *class weights*, each a declared dyadic rational on the declared lattice (Definition 9.1 (b)), with $q_t > 0$ and $\sum_{t\in\mathcal{T}_{\text{set}}} q_t = 1$ *exactly*.

(a) *(the joint alphabet, its classes and weights)* The *joint alphabet* is $\mathcal{V} \coloneqq \mathcal{Y} \times \mathcal{T}_{\text{set}}$, ordered with y varying fastest, which is its declared order; its *classes* are $\mathcal{V}_t \coloneqq \mathcal{Y} \times \{t\}$, $t \in \mathcal{T}_{\text{set}}$, a partition $\mathcal{V} = \bigsqcup_{t\in\mathcal{T}_{\text{set}}} \mathcal{V}_t$, each class ordered as $\mathcal{Y}$ is; and the triple $\big(\mathcal{V}, \{\mathcal{V}_t\}_{t\in\mathcal{T}_{\text{set}}}, \{q_t\}_{t\in\mathcal{T}_{\text{set}}}\big)$ is the *joint alphabet with classes and weights*. When no private setting is declared, $\mathcal{T}_{\text{set}}$ is a singleton $\{*\}$ with $q_* = 1$, and $\mathcal{V} = \mathcal{Y} \times \{*\}$ is identified with $\mathcal{Y}$: one class, of weight one.

(b) *(the joint family)* Let B be a system and, for each $t \in \mathcal{T}_{\text{set}}$, let $\{\widetilde{M}_{y|t}\}_{y\in\mathcal{Y}}$ be a *conditional* POVM on B — complete at each fixed t, $\sum_y \widetilde{M}_{y|t} = \mathbb{1}_B$. Their *joint family* is

$$M_{(y,t)} \;\coloneqq\; q_t\, \widetilde{M}_{y|t}, \qquad (y,t) \in \mathcal{V}, \tag{4.2}$$

formed from the conditional elements and the weights by one product each, exactly. It is a POVM on B over $\mathcal{V}$ by Lemma 4.3 (ii) — $\sum_{(y,t)} q_t \widetilde{M}_{y|t} = \sum_t q_t \mathbb{1}_B = \mathbb{1}_B$ — and it satisfies the *class-marginal identity*

$$\sum_{v\in\mathcal{V}_t} M_v \;=\; q_t\, \mathbb{1}_B \qquad \text{for every } t \in \mathcal{T}_{\text{set}}, \tag{4.3}$$

an operator identity on B, by completeness of the conditional POVM at t.

Two consequences are immediate and are used throughout. First, given conditional relations $\widetilde{M}_{|t} \preceq_\Lambda \widetilde{F}_{|t}$, $t \in \mathcal{T}_{\text{set}}$, with one common channel Λ, Lemma 4.3 (ii) gives the relation $\{q_t \widetilde{M}_{y|t}\} \preceq_\Lambda \{q_t \widetilde{F}_{y|t}\}$ for the joint families: the setting never enters the target space, no block is opened for it, and the weights are carried by the target elements themselves. A relation for a joint family need not have a target of this form. Second, where every $\widetilde{M}_{y|t}$ is block diagonal for an orthogonal decomposition of $\mathcal{H}(B)$, so is every $M_{(y,t)}$, and (4.3) holds block by block; where such a joint family is itself a target, the decomposition being finite and the carried operators block diagonal for it as well, Corollary 4.7 applies to it.

Proposition 4.6 (Sector pinching as a squashing)**.** Let $\{\Pi_s\}_{s\in\mathbb{S}}$ be finitely many mutually orthogonal projections on $\mathcal{H}(Q)$ with $\sum_s \Pi_s = \mathbb{1}_Q$, and let

$\mathcal{P}_{\mathbb{S}}(\sigma) \coloneqq \sum_s \Pi_s \sigma \Pi_s$. Then $\mathcal{P}_{\mathbb{S}}$ is a channel with $\mathcal{P}_{\mathbb{S}}^\dagger = \mathcal{P}_{\mathbb{S}}$, and if every F_v is block diagonal, $F_v = \sum_s \Pi_s F_v \Pi_s$, then $F \preceq_{\mathcal{P}_{\mathbb{S}}} F$.

Corollary 4.7 (Restriction to block-diagonal squashed states)**.** Let $(M;O) \preceq_\Lambda (F;\hat{O})$ and suppose every F_v and every $\hat{O}_k$ is block diagonal for a finite orthogonal decomposition $\{\Pi_s\}_{s\in\mathbb{S}}$ of $\mathcal{H}(Q)$. Then

(i) $(M;O) \preceq_{\mathcal{P}_{\mathbb{S}}\circ\Lambda} (F;\hat{O})$, with the same target, the same tuple and the same pull-backs;

(ii) every state reachable through $\mathcal{P}_{\mathbb{S}} \circ \Lambda$ is block diagonal in each round, so "σ is block diagonal in every round's sector decomposition" is a sound constraint for that channel (Definition 3.10);

(iii) consequently the squashed analysis may be restricted to block-diagonal states without excluding a reachable one.

Proposition 4.8 (Tensor products of squashing relations)**.** If $M \preceq_\Lambda F$ and $M' \preceq_{\Lambda'} F'$ then $\{M_v \otimes M'_{v'}\}_{(v,v')} \preceq_{\Lambda\otimes\Lambda'} \{F_v \otimes F'_{v'}\}_{(v,v')}$.

Lemma 4.9 (Necessary conditions for a squashing relation)**.** Let $M \preceq F$, let $X = \sum_v a_v F_v$ be a real linear combination of target effects and let $X' \coloneqq \sum_v a_v M_v$. Then:

(i) *(spectral inclusion)* $\operatorname{spec}(X') \subseteq [\inf \operatorname{spec} X,\ \sup \operatorname{spec} X]$; in particular $\operatorname{spec}(M_v) \subseteq [\inf \operatorname{spec} F_v,\ \sup \operatorname{spec} F_v]$ and $\|M_v\| \le \|F_v\|$ for every v;

(ii) *(span monotonicity)* $\dim \operatorname{span}\{M_v\} \le \dim \operatorname{span}\{F_v\} \le (\dim \mathcal{H}(Q))^2$, and every linear relation $\sum_v a_v F_v = c\,\mathbb{1}_Q$ forces $\sum_v a_v M_v = c\,\mathbb{1}_B$.

5 Branching and flag targets

Lemma 5.1 (Flag PVM as a top element)**.** Let M be any POVM on any B with finite alphabet $\mathcal{V}$, and let $P^{\mathrm{Flg}} = \{|v\rangle\langle v|\}_{v\in\mathcal{V}}$ be the *flag PVM* on $\mathcal{H}(\mathrm{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ with orthonormal basis $\{|v\rangle\}_{v\in\mathcal{V}}$. Then

$$M \ \preceq_{\Lambda_{\mathrm{mp}}}\ P^{\mathrm{Flg}}, \qquad \Lambda_{\mathrm{mp}}(\rho) \coloneqq \sum_{v\in\mathcal{V}} \operatorname{tr}[M_v \rho]\ |v\rangle\langle v|\,. \tag{5.1}$$

More generally, if P is any PVM on a space Q with the same alphabet and $P_v = 0 \Rightarrow M_v = 0$, then $M \preceq P$.

Definition 5.2 (Generalized instrument)**.** A *generalized instrument* on B is a finite family $\{\mathcal{I}_j\}_{j\in J}$ of CP maps $\mathcal{I}_j \colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$ whose traces add up to the trace, $\sum_j \operatorname{tr}\circ\mathcal{I}_j = \operatorname{tr}$: branch j occurs with probability $\operatorname{tr}\mathcal{I}_j(\rho)$ and leaves the subnormalized state $\mathcal{I}_j(\rho)$ on the system B'_j, which may differ from branch to branch.

Lemma 5.3 (Instrument-conditioned squashing)**.** Let J be a finite set and let $\mathcal{I}_j \colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$ be CP maps with $\sum_j \operatorname{tr} \circ \mathcal{I}_j = \operatorname{tr}$, that is, a generalized instrument (Definition 5.2). For each j let $M^{(j)}$ be a POVM on B'_j over a common finite alphabet $\mathcal{V}$ with $(M^{(j)}; O^{(j)}) \preceq_{\Lambda_j} (F^{(j)}; \hat{O}^{(j)})$, where $F^{(j)}$ is a POVM on a finite-dimensional Q_j. Each $\mathcal{I}_j$ is trace non-increasing; write $\mathcal{I}_j^\dagger$ for its adjoint in the sense of Lemma 1.2, the completely positive map fixed by $\operatorname{tr}[X \mathcal{I}_j(\rho)] = \operatorname{tr}\left[\mathcal{I}_j^\dagger(X)\rho\right]$ for $X \in \mathcal{L}(B'_j)$ and $\rho \in \mathcal{T}(B)$. Then:

(i) $M_{v,j} \coloneqq \mathcal{I}_j^\dagger\big(M_v^{(j)}\big)$ defines a POVM on B with joint outcome $(v, j) \in \mathcal{V} \times J$;

(ii) with $Q \coloneqq \bigoplus_{j \in J} Q_j$ and $F_{v,j}$ the operator equal to $F_v^{(j)}$ in the j-th block and zero elsewhere,

$$\{M_{v,j}\} \preceq_\Lambda \{F_{v,j}\}, \qquad \Lambda(\rho) \coloneqq \bigoplus_{j \in J} \Lambda_j\big(\mathcal{I}_j(\rho)\big); \tag{5.2}$$

(iii) *(coarse graining)* with the same channel, $\big\{\sum_j M_{v,j}\big\}_v \preceq_\Lambda \big\{\bigoplus_j F_v^{(j)}\big\}_v$;

(iv) *(branch weights)* $\Lambda^\dagger(\Pi_j) = \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j})$, where Π_j projects onto the j-th block. In particular, if $\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = q_j\, \mathbb{1}_B$ for constants q_j, then for every finite-dimensional side system R and every reachable σ, $\operatorname{tr}_Q[(\Pi_j \otimes \mathbb{1}_R)\sigma] = q_j\, \sigma_R$, and in particular $\operatorname{tr}[\Pi_j \sigma] = q_j$;

(v) *(transported operators)* the composite carries the operator equal to $\hat{O}_k^{(j)}$ in block j and zero elsewhere, with pull-back $\mathcal{I}_j^\dagger\big(O_k^{(j)}\big)$.

Corollary 5.8 (Three special cases of instrument-conditioned squashing)**.** Let Λ_j and $F^{(j)}$ be branch channels and branch targets as in Lemma 5.3.

(i) *(convex mixture with a known weight)* If $M_v = (1-q)G_v + qR_v$ for POVMs G, R on B and $q \in [0,1]$, with $G \preceq_{\Lambda_G} F^{(G)}$ and R *arbitrary*, then $M \preceq \big\{F_v^{(G)} \oplus |v\rangle\langle v|_{\mathrm{Flg}}\big\}_v$, with the sound branch constraints $\operatorname{tr}[\Pi_1 \sigma] = 1 - q$ and $\operatorname{tr}[\Pi_{\mathrm{Flg}} \sigma] = q$, in the operator form of Lemma 5.3 (iv).

(ii) *(block-wise squashing)* Let $\mathcal{H}(B) = \bigoplus_j \mathcal{H}(B_j)$ be a finite orthogonal decomposition and let every M_v be block diagonal, $M_v = \bigoplus_j M_v^{(j)}$, with $M^{(j)} \preceq_{\Lambda_j} F^{(j)}$. Then $M \preceq \bigoplus_j F^{(j)}$.

(iii) *(POVM-conditioned squashing)* If $B = B_0 D$ with a POVM $\{N_j\}_{j \in J}$ on D and $M_v = \sum_j M_{v|j} \otimes N_j$ with $M_{|j} \preceq_{\Lambda_j} F_{|j}$ on Q_j, then

$\{M_v\}_v \preceq_\Lambda \{\bigoplus_j F_{v|j}\}_v$ with

$$\Lambda(\rho) = \bigoplus_{j\in J} \Lambda_j\Big(\operatorname{tr}_D\left[(\mathbb{1}_{B_0}\otimes N_j^{1/2})\,\rho\,(\mathbb{1}_{B_0}\otimes N_j^{1/2})\right]\Big). \tag{5.4}$$

Proposition 5.9 (Common and dominating targets)**.** In the setting of Lemma 5.3, suppose additionally $F^{(j)} \preceq_{\Psi_j} F$ for a single POVM F on a space Q_* (a *dominating target*; $\Psi_j = \mathrm{id}$ if all branch targets already coincide). Then

$$\Big\{\sum_j M_{v,j}\Big\}_v \preceq_{\Lambda'} F, \qquad \Lambda' := \sum_{j\in J} \Psi_j \circ \Lambda_j \circ \mathcal{I}_j.$$

Corollary 5.11 (Arbitrary unpopulated block in a flag-extended target)**.** Let $\{G_v\}_{v\in\mathcal{V}}$ be any POVM on a finite-dimensional space B_0 and let $\{H_v\}_{v\in\mathcal{V}}$ be any POVM on B_0. Then, on $\mathcal{H}(B_0)\oplus\mathcal{H}(\mathrm{Flg})$,

$$\left\{G_v\oplus|v\rangle\langle v|_{\mathrm{Flg}}\right\}_v \preceq_\Lambda \left\{H_v\oplus|v\rangle\langle v|_{\mathrm{Flg}}\right\}_v, \qquad \Lambda(\rho) := 0_{B_0}\oplus\sum_v \operatorname{tr}\left[(G_v\oplus|v\rangle\langle v|)\rho\right]\,|v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{5.5}$$

In particular $H_v = \frac{1}{|\mathcal{V}|}\mathbb{1}_{B_0}$ is admissible, and the squashing channel depends on G while the target does not.

Theorem 5.13 (Register-conditioned squashing)**.** Let $B = B_0 D$, let $\{N_j\}_{j\in J}$ be a POVM on D with J finite, and let

$$M_v = \sum_{j\in J} M_{v|j}\otimes N_j, \qquad v\in\mathcal{V}, \tag{5.6}$$

where $\{M_{v|j}\}_v$ is a POVM on B_0 for each j. Suppose $(M_{|j}; O^{(j)}) \preceq_{\Lambda_j} (F_{|j}; \hat{O}^{(j)})$ for each j, with $F_{|j}$ a POVM on a finite-dimensional Q_j. Put $\mathcal{H}(Q) = \bigoplus_{j\in J}\mathcal{H}(Q_j)$ with block projections Π_j and

$$\Lambda(\rho) := \bigoplus_{j\in J} \Lambda_j\Big(\operatorname{tr}_D\left[(\mathbb{1}_{B_0}\otimes N_j^{1/2})\,\rho\,(\mathbb{1}_{B_0}\otimes N_j^{1/2})\right]\Big). \tag{5.7}$$

Then Λ is a channel and:

(i) $\{M_v\}_v \preceq_\Lambda \{\bigoplus_j F_{v|j}\}_v$; in particular $\{M_v\}$ is a POVM on B;

(ii) the composite transports the operator $\hat{O}_k^{(j)}$ placed in block j to $O_k^{(j)}\otimes N_j$;

(iii) $\Lambda^\dagger(\Pi_j) = \mathbb{1}_{B_0} \otimes N_j$, so on every reachable σ and every finite-dimensional side system R,

$$\mathrm{tr}_Q\left[(\Pi_j \otimes \mathbb{1}_R)\,\sigma\right] = \mathrm{tr}_B\left[\left((\mathbb{1}_{B_0} \otimes N_j) \otimes \mathbb{1}_R\right)\rho\right]; \tag{5.8}$$

this is an identity, not a constraint, unless the right-hand side is a declared multiple of ρ_R;

(iv) if every $F_{|j}$ and every $\hat{O}_k^{(j)}$ is block diagonal for a finite orthogonal decomposition of $\mathcal{H}(Q_j)$, then every $\bigoplus_j F_{v|j}$ and every carried operator is block diagonal for the union decomposition of $\mathcal{H}(Q)$, and $\mathcal{P}_{\mathbb{S}} \circ \Lambda$ realizes the same relation with block-diagonal reachable states (Corollary 4.7).

6 Event-level device facts, attack sets and the licence

Proposition 6.1 (Obstruction to encoding event bounds in classical targets)**.** Call a target POVM F on Q *classical* if every F_v is diagonal in one fixed orthonormal basis $\{|f\rangle\}_f$ of $\mathcal{H}(Q)$. There exist a POVM M on $\mathcal{H}(B) = \mathbb{C}^2$, an event $\mathcal{O}_* \subseteq \mathcal{V}$ and a constant $p_* > 0$ with $\sum_{v\in\mathcal{O}_*} M_v \succeq p_* \mathbb{1}_B$ such that *no* pair (F, Λ) exists with F classical, $M \preceq_\Lambda F$ and $\sum_{v\in\mathcal{O}_*} F_v \succeq p_* \mathbb{1}_Q$. Concretely,

$$M_1 = \tfrac{1}{2}\,|0\rangle\langle 0|\,, \quad M_2 = \tfrac{1}{2}\,|+\rangle\langle +|\,, \quad M_3 = \mathbb{1}_B - M_1 - M_2, \quad \mathcal{O}_* = \{1,2\}, \quad p_* = \tfrac{1}{2}\Big(1 - \tfrac{1}{\sqrt{2}}\Big), \tag{6.1}$$

with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Definition 6.3 (Admissible attack sets in the consumer's format)**.** Let $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, \Theta_k)\}_{k=1}^m$ be a declared list of Hermitian $\hat{O}_k \in \mathcal{L}(Q)$, relations $\bowtie_k \in \{\preceq, =, \succeq\}$ and right-hand sides typed by the declared source mode. For a list whose entries carry real constants c_k, as every list this module emits does (Definition 6.5), the admissible sets are the attack sets of $\mathfrak{L}$ of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) at the per-round system $S = Q$, which the consuming contract states at the same system (Definition "Round model and licensed attack sets" (meat-core/def:roundmodel)); their displays at that system are, verbatim,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, QE'') := \Big\{\mathcal{A} \in \mathrm{CPTP}(E', QE'') :\ \mathcal{A}^\dagger\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k\Big\} \qquad \text{(trusted-source mode)}, \tag{6.2}$$

and, with the constraint operator lifted to the larger supplied system,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, AQE'') := \Big\{\mathcal{A} \in \mathrm{CPTP}(E', AQE'') :\ \mathcal{A}^\dagger\big(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k\Big\} \tag{6.3}$$

in untrusted-source mode, the two distinguished by the output system named in the argument. This contract writes $\bowtie_k$ in the operator spelling of (6.2) and (6.3) in both source modes, and everything it emits is admitted under either spelling. An entry that declares a Hermitian right-hand side Θ_k on Alice's system, rather than a real constant, is a *general* entry: it is admissible input to the consuming contract but is never licensed (Definition "Licensed constraint list" (protocol-compiler/def:homog)). Every entry this module emits carries a real scalar c_k and is written $(\hat{O}_k, \bowtie_k, c_k)$ (Definition 6.5, Theorem 6.7 (iv)).

Party-local entries. Where the measured system is a product $Q = \bigotimes_f Q_f$ and $\mathfrak{L}$ carries entries tagged party-local (Definition 6.5), the admissible set is cut out, for each such entry $(\hat{O}_k, \bowtie_k, c_k; f)$, by the condition

$$\big(\mathrm{id}_{R'} \otimes \mathcal{A}\big)^{\dagger}\Big(X \otimes \big(\hat{O}_k - c_k\, \mathbb{1}_{Q_f}\big) \otimes \mathbb{1}_{E''}\Big) \;\bowtie_k\; 0 \qquad \text{for every finite-dimensional } R' \text{ and every } X \in \mathrm{Pos}(R' \tag{6.4}$$

in place of the scalar condition of (6.2) or (6.3), where Q_{f^c} is the product of the other factors and R is A in untrusted-source mode and trivial in trusted-source mode; the untagged entries keep their conditions. The reference system R' is what carries Alice's retained system in trusted-source mode, where it is no output of $\mathcal{A}$. Taking R' trivial and $X = \mathbb{1}$ in (6.4) gives the scalar condition for the lifted operator $\mathbb{1}_{Q_{f^c}} \otimes \hat{O}_k$, so the set with party-local entries is contained in the set with the same entries read as scalars. The condition (6.4) is that of an entry tagged with the factor f in Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) (iii) at $S = Q$, which the consuming contract states at the same system. An attack sequence is $\mathfrak{L}$-admissible (an admissible attack sequence for $\mathfrak{L}$) if $\mathcal{A}_j$ lies in the corresponding set with $E' = E_{j-1}$ and $E'' = E'_j$ for every j.

Definition 6.5 (Licensed entries and the party-local form)**.** The list $\mathfrak{L}$ is *licensed* for the declared source mode when each of its entries either is untagged with its right-hand side in the form that mode admits, or is in the party-local form below. The untagged forms are $\Theta_k = c_k\, \tau_A$ with $c_k \in \mathbb{R}$ in trusted-source mode, the state-level entry reading $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\, \omega] \bowtie_k c_k\, \tau_A$; and $\Theta_k = c_k \in \mathbb{R}$ in untrusted-source mode, the entry reading $\mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\, \omega\Big] \bowtie_k c_k$, a comparison of real numbers (Definition "Licensed constraint list" (protocol-compiler/def:homog)). In both cases the untagged entries are written $(\hat{O}_k, \bowtie_k, c_k)$.

The *party-local form.* Where the measured system is declared as a product $Q = \bigotimes_f Q_f$ of party-scoped factors, an entry may in addition be *scoped to a factor* f. It is written $(\hat{O}_k, \bowtie_k, c_k; f)$ with $\hat{O}_k \in \mathcal{L}(Q_f)$ Hermitian and $c_k \in \mathbb{R}$, it carries the party-local tag, and its state-level reading, in either

source mode, is

$$\operatorname{tr}_{Q_f}\Big[\big(\mathbb{1}\otimes(\hat{O}_k - c_k\,\mathbb{1}_{Q_f})\big)\,\omega\Big] \;\bowtie_k\; 0 \qquad \text{as an operator on the systems of } \omega \text{ other than } Q_f, \tag{6.5}$$

those systems being A, where the declaration carries one, and the factors $Q_{f'}$, $f' \neq f$. Equivalently $\operatorname{tr}_{Q_f}[(\mathbb{1}\otimes\hat{O}_k)\omega] \bowtie_k c_k \operatorname{tr}_{Q_f}\omega$: the entry is homogeneous with respect to the marginal that is actually there. The attack-set licence of Theorem 6.7 covers a party-local entry only through a certificate on the factor's own squashing channel (its part (v)); the product structure that makes it sound is Lemma 6.6.

What the tag does not change. An entry without the tag is read in the form of its source mode, exactly as above. With one factor, the party-local reading in trusted-source mode is $\operatorname{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\omega] \bowtie_k c_k\,\omega_A$, which under the consumer's marginal condition $\omega_A = \tau_A$ is the trusted-source form; with one factor and $d_A = 1$ it is the untrusted-source comparison of real numbers. *With one factor, in untrusted-source mode and $d_A > 1$ it is neither*: it reads $\operatorname{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\omega] \bowtie_k c_k\,\omega_A$, an operator inequality on A against the marginal that is actually there, and the untrusted-source scalar entry is its trace (Lemma 6.6 (iii)), implied by it and not equivalent to it. With more than one factor it implies the untagged reading of the same $(\hat{O}_k, \bowtie_k, c_k)$ for the lifted operator $\mathbb{1}\otimes\hat{O}_k$, which is obtained from it by the partial trace over the other factors (and over A in untrusted-source mode; in trusted-source mode under the consumer's marginal condition $\omega_A = \tau_A$), and in general it is not implied by it.

Lemma 6.6 (Party-local transport)**.** Let the measured system be a tensor product $B = B_f \otimes B_{f^c}$ of a *part* B_f and the product B_{f^c} of all other parts, let $\Lambda_f\colon \mathcal{T}(B_f) \to \mathcal{T}(Q_f)$ and $\Lambda_{f^c}\colon \mathcal{T}(B_{f^c}) \to \mathcal{T}(Q_{f^c})$ be channels, and put $\Lambda \coloneqq \Lambda_f \otimes \Lambda_{f^c}$ and $Q \coloneqq Q_f \otimes Q_{f^c}$. Let $\hat{O} \in \mathcal{L}(Q_f)$ be Hermitian with a spectral pull-back certificate *on the part*,

$$\Lambda_f^\dagger(\hat{O}) \;\bowtie\; c\,\mathbb{1}_{B_f}, \qquad \bowtie \in \{\preceq, =, \succeq\}, \quad c \in \mathbb{R}. \tag{6.6}$$

Let R and R' be finite-dimensional systems, E', E'' finite-dimensional, and $\mathcal{A} \in \mathrm{CPTP}(E', R\,B\,E'')$ *any* channel, with $\mathcal{A}^{\mathrm{sq}} \coloneqq (\mathrm{id}_R \otimes \Lambda \otimes \mathrm{id}_{E''}) \circ \mathcal{A}$. Write $\hat{D} \coloneqq \hat{O} - c\,\mathbb{1}_{Q_f}$.

(i) *(channel level)* for every $X \in \mathrm{Pos}(R'R\,Q_{f^c})$,

$$\big(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}}\big)^\dagger\big(X \otimes \hat{D} \otimes \mathbb{1}_{E''}\big) \;\bowtie\; 0 \qquad \text{as an operator on } R'E', \tag{6.7}$$

the tensor factors written in the order $R'R\,Q_{f^c}$, Q_f, E''.

(ii) *(state level)* for every $\varrho \in \mathrm{Pos}(R'E')$, the operator $\omega \coloneqq \operatorname{tr}_{E''}\big[(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}})(\varrho)\big]$ on $R'R\,Q$ satisfies

$$\operatorname{tr}_{Q_f}\big[(\mathbb{1}_{R'R\,Q_{f^c}} \otimes \hat{D})\,\omega\big] \;\bowtie\; 0 \qquad \text{as an operator on } R'R\,Q_{f^c}, \tag{6.8}$$

equivalently $\mathrm{tr}_{Q_f}[(\mathbb{1}\otimes\hat{O})\,\omega] \bowtie c\,\mathrm{tr}_{Q_f}\omega$: the right-hand side is c times the marginal of ω *that is actually there*, and no marginal is assumed.

(iii) *(the scalar entry is its trace)* taking the trace of (6.8) gives $\mathrm{tr}\Big[(\mathbb{1}\otimes\hat{O})\,\omega\Big] \bowtie c\,\mathrm{tr}\,\omega$, the scalar form. With R' trivial and $R = A$, Q_{f^c} trivial (one factor), (6.8) reads $\mathrm{tr}_Q[(\mathbb{1}_A\otimes\hat{O})\omega] \bowtie c\,\omega_A$.

(iv) *(the product structure is needed)* if Λ is not a product the conclusion can fail while the scalar one holds. Let Q_f and Q_{f^c} be qubits, R, R' trivial, $\hat{O} = |0\rangle\langle 0|$, $c = \frac{1}{2}$, and let Λ discard its input and prepare $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Then $\Lambda^\dagger(\mathbb{1}\otimes\hat{O}) = \frac{1}{2}\mathbb{1}_B \succeq c\,\mathbb{1}$, so every squashed attack satisfies the scalar entry with $\succeq$; but $\mathrm{tr}_{Q_f}[(\mathbb{1}\otimes\hat{D})\omega] = \frac{1}{4}\,|0\rangle\langle 0| - \frac{1}{4}\,|1\rangle\langle 1|$, which is not positive semidefinite.

Theorem 6.7 (The attack-set licence). Let $(M;O) \preceq_\Lambda (F;\hat{O})$ with $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$, and suppose each carried operator has a *spectral pull-back certificate*: for each k there are $\bowtie_k \in \{\preceq, =, \succeq\}$ and $c_k \in \mathbb{R}$ with

$$O_k \;=\; \Lambda^\dagger(\hat{O}_k) \;\bowtie_k\; c_k\,\mathbb{1}_B \qquad \text{as operators on } \mathcal{H}(B). \tag{6.9}$$

Then, with $\mathfrak{L} := \{(\hat{O}_k, \bowtie_k, c_k)\}_{k=1}^m$:

(i) *(trusted-source mode)* for all finite-dimensional E', E'' and *every* $\mathcal{A} \in \mathrm{CPTP}(E', BE'')$, the composite $\mathcal{A}^{\mathrm{sq}} := (\Lambda\otimes\mathrm{id}_{E''})\circ\mathcal{A}$ lies in $\mathfrak{A}_{\mathfrak{L}}(E', QE'')$ of (6.2);

(ii) *(untrusted-source mode)* for every $\mathcal{A} \in \mathrm{CPTP}(E', ABE'')$, the composite $\mathcal{A}^{\mathrm{sq}} := (\mathrm{id}_A\otimes\Lambda\otimes\mathrm{id}_{E''})\circ\mathcal{A}$ lies in $\mathfrak{A}_{\mathfrak{L}}(E', AQE'')$ of (6.3);

(iii) *(the licensed reduction, in the declared mode)* consequently, for every attack sequence $(\mathcal{A}_j)_{j=1}^n$ against the physical protocol of the type (3.3) for the declared source mode, the sequence $(\mathcal{A}_j^{\mathrm{sq}})_j$ of the *same* branch of (3.4) is $\mathfrak{L}$-admissible — by (i) with the trusted branch of (3.3), by (ii) with the untrusted branch — and produces the *same* protocol output state, by the correspondingly named branch of Corollary 3.3; so security of the squashed round model against $\mathfrak{L}$-admissible sequences implies security of the physical protocol against all sequences of that type, with the same ε;

(iv) *(licensed entries)* $\mathfrak{L}$ is licensed for either source mode in the sense of Definition 6.5, with the same constants c_k; on every reachable state σ the transported entries hold in the form with the marginal that is present, $\mathrm{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\sigma] \bowtie_k c_k\,\sigma_A$, and so in the untrusted-source form $\mathrm{tr}\Big[(\mathbb{1}_A\otimes\hat{O}_k)\sigma\Big] \bowtie_k c_k$; and the trusted-source form $\mathrm{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\sigma] \bowtie_k c_k\,\tau_A$ holds on every reachable state whose marginal on A is τ_A.

(v) *(party-local entries, product measured system) in place of the hypotheses above*, which type $\hat{O}_k$ on Q and certify it against Λ, let $B = \bigotimes_f B_f$, let $M^f \preceq_{\Lambda_f} F^f$ for every factor with $\Lambda_f \colon \mathcal{T}(B_f) \to \mathcal{T}(Q_f)$, and put $M \coloneqq \{\bigotimes_f M^f_{v_f}\}$, $F \coloneqq \{\bigotimes_f F^f_{v_f}\}$ and $\Lambda \coloneqq \bigotimes_f \Lambda_f$, so that $M \preceq_\Lambda F$ (Proposition 4.8). Let each entry of $\mathfrak{L}$ be scoped to a factor f_k with $\hat{O}_k \in \mathcal{L}(Q_{f_k})$ and a certificate *on that factor's channel*, $\Lambda^\dagger_{f_k}(\hat{O}_k) \bowtie_k c_k \mathbb{1}_{B_{f_k}}$. Then for every attack $\mathcal{A}$ of either type of (3.3), the squashed attack of the same branch of (3.4) satisfies (6.4) for every k, and also the scalar condition of (6.2) or (6.3) for the lifted operator $\mathbb{1}_{Q_{f_k^c}} \otimes \hat{O}_k$ with the same c_k; every squashed sequence is admissible for $\hat{\mathfrak{L}}$ read with any subset of its entries tagged party-local, it produces the same protocol output state as the physical sequence, and the security implication of (iii) holds with the same ε. On every reachable state the entries hold in the form (6.5).

7 Typed links and reduction chains

Definition 7.1 (Typed link)**.** A *typed link* carries a stage $\mathsf{stage} \in \{\mathsf{source}, \mathsf{receiver}\}$. A link at the stage source is a source link in the sense of Definition "Source link" (transmitter/def:source-link): its descriptors are Alice-side, a transmitted system with a signal ensemble on either end and a channel witness between them, and no condition on it mentions the receiver's systems, outcome alphabets or measurements. *This module reads that type as the transmitter states it*: the channel witness is supplied either *as data*, in which case the source identity is decided on declared numbers, or *as a theorem constructor*, in which case the link carries an obligation record, namely the triple of the kind, the id of the soundness claim and the assumptions the obligation carries, which Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (A1) defines and which this module neither computes nor re-decides, together with the member of its declared descriptor family that its descriptors are; and a source link carries in addition that family, an optional retained-register sector map, its transported constraint entries with its licence slot, and its status. Which kinds are admitted at all, and which components a consumer decides *on data* (the emitted soundness registry, the triple compared against it entry by entry with those assumptions demanded in the ledger, and the provenance rule for a non-empty segment), is Definition "Source segment and its interface conditions" (transmitter/def:source-segment), conditions (A1) to (A3); this module re-decides none of them, adds none, and adds no source-stage kind. A *receiver-stage typed link* ℓ is a tuple

$$\ell = \Big(\mathsf{stage},\ \mathsf{kind},\ \big(B^{\mathrm{in}}, M^{\mathrm{in}}\big),\ \big(B^{\mathrm{out}}, M^{\mathrm{out}}\big),\ \Lambda_\ell,\ \big(\hat{O}^{\mathrm{out}}, O^{\mathrm{in}}\big),\ S^{\mathrm{rch}}_\ell,\ \mathfrak{L}_\ell,\ \mathsf{licence},\ \mathsf{status}\Big) \tag{7.1}$$

in which: stage = receiver; kind is one of the receiver-stage link kinds the module's interface declares (the core's kinds, and those a submodule's interface adds); $(B^{\text{in}}, M^{\text{in}})$ is the *source family* — the system and POVM the link starts from — and $(B^{\text{out}}, M^{\text{out}})$ its *target*; Λ_ℓ is the *channel witness*, with $\big(M^{\text{in}}; O^{\text{in}}\big) \preceq_{\Lambda_\ell} \big(M^{\text{out}}; \hat{O}^{\text{out}}\big)$; S_ℓ^{rch} is the *reachable-set statement* of Definition 3.10 for Λ_ℓ and the declared family of inputs; $\mathfrak{L}_\ell$ is the list of transported constraint entries $(\hat{O}_k, \bowtie_k, c_k)$ with, per entry, the certificate case of Corollary 1.7 that discharges it; licence is the *licence record* of the module's outputs, present exactly when $\mathfrak{L}_\ell \neq \varnothing$; and status $\in \{\text{exact}, \text{certified_outer}, \text{exploratory_numeric}\}$, with exact meaning that $\Lambda_\ell^\dagger(M_v^{\text{out}}) = M_v^{\text{in}}$ holds with no error term.

The count. A receiver-stage link has ten slots. Six of them are the required fields — source family, target, channel witness, reachable-set statement, transported constraint entries, licence record — and the remaining four are stage, kind, the carried operator pair, and status. Wherever this contract says "the six required fields" it means the first group; the tuple (7.1) is the whole of what an implementer binds.

Definition 7.2 (Reduction chain)**.** A *reduction chain* is a pair $\mathfrak{C} = (\mathfrak{C}_{\text{src}}, \mathfrak{C}_{\text{rec}})$ of a source segment and a *receiver segment*. The source segment $\mathfrak{C}_{\text{src}} = (\Xi_1, \dots, \Xi_{L_s})$, $L_s \geq 0$, is a source segment in the sense of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (written $\mathfrak{R}_{\text{src}}$ there): a list of source links (Definition "Source link" (transmitter/def:source-link)) satisfying that definition's interface conditions (S1)–(S4) — descriptor match, index preservation, the source identity where checkable, and the slots — every descriptor of which is Alice-side. The receiver segment $\mathfrak{C}_{\text{rec}} = (\ell_1, \dots, \ell_k)$, $k \geq 1$, is a list of receiver-stage typed links (Definition 7.1) that is

(C1) *composable*: $(B_i^{\text{out}}, M_i^{\text{out}}) = (B_{i+1}^{\text{in}}, M_{i+1}^{\text{in}})$ for $1 \leq i < k$;

(C2) *tuple-chained*: $\hat{O}_i^{\text{out}} = O_{i+1}^{\text{in}}$ for $1 \leq i < k$, so that each declared operator on the final target has a pull-back at every intermediate stage;

(C3) *stage-ordered*: every source-stage link precedes every receiver-stage link, which the pair form makes structural.

Its *composite* is the *pair*

$$\big(\Xi_{\mathfrak{C}},\ \Lambda_{\mathfrak{C}}\big), \qquad \Xi_{\mathfrak{C}} \coloneqq \Xi_{L_s} \circ \cdots \circ \Xi_1, \qquad \Lambda_{\mathfrak{C}} \coloneqq \Lambda_{\ell_k} \circ \cdots \circ \Lambda_{\ell_1}, \tag{7.2}$$

of the *source composite* — the composite the transmitter's (S3) writes Ξ, the identity when $L_s = 0$ — and the *receiver composite*. The two act on different systems and no condition above relates them: $\Xi_{\mathfrak{C}}$ carries the virtual

transmitted system to the transmitted one, on Alice's side, and $\Lambda_{\mathfrak{C}}$ the measured system to the target, on the receiver's. What the pair composes to *as a reduction* is Theorem 8.6, and Theorem 7.4 is what the receiver composite alone delivers.

Which chains are admitted. The rule is a condition on a *declaration* that carries both a non-empty source segment and a constraint list: such a declaration is *admitted* only where every entry of the list is licensed in the sense of Definition 6.5 — a real constant c_k as the right-hand side, the entry tagged or untagged — *and only in trusted-source mode*, Theorem 8.6 being stated there alone, so that a declaration pairing a non-empty source segment with untrusted-source mode is refused rather than admitted in silence. An entry declaring a Hermitian right-hand side on Alice's retained system, a *general* entry in the sense of Definition 6.3, is *refused* beside one. The rule is decided by the consumer that assembles those two.

Where the source segment is empty, $L_{\mathrm{s}} = 0$, the source composite is the identity, $A'' = A'$, the virtual ensemble is the specified one, and the pair (7.2) carries exactly the receiver composite: Alice's declared states are the ones the security statement is about, and the receiver segment is the whole of the chain's content (Theorem 7.4).

Theorem 7.4 (Composition of a chain)**.** Let $\mathfrak{C} = (\mathfrak{C}_{\mathrm{src}}, \mathfrak{C}_{\mathrm{rec}})$ be a reduction chain with receiver segment $\mathfrak{C}_{\mathrm{rec}} = (\ell_1, \ldots, \ell_k)$ and receiver composite $\Lambda_{\mathfrak{C}}$ of (7.2). Then:

(i) $\left(M_1^{\mathrm{in}}; O_1^{\mathrm{in}}\right) \preceq_{\Lambda_{\mathfrak{C}}} \left(M_k^{\mathrm{out}}; \hat{O}_k^{\mathrm{out}}\right)$, and $\Lambda_{\mathfrak{C}}^{\dagger} = \Lambda_{\ell_1}^{\dagger} \circ \cdots \circ \Lambda_{\ell_k}^{\dagger}$;

(ii) Theorem 3.1, Theorem 3.11 and Corollary 3.3 apply to the composite: the physical protocol with M_1^{in} and the final squashed protocol with M_k^{out} have the same output state, round by round and adaptively;

(iii) an entry $(\hat{O}_k, \bowtie_k, c_k)$ declared on the *final* target is sound for the receiver composite when the certificate (6.9) holds for its pull-back $\Lambda_{\mathfrak{C}}^{\dagger}(\hat{O}_k)$, which (C2) computes link by link; and Theorem 6.7 then applies with $\Lambda_{\mathfrak{C}}$;

(iv) the composite status is exact whenever every link's is, and no other composite value is asserted.

Nothing above reads the source segment: (i)–(iv) are statements about the receiver composite $\Lambda_{\mathfrak{C}}$ and hold at every length $L_{\mathrm{s}} \geq 0$. What the *pair* (7.2) composes to as a reduction is Theorem 8.6, under the source segment's own interface conditions and the admission rule of Definition 7.2, and this theorem does not restate it. Where the source segment is empty the source composite is the identity and the receiver composite is the whole of the chain's reduction.

8 The source-side reduction and the two-sided theorem

Lemma 8.1 (Licensed admissibility survives precomposition with a source channel)**.** Let $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, c_k)\}_{k=1}^{m}$ be a list of Hermitian $\hat{O}_k$, relations $\bowtie_k \in \{\preceq, =, \succeq\}$ and *real* constants c_k. Let $\mathcal{H}(E_{\mathrm{pre}})$ and $\mathcal{H}(E')$ be separable, let Q, E'', A, R and R' be finite-dimensional, and let $\Gamma \in \mathrm{CPTP}(E_{\mathrm{pre}}, E')$ be *any* channel. The sets below are the ones the displays (6.2), (6.3) and (6.4) of Definition 6.3 cut out, read at these registers with the adjoints of Lemma 1.2; where E_{pre} and E' are finite-dimensional they are the attack sets that definition names.

(i) *(trusted-source mode)* if $\mathcal{A} \in \mathfrak{A}_{\mathfrak{L}}(E', QE'')$ of (6.2), then $\mathcal{A} \circ \Gamma \in \mathfrak{A}_{\mathfrak{L}}(E_{\mathrm{pre}}, QE'')$;

(ii) *(untrusted-source mode)* if $\mathcal{A} \in \mathfrak{A}_{\mathfrak{L}}(E', AQE'')$ of (6.3), then $\mathcal{A} \circ \Gamma \in \mathfrak{A}_{\mathfrak{L}}(E_{\mathrm{pre}}, AQE'')$;

(iii) *(party-local entries)* let $Q = \bigotimes_f Q_f$ and let the entry $(\hat{O}_k, \bowtie_k, c_k; f_k)$ carry the party-local tag (Definition 6.5), so that its condition on $\mathcal{A}$ is (6.4). If $\mathcal{A}$ satisfies (6.4) for every finite-dimensional R' and every $X \in \mathrm{Pos}(R'RQ_{f_k^c})$, then so does $\mathcal{A} \circ \Gamma$, with E_{pre} in place of E'. Consequently the admissible set of Definition 6.3, with *any* subset of its entries tagged and the rest read as scalars, is closed under precomposition with a channel into its input register;

(iv) *(the real constant is what carries it)* replace the right-hand side $c_k \mathbb{1}_{E'}$ of (6.2) by a Hermitian $\Theta \in \mathcal{L}(E')$ that is not a real multiple of $\mathbb{1}_{E'}$. Then the condition $\mathcal{A}^\dagger(\hat{O} \otimes \mathbb{1}_{E''}) \succeq \Theta$ is not preserved by precomposition: take $E_{\mathrm{pre}} = E' = Q = \mathbb{C}^2$ with declared basis $\{|0\rangle, |1\rangle\}$, E'' one-dimensional, $\mathcal{A} = \mathrm{id}$ the identity channel $E' \to Q$, $\hat{O} = \Theta = |0\rangle\langle 0|$, and $\Gamma(\rho) := \mathrm{tr}[\rho]\ |1\rangle\langle 1|$. The condition holds for $\mathcal{A}$ with equality, and $(\mathcal{A} \circ \Gamma)^\dagger(\hat{O}) = 0 \not\succeq |0\rangle\langle 0| = \Theta$.

Lemma 8.3 (Exactness of the source map in trusted-source mode)**.** Let a source specification in *trusted-source mode* (Definition "Source specification" (transmitter/def:source-spec)) declare an alphabet $\mathcal{X}$ with weights $(p_x)_{x \in \mathcal{X}}$, the *specified* ensemble $\{p_x, \rho_x\}$ on the transmitted system A' and the virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$ on the virtual transmitted system A'', with $\mathcal{H}(A')$ separable and $\mathcal{H}(A'')$ finite-dimensional, as Definition "Source specification" (transmitter/def:source-spec) declares the virtual transmitted system, and let its source segment satisfy the interface conditions (S1)–(S4) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment), so that its composite $\Xi\colon \mathcal{T}(A'') \to \mathcal{T}(A')$ satisfies the *source identity*

$$\Xi\big(\rho_x^{\mathrm{v}}\big) = \rho_x \qquad \text{for every } x \in \mathcal{X}. \tag{8.1}$$

Fix n, a finite-dimensional adversary register E_0 and a protocol in the sense of Definition 2.1, and put

$$\rho^{\mathrm{sp}} \coloneqq \bigotimes_{j=1}^{n} \Big(\sum_{x \in \mathcal{X}} p_x \, |x\rangle\langle x|_{X_j} \otimes \rho_x \Big), \qquad \rho^{\mathrm{v}} \coloneqq \bigotimes_{j=1}^{n} \Big(\sum_{x \in \mathcal{X}} p_x \, |x\rangle\langle x|_{X_j} \otimes \rho_x^{\mathrm{v}} \Big), \tag{8.2}$$

the j-th factor living on $X_j A_j'$ and on $X_j A_j''$ respectively.

The two presentations. A *strategy* is a pair $a = (\zeta, (\mathcal{A}_j)_{j=1}^{n})$ of an initial adversary state $\zeta \in S_{=}(E_0)$ and channels

$$\mathcal{A}_1 \in \mathrm{CPTP}\big(A'^n E_0,\ B_1 E_1'\big), \qquad \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1},\ B_j E_j'\big) \quad (2 \le j \le n), \tag{8.3}$$

with E_j', E_j as in Definition 3.2: the *trusted* branch of (3.3), the only branch this lemma is about, in which A_j is Alice's and is no output of $\mathcal{A}_j$. A strategy of the *virtual* type is the same with A''^n in place of A'^n in (8.3). The *specified-source* presentation P^{sp} has these strategies and the output map $\mathsf{out}^{\mathrm{sp}}(a) \coloneqq$ the state (2.1) produces on the output registers of K together with E_n, from the input $\rho^{\mathrm{sp}} \otimes \zeta$, with B_j supplied by $\mathcal{A}_j$ and the classical X_j carried into $\bar{\mathsf{G}}_j$; the *virtual-source* presentation P^{v} has the strategies of the virtual type and the output map built the same way from $\rho^{\mathrm{v}} \otimes \zeta$, the two protocols agreeing in *every* other component: the measured system and its POVM, $\bar{\mathsf{G}}_j$ and K. The output registers are literally the same, which is condition (R1) of Definition 2.3.

(i) *(the source map)* define $r_{\mathrm{src}}(\zeta, (\mathcal{A}_j)_j) \coloneqq (\zeta, (\mathcal{A}_j^{\mathrm{v}})_j)$ by

$$\mathcal{A}_1^{\mathrm{v}} \coloneqq \mathcal{A}_1 \circ \big(\Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big), \qquad \mathcal{A}_j^{\mathrm{v}} \coloneqq \mathcal{A}_j \quad (2 \le j \le n). \tag{8.4}$$

Then r_{src} is an exact operational reduction from P^{sp} to P^{v} in the sense of Definition 2.3, with the whole of the virtual type as the admissible strategy set;

(ii) *(the two descriptions of the virtual protocol)* let the virtual protocol be described *entanglement-based* instead: Alice's retained systems $A_1, \dots, A_n$ carry, jointly with the virtual transmitted systems, the preparation $\Psi^{\otimes n}$, where Ψ is a source-replacement certificate for the declared data $(A, \mathbb{T}, \tau_A, \{M_x\})$ against $\{p_x, \rho_x^{\mathrm{v}}\}$ in any of the declared forms of Definition "Source-replacement certificate" (transmitter/def:srcert), satisfying (SR1)–(SR3); and x_j is produced by her declared measurement channel $\mathcal{N}_{\mathsf{A}}$, the measurement map of part (iii) of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency), applied to A_j at its declared time in round j. Write $\mathsf{out}^{\mathrm{eb}}$ for the resulting output map on the virtual type. Then, under A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), $\mathsf{out}^{\mathrm{eb}}(a) = \mathsf{out}^{\mathrm{v}}(a)$ for every

strategy a of that type, so the identity map on strategies is an exact operational reduction between the two presentations in either direction;

(iii) *(the composite)* consequently r_{src} is an exact operational reduction from P^{sp} to the virtual protocol in its entanglement-based description, the object the rest of this contract and the consuming engine analyse;

(iv) *(one channel for every label is essential)* (8.1) with *one* Ξ cannot be weakened to a family $(\Xi_x)_x$ with $\Xi_x(\rho_x^{\mathrm{v}}) = \rho_x$. Take $n = 1$, $\mathcal{X} = \{0,1\}$, $p_0 = p_1 = \frac{1}{2}$, $A' = A'' = \mathbb{C}^2$, $\rho_0^{\mathrm{v}} = \rho_1^{\mathrm{v}} = |0\rangle\langle 0|$, $\rho_0 = |0\rangle\langle 0|$, $\rho_1 = |1\rangle\langle 1|$, and $\Xi_x(\cdot) \coloneqq \mathrm{tr}[\cdot]\,\rho_x$. Every identity $\Xi_x(\rho_x^{\mathrm{v}}) = \rho_x$ holds, yet for the attack that measures the received signal in the basis $\{|0\rangle, |1\rangle\}$ and keeps the outcome in E_1', the marginal on X_1E_1' is perfectly correlated in the specified-source protocol and, for *every* attack of the virtual type, a product in the virtual one, so no map of strategies can make the two full outputs equal; the protocol exhibited in the proof announces nothing in round one, so that E_1 is E_1' there.

Theorem 8.6 (The two-sided reduction in trusted-source mode). Assume the hypotheses of Lemma 8.3: a trusted-source specification with virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$, a source segment whose composite Ξ satisfies the source identity (8.1), a source-replacement certificate satisfying (SR1)–(SR3), and A-SEQ. Assume in addition $M \preceq_\Lambda F$ together with a spectral pull-back certificate (6.9) for the list $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, c_k)\}$, or the factorwise hypotheses of Theorem 6.7 (v) for a list with entries carrying the party-local tag. Let P^{sp} be the specified-source presentation of Lemma 8.3, its receiver measuring M on B_j, and let $\mathsf{P}_{\mathfrak{L}}^{\mathrm{v,sq}}$ be the presentation whose protocol has the virtual source and the target F on Q_j and whose strategies are the $\mathfrak{L}$-admissible sequences of the virtual type (Definition 6.3). Define

$$r_{\mathrm{two}}\big(\zeta, (\mathcal{A}_j)_j\big) \coloneqq \big(\zeta, (\mathcal{A}_j^{\mathrm{v,sq}})_j\big), \qquad \mathcal{A}_1^{\mathrm{v,sq}} \coloneqq \lambda_1 \circ \mathcal{A}_1 \circ \big(\Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big), \quad \mathcal{A}_j^{\mathrm{v,sq}} \coloneqq \lambda_j \circ \mathcal{A}_j \ \ (j \geq 2), \tag{8.6}$$

where $\lambda_j \coloneqq \Lambda \otimes \mathrm{id}_{E_j'}$ is the trusted-source branch of (3.4), the only branch the strategies (8.3) of P^{sp} are typed in.

(i) *(one map, and the order of the two replacements is immaterial)* $r_{\mathrm{two}} = r_{\mathrm{sq}} \circ r_{\mathrm{src}} = r_{\mathrm{src}} \circ r_{\mathrm{sq}}$, where r_{src} is the source map (8.4) and r_{sq} the squashing map $(\mathcal{A}_j)_j \mapsto (\mathcal{A}_j^{\mathrm{sq}})_j$ of (3.4); both composites are defined on the same strategies and agree there. And r_{two} is an exact operational reduction from P^{sp} to the virtual, squashed protocol taken with *all* strategies of its type;

(ii) *(the image is admissible)* let E_0, every E_j' and Q be finite-dimensional, as the round model of Definition 3.2 makes the adversary's registers, every register of it being finite-dimensional, and Definition 1.1 makes

the target space, and let $\mathcal{H}(A'')$ be finite-dimensional, as it is in both cases of Definition "Source specification" (transmitter/def:source-spec): trivially for an ensemble declared as matrices, and by its clause (b) for a separable specified system. Then every $\mathcal{A}_j^{\mathrm{v,sq}}$ of (8.6) lies in the set Definition 6.3 assigns to its round, so r_{two} has its image in the strategy set of $\mathsf{P}_{\mathfrak{L}}^{\mathrm{v,sq}}$ and is an exact operational reduction into it. *No hypothesis on the specified transmitted system is needed*: $\mathcal{H}(A')$ may be separable and infinite-dimensional, as the transmitted system of a source model is, and it is not a register Definition 1.1 or the round model ranges over but a system in transit;

(iii) *(transfer, with the same ε)* consequently, if the virtual, squashed protocol is ε-secure against every $\mathfrak{L}$-admissible sequence of the virtual type, a security criterion in the sense of Definition 2.2 taking a value at most ε at each of them, then the specified-source physical protocol is ε-secure against *every* attack sequence of its type, with the *same* ε and simultaneously for every such criterion;

(iv) *(in the description the certificate presents)* (i) to (iii) hold with the virtual protocol read entanglement-based, Alice retaining A_j, which trusted-source mode is the mode that gives her, and measuring $\{M_x\}$, this being the description the consuming engine analyses;

(v) *(the empty source segment)* where the source segment is empty, Ξ is the identity, $A'' = A'$, $\rho_x^{\mathrm{v}} = \rho_x$, r_{src} is the identity map on strategies and r_{two} is r_{sq}; (i) to (iii) are then Corollary 3.3 and Theorem 6.7 (iii) word for word, and the theorem states nothing new about the case this module emits.

9 The number model and exact decisions

Definition 9.1 (The number model, the declared lattice and exact admissibility)**.** *(a) The model.* Every numerical datum this module reads or emits, the displacement record below aside, is a binary64 value on the wire and *is* that value: no exact object is posited behind it and none is approximated by it. A binary64 value is a dyadic rational, and *this contract reads it as that rational* (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). Every predicate this contract decides is decided, and every emitted number but one is formed, by finitely many operations in the substrate of Definition 9.2 and Lemma 9.3. *The exception is the emission displacement Δ_s of* (11.4): an entrywise maximum *modulus*, hence the square root of an exact rational and in general irrational, emitted as a binary64 rendering of a record of what the normalizer of Definition 11.1 did, in a direction nothing records; it decides nothing, and no statement of this contract reads it. That is the reading the exact island (Definition

"The exact island" (foundations/def:exact-island)) gives, which contains both the admission decisions on declared blocks and a producer's construction of what it emits. The only non-numerical data are combinatorial (alphabets, orders, block identifiers, a coarse-graining map, a branch index set, a cutoff) and carry no arithmetic.

(b) The declared lattice. The request declares one integer exponent κ, and every declared and every emitted operator entry, together with every *real scalar* the request declares of the following kinds — the class weights q_t, any certified weight, the entries of a declared stochastic matrix, and any declared device constant — satisfies

$$2^{\kappa}x \in \mathbb{Z}, \qquad |x| \le 1, \qquad 1 \le \kappa \le 53. \tag{9.1}$$

Such an x is exactly a binary64 number (Lemma "Binary64 numbers on dyadic lattices" (foundations/lem:binary64-lattice) (i)), its significand carrying at most $\kappa \le 53$ bits. The list is exhaustive: the constant a caller declares for a requested constraint entry is not among them, and where it is off the lattice it is rounded onto it or refused (Corollary 11.3 (iii)). It is a condition on a *set* of values: it says which numbers may be declared and nothing about how near one number is to another. A complex entry satisfies (9.1) when its real and imaginary parts do.

(c) Exact admissibility on the declared lattice. Let $\{A_v\}_{v\in I}$ be a finite family of matrices presented by its blocks over declared bases, indexed by the finite set I the invoking rule names (the outcome alphabet at a fixed setting for a declared block family, a class $\mathcal{V}_t$ of the datum of a reduction instance (Definition 10.1), $\mathcal{V}$ for the family this module emits), and let c be the constant the rule names for its completeness identity: 1 for a POVM, the class weight q_t for a class of a datum. The family is *exactly admissible on the declared lattice* when

(a) *(the sector map)* the block kind, the block identifiers and their order are the ones the consumer holds, a comparison of declared labels, decided and never measured;

(b) *(Hermiticity)* $A_v = A_v^{\dagger}$, entry for entry;

(c) *(positivity)* $A_v \succeq 0$, decided by (9.3);

(d) *(completeness)* $\sum_{v\in I} A_v = c\,\mathbb{1}$ on each block, as an identity of the entries;

(e) *(the sum rounds nowhere)* every entrywise binary64 addition of that accumulation, taken in the declared order of I, is exact, so that the *computed* sum *is* the sum of (d);

and every entry satisfies (9.1). Conditions (a)–(e) say that the family is exactly admissible with constant c for the sector map the consumer holds

(Definition "Exact admissibility of a declared operator family" (foundations/def:exact-admissibility)), each block being presented at the dimension the consumer's map declares for it. Condition (e) is a condition in its own right and is stated on the *additions*: a vanishing completeness *defect* says that the computed sum is the identity, not that $\sum_v A_v = c\,\mathbb{1}$ holds of the numbers. Lemma 9.5 shows that (e) is satisfiable on this contract's own emission, rather than a condition that would refuse every family.

(d) Scope, stated once and binding wherever this definition is invoked. This definition and the rules that invoke it are about data this contract requires to be presented as a finite list of matrices of declared dimension: the data sectors of a reduction instance (Definition 10.1 (I1)) and the declared blocks they are formed from; the declared classical scalars and matrices a submodule's route reads (a stochastic matrix and its descended form, the class weights, each certified weight, any declared device constant); and the family this module emits. They are *not* available for the objects an instance's hypotheses (H) are about: operators on a part of $\mathcal{H}(B)$ that no finite list of matrices presents, an above-cutoff family, a register POVM, an uncharacterised branch. There is nothing on those to decide, and their conditions remain hypotheses of the instance (Definition 10.1 (I1)).

Definition 9.2 (The arithmetic substrate)**.** The *substrate* of this contract is an ordered field $\mathbb{F}$ with $\mathbb{Q} \subseteq \mathbb{F} \subseteq \mathbb{R}$ together with its Gaussian extension $\mathbb{F}_i := \mathbb{F} + i\mathbb{F}$, which is closed under complex conjugation. What is fixed is the *predicate* each rule decides and the requirement that it be decided exactly; which such field carries the arithmetic is not. The choice cannot change any decision. Binary64 values are dyadic rationals (Definition 9.1 (a)), and every number a rule reads is formed from the declared entries and from rational constants by field operations, floors of rationals and roundings of rationals to a lattice, so it is rational; since $\mathbb{Q}$ is an ordered subfield of every admissible $\mathbb{F}$ with the induced order, and $\mathbb{Q} + i\mathbb{Q}$ a subfield of $\mathbb{F}_i$, each decision, a comparison or an equality of such numbers, is the same decision taken over $\mathbb{Q}$. In particular $\mathbb{F} = \mathbb{Q}$ always suffices. The latitude is bounded: it is a choice of arithmetic substrate, not of mathematics.

Lemma 9.3 (Exact decision of positivity and of an operator-norm bound)**.** Let $\mathbb{F}, \mathbb{F}_i$ be the substrate of Definition 9.2 and let $d, d' \geq 1$.

(i) Let $X = X^\dagger \in \mathcal{L}(\mathbb{C}^d)$ have all entries in $\mathbb{F}_i$ and write

$$\det\left(\lambda\,\mathbb{1}_d + X\right) = \sum_{k=0}^{d} e_k(X)\,\lambda^{d-k}, \qquad e_0(X) = 1, \tag{9.2}$$

so that $e_k(X)$ is the sum of the $k \times k$ principal minors of X. Then every $e_k(X)$ lies in $\mathbb{F}$, and

$$X \succeq 0 \quad\Longleftrightarrow\quad e_k(X) \geq 0 \ \text{ for every } k = 1, \ldots, d. \tag{9.3}$$

(ii) Let $Z \in \mathcal{L}(\mathbb{C}^{d'}, \mathbb{C}^{d})$ have all entries in $\mathbb{F}_i$ and let $\delta \in \mathbb{F}$, $\delta \geq 0$. Then

$$\|Z\| \leq \delta \quad \Longleftrightarrow \quad \delta^2\, \mathbb{1}_{d'} - Z^\dagger Z \;\succeq\; 0, \tag{9.4}$$

the right-hand side being a test of the form (i) on a Hermitian matrix with entries in $\mathbb{F}_i$.

Both predicates are therefore decided by finitely many field operations and sign comparisons in $\mathbb{F}$: no rounding, no threshold, and *no eigenvalue is computed*.

10 Reduction instances

Definition 10.1 (Reduction instance)**.** Fix the declared lattice exponent κ of Definition 9.1 (b) and a joint alphabet with classes and class weights $\big(\mathcal{V}, \{\mathcal{V}_t\}_{t\in\mathcal{T}_{\mathrm{set}}}, \{q_t\}\big)$ (Definition 4.4 (a)). Let $\mathbb{S}_{\mathrm{dat}}$ be a finite ordered list of *data sector*s, each s with a declared dimension $d_s \geq 1$ and a declared *support* $\mathcal{V}^{(s)} \subseteq \mathcal{V}$ that meets every class. A *datum* for these sectors is a family $A = \{A_v^{(s)}\}_{s\in\mathbb{S}_{\mathrm{dat}},\, v\in\mathcal{V}}$ of $d_s \times d_s$ matrices with entries in $\mathbb{F}_i$ and $A_v^{(s)} = 0$ for $v \notin \mathcal{V}^{(s)}$: these elements lie *outside the sector support*, and no step of this contract reads, rounds or repairs them. It is an *admissible datum*, written $A \in \mathcal{D}_{\mathrm{adm}}$, when for every s: (a) $A_v^{(s)} = (A_v^{(s)})^\dagger$ entry for entry; (b) $A_v^{(s)} \succeq 0$, decided by (9.3); (c) it is *class complete*,

$$\sum_{v\in\mathcal{V}_t} A_v^{(s)} \;=\; q_t\, \mathbb{1}_{d_s} \qquad \text{exactly, for every } t \in \mathcal{T}_{\mathrm{set}}; \tag{10.1}$$

and it is an *exactly admissible datum*, written $A \in \mathcal{D}_\kappa$, when in addition (d) every entry satisfies (9.1). These are conditions (b)–(d) of Definition 9.1 (c) and the lattice invariant, read per data sector with the class weight q_t as the constant of the completeness identity.

A *reduction instance* $\mathcal{J}$ over $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ consists of the following four parts.

(I1) *(the handed model and its datum)* A system B with $\mathcal{H}(B)$ separable (Definition 1.1); data sectors $\mathbb{S}_{\mathrm{dat}}$ with their declared dimensions and supports; a *handed datum* $A \in \mathcal{D}_{\mathrm{adm}}$, every entry of which is formed from the numbers the instance was handed by finitely many operations in $\mathbb{F}_i$; a finite list of *hypotheses (H)* about the rest of the model (operators without a finite presentation, facts about a part of $\mathcal{H}(B)$ the datum does not describe, a register POVM), each of which is either decided exactly on declared data, the request being refused where it fails, or carried as an assumption of the emitted ledger; and a *model map* $A' \mapsto M[A'] = \{M_v[A']\}_{v\in\mathcal{V}}$ that assigns to every admissible datum $A' \in \mathcal{D}_{\mathrm{adm}}$ a POVM on B over $\mathcal{V}$, under (H). The

handed model is $M[A]$, the model map at the handed datum. The *stability clause* requires that no hypothesis in (H) read an entry of the datum, so that (H) holds of $M[A']$ for every $A' \in \mathcal{D}_{\mathrm{adm}}$ as soon as it holds of the handed model; and that parts (I2)–(I4) be established for *every* $A' \in \mathcal{D}_{\mathrm{adm}}$ under (H), not only at the handed datum.

(I2) *(the target)* A finite ordered *sector set* $\mathbb{S} = \mathbb{S}_{\mathrm{dat}} \sqcup \mathbb{S}_{\mathrm{sc}}$, the second summand the *scalar sector*s, each $s \in \mathbb{S}$ with a declared dimension $d_s \geq 1$; for each scalar sector s and each $v \in \mathcal{V}$ a *scalar value* $\phi_s(v) \in [0,1]$ satisfying (9.1) with $\sum_{v \in \mathcal{V}} \phi_s(v) = 1$ exactly, each formed from the numbers the instance was handed by finitely many operations in $\mathbb{F}$ and, where a rounding onto the lattice is needed, by the *downward* rounding of a quantity the value is a certified lower bound of, any complementary value being recomputed from the rounded ones and never rounded on its own; the target space $\mathcal{H}(Q) \coloneqq \bigoplus_{s \in \mathbb{S}} \mathbb{C}^{d_s}$ with sector projectors Π_s, so that $\dim \mathcal{H}(Q) = \sum_s d_s$; and the *target family*

$$F_v[A'] \coloneqq \bigoplus_{s \in \mathbb{S}} F_v^{(s)}[A'], \qquad F_v^{(s)}[A'] \coloneqq \begin{cases} A_v'^{(s)} & s \in \mathbb{S}_{\mathrm{dat}}, \\ \phi_s(v)\, \mathbb{1}_{d_s} & s \in \mathbb{S}_{\mathrm{sc}}, \end{cases} \tag{10.2}$$

assigned one sector block at a time, so that $F_v[A']$ is sector-block-diagonal *by construction*: no off-sector entry is written and none is to be verified. The sector map of the instance is the list $\mathbb{S}$ with its order, each sector's dimension and its kind (data or scalar), and it is what a consumer compares (Definition 9.1 (c), condition (a)). The *default party partition* of the target assigns every factor of every $\mathcal{H}(Q_s)$, $s \in \mathbb{S}$, to *Bob*: *every sector of the target is Bob's.* That is the assignment the physical model makes: the target describes what reaches Bob's receiver, the handed model living on his system B. An instance may in addition be declared with *another party's label*, one label for the whole instance, and then the same assignment is made with that label in its place: every factor of every $\mathcal{H}(Q_s)$ is that party's, because the handed model lives on the system *that* party receives. Where no label is declared the label is Bob's. Nothing else in the instance depends on the label: it is a declared identifier, compared and never computed, and no condition of (I1)–(I4) reads it.

(I3) *(the existence theorem)* A channel $\Lambda[A'] = \mathcal{P}_{\mathbb{S}} \circ \Lambda_0[A'] \colon \mathcal{T}(B) \to \mathcal{T}(Q)$, given by a constructor and recorded as one, with $\mathcal{P}_{\mathbb{S}}$ the sector pinching of Proposition 4.6 for $\{\Pi_s\}_{s \in \mathbb{S}}$; and a finite *transported set* $\hat{O}[A']$ of sector-block-diagonal Hermitian operators on Q containing

every sector projector Π_s, with pull-backs $O[A']$; such that

$$\big(M[A'];\, O[A']\big) \;\preceq_{\Lambda[A']}\; \big(F[A'];\, \hat{O}[A']\big) \qquad \text{for every } A' \in \mathcal{D}_{\mathrm{adm}} \text{ under (H).} \tag{10.3}$$

By linearity of $\Lambda[A']^\dagger$, every real linear combination of transported operators is transported to the same combination of their pull-backs.

(I4) *(the certified entries)* A finite list $\mathfrak{L}[A'] = \{(\hat{O}_k, \bowtie_k, c_k)\}_k$ of *certified entries*, each $\hat{O}_k$ a real linear combination of transported operators of the form

$$\hat{O}_k \;=\; \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{k,v,s}\, \Pi_s F_v[A'] \Pi_s \;+\; Z_k, \tag{10.4}$$

with real coefficients $\alpha_{k,v,s}$ *on the lattice*, each an integer multiple of $2^{-\kappa}$, and Z_k a real combination of sector projectors with coefficients of the same kind, $\bowtie_k \in \{\preceq, =, \succeq\}$ and c_k a declared constant on the lattice, whose certificate (6.9), $\Lambda[A']^\dagger(\hat{O}_k) \bowtie_k c_k\, \mathbb{1}_B$, is *discharged* for every $A' \in \mathcal{D}_{\mathrm{adm}}$ under (H); and a *class-marginal flag* recording whether the class-marginal identity

$$\sum_{v\in\mathcal{V}_t} M_v[A'] \;=\; q_t\, \mathbb{1}_B \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}} \tag{10.5}$$

is *discharged*, that is, established by the construction for every $A' \in \mathcal{D}_{\mathrm{adm}}$ under (H), or not. The modulus bound of (9.1) is not imposed on a coefficient: $|x| \le 1$ is a condition on a declared or emitted *entry* (a value of the datum, a scalar value, a declared scalar or constant), where a POVM element and a completeness identity give it, and not on the coefficients of a linear combination, which a branching instance forms as the difference $\beta_s - c_k$ of two lattice multiples (Theorem 10.4), again a lattice multiple; *being on the lattice is all that is asked of a coefficient.* What (9.1) does bind is the entries of the operator the combination produces (Corollary 11.3 (iii)).

The instance *at* a datum $A' \in \mathcal{D}_{\mathrm{adm}}$ is the same B, (H), sectors, supports, scalar values, constructor and entry rule with A' in place of the handed datum; by the stability clause it is a reduction instance whenever $\mathcal{J}$ is.

Five consequences, each immediate. First, for every $A' \in \mathcal{D}_{\mathrm{adm}}$ the target $\{F_v[A']\}_{v\in\mathcal{V}}$ is a POVM on Q: every sector block is Hermitian and positive semidefinite, by (a) and (b) on a data sector and by $\phi_s(v) \in [0,1]$ on a scalar one; and it is complete, on a data sector because $\sum_{v\in\mathcal{V}} A_v'^{(s)} = \sum_t \sum_{v\in\mathcal{V}_t} A_v'^{(s)} = \sum_t q_t \mathbb{1}_{d_s} = \mathbb{1}_{d_s}$ by (10.1) and $\sum_t q_t = 1$, on a scalar sector because $\sum_v \phi_s(v) = 1$. Its data sectors *are* the datum, entry by entry, so the emitted target carries the datum and nothing else of the model.

Second, every $F_v[A']$, every Π_s and every transported operator is sector-block-diagonal, so Corollary 4.7 (i) applies: a relation established for $\Lambda_0[A']$ holds for $\mathcal{P}_{\mathbb{S}} \circ \Lambda_0[A']$ with the same target, the same transported set and the same pull-backs, which is why (I3) may be stated with the pinching in front, and the squashed analysis may be restricted to block-diagonal states without excluding a reachable one (Corollary 4.7 (iii)). *Third*, the pull-back of every sector projector is positive semidefinite, $\Lambda[A']^\dagger(\Pi_s) \succeq 0$, because the adjoint of a channel is a positive map, and $\sum_s \Lambda[A']^\dagger(\Pi_s) = \mathbb{1}_B$ by unitality; so $(\Pi_s, \succeq, 0)$ carries the certificate of Corollary 1.7 (ii) for *every* instance and needs no part of (I4) to say so. *That certificate travels with the transported set, whatever the request carries*, which is where (I3) puts every sector projector and where the artefact records the certificate case per entry; it becomes an *entry* of $\mathfrak{L}$ only where a caller requests it *as an entry* (Corollary 11.3 (i)), the emitted list being empty where the request carries no entry. *Fourth*, the relation pulls a class sum of the target back to the class sum of the model, $\Lambda[A']^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v[A']\big) = \sum_{v\in\mathcal{V}_t} M_v[A']$, by linearity; so (10.5) holds of the model exactly when the pull-back identity $\Lambda[A']^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v[A']\big) = q_t\mathbb{1}_B$ does, which is the expectation-form entry Corollary 11.3 (ii) emits where the flag says it is discharged, and never the operator identity $\sum_{v\in\mathcal{V}_t} F_v[A'] = q_t\mathbb{1}_Q$, which fails on every scalar sector whose values do not sum to q_t over $\mathcal{V}_t$. *Fifth*, $\dim \mathcal{H}(Q) = \sum_s d_s$ is a sum of declared dimensions, finite, and the sector map is a comparison of declared labels; both are read off the declaration with no arithmetic performed.

The link of Definition 7.1 an instance produces is the relation (10.3) at the declared datum, with the transported set and the entry list as its slots, and its chain (Definition 7.2) is that link preceded by whatever post-processing the instance's submodule applies before it.

Corollary 10.2 (The identity instance)**.** Let $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ be a joint alphabet with classes and weights, let $\mathbb{S}_0$ be a declared finite ordered sector set with dimensions d_s, and let $\{M_v\}_{v\in\mathcal{V}}$ be a POVM on $\mathcal{H}(B) = \bigoplus_{s\in\mathbb{S}_0} \mathbb{C}^{d_s}$ presented as the sector-block list $M_v = \bigoplus_s A_v^{(s)}$ of an admissible datum $A \in \mathcal{D}_{\mathrm{adm}}$ for the sectors $\mathbb{S}_0$ with the default supports $\mathcal{V}^{(s)} = \mathcal{V}$, that is, a declared family every block of which is decided exactly admissible on the declared lattice on entry with the class sums at q_t (Definition 9.1 (c)). Then the *identity instance* $\mathcal{J}_{\mathrm{id}}$, given by

(I1) B as given; $\mathbb{S}_{\mathrm{dat}} = \mathbb{S}_0$; the handed datum A; (H) empty; and the model map $M_v[A'] \coloneqq \bigoplus_s A_v'^{(s)}$;

(I2) $\mathbb{S}_{\mathrm{sc}} = \varnothing$, so that $Q = B$ and $F[A'] = M[A']$;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}}$ for the decomposition $\mathbb{S}_0$ (that is, $\Lambda_0 = \mathrm{id}$), independent of the datum, with the transported set $\{\Pi_s\}_{s\in\mathbb{S}_0}$ and pull-backs Π_s;

(I4) an empty entry list, and the class-marginal identity discharged,

is a reduction instance. Its chain is the identity link: one receiver-stage link of kind identity with $\Lambda_\ell = \mathcal{P}_{\mathbb{S}}$ (the identity on every state block diagonal for $\mathbb{S}_0$, and id outright when $\mathbb{S}_0$ has one sector), with $M^{\text{out}} = M^{\text{in}}$, the sector projectors carried to themselves, the reachable-set statement the declared input family, pinched, no entries, no licence and status = exact.

Corollary 10.3 (The flag instance of an uncharacterised branch)**.** Let $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ be a joint alphabet with classes and weights and let $\{M_v\}_{v\in\mathcal{V}}$ be any POVM on any system B with $\mathcal{H}(B)$ separable, about which nothing beyond that is declared. Then the *flag instance* $\mathcal{J}_{\text{flag}}[M]$, given by

(I1) B as given; no data sector, $\mathbb{S}_{\text{dat}} = \varnothing$, so that the datum is empty and $\mathcal{D}_{\text{adm}} = \{\varnothing\}$; (H) the single hypothesis that $\{M_v\}$ is a POVM on B; and the constant model map $M[\varnothing] \coloneqq M$;

(I2) one scalar sector (f) of dimension 1 for each $f \in \mathcal{V}$, in the order of $\mathcal{V}$, with $\phi_f(v) \coloneqq \delta_{vf}$, so that $\mathcal{H}(Q) = \mathcal{H}(\text{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ and $F_v = |v\rangle\langle v|_{\text{Flg}}$, the flag PVM P^{Flg} of Lemma 5.1;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}} \circ \Lambda_{\text{mp}} = \Lambda_{\text{mp}}$, the measure-and-prepare channel (5.1) of M, with the transported set $\{\Pi_f = |f\rangle\langle f|\}_{f\in\mathcal{V}}$ and pull-backs M_f;

(I4) an empty entry list, and the class-marginal identity discharged exactly when (H) is extended by the declared fact $\sum_{v\in\mathcal{V}_t} M_v = q_t \mathbb{1}_B$ for every t, as it is when the branch is declared to be a joint family (Definition 4.4 (b)), and not otherwise,

is a reduction instance. The normalizer of Definition 11.1 has nothing to act on: the declared datum is the handed one, no Δ_s exists, and the declared model is M itself.

Theorem 10.4 (The branching instance: an instrument followed by branch instances)**.** Let J be a finite ordered set, let $\{\mathcal{I}_j\}_{j\in J}$, $\mathcal{I}_j \colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$, be a generalized instrument (Definition 5.2), $\sum_j \text{tr} \circ \mathcal{I}_j = \text{tr}$, given independently of any datum, and for each j let $\mathcal{J}_j$ be a reduction instance on B'_j over one common joint alphabet with classes and weights $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$, with handed datum A_j, data sectors $\mathbb{S}_{j,\text{dat}}$ with supports $\mathcal{V}^{(j,s)}$, scalar sectors $\mathbb{S}_{j,\text{sc}}$ with values $\phi_{j,s}$, hypotheses (H_j), channel $\Lambda_j[\cdot]$, transported set $\hat{O}^{(j)}[\cdot]$ and entry list $\mathfrak{L}_j[\cdot]$. Write $\mathbb{S} \coloneqq \bigsqcup_{j\in J}\{j\} \times \mathbb{S}_j$, ordered by j and then as $\mathbb{S}_j$ is, $\Pi_{(j,s)}$ for its sector projectors, $\Pi_j \coloneqq \sum_{s\in\mathbb{S}_j} \Pi_{(j,s)}$ for the *branch projectors*, $\hat{X}|_j$ for the operator equal to $\hat{X}$ in block j and zero elsewhere, and $A \coloneqq (A_j)_{j\in J}$ for the joint datum. Then each of the two constructions (i) and (ii) is a reduction instance, with

- the channel
$$\Lambda[A'] \;\coloneqq\; \mathcal{P}_{\mathbb{S}} \circ \bigoplus_{j\in J} \Lambda_j[A'_j] \circ \mathcal{I}_j, \tag{10.6}$$

- the sector set $\mathbb{S}$, the hypotheses $(\mathrm{H}) = \bigcup_j (\mathrm{H}_j)$ together with the hypothesis that $\{\mathcal{I}_j\}$ is a generalized instrument and, for every j in a declared set $J_{\mathrm{w}} \subseteq J$ of branches with declared lattice constants q_j, the *branch-weight hypothesis* $\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = q_j \mathbb{1}_B$,

- the transported set consisting of the sector projectors and, for every j and every $\hat{X} \in \hat{O}^{(j)}$, the operator $\hat{X}|_j$, with pull-back $\mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\hat{X})\big)$, in particular
$$\Lambda^\dagger\big(\Pi_{(j,s)}\big) = \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\Pi_s)\big), \qquad \Lambda^\dagger\big(\Pi_j\big) = \mathcal{I}_j^\dagger\big(\mathbb{1}_{B'_j}\big), \tag{10.7}$$

- the entry list consisting of, for every branch j and every entry $(\hat{O}_k, \bowtie_k, c_k) \in \mathfrak{L}_j$, the entry
$$\big(\hat{O}_k|_j - c_k\,\Pi_j,\ \bowtie_k,\ 0\big), \tag{10.8}$$
together with the *branch-weight entry* $(\Pi_j, =, q_j)$ for every $j \in J_{\mathrm{w}}$, where that set is not empty:

(i) *(the joint-outcome form)* over the joint alphabet $\mathcal{V} \times J$ of Definition 4.4 (a) whose outcome alphabet is $\mathcal{Y} \times J$, ordered with the branch label major and the outcome minor, and whose settings are $\mathcal{T}_{\mathrm{set}}$: writing its element (y, j, t) as (v, j) with $v = (y, t) \in \mathcal{V}$, its declared order has $(v, j) < (v', j')$ iff $t < t'$, or $t = t'$ and $j < j'$, or $t = t'$, $j = j'$ and $y < y'$, so that every class $\mathcal{V}_t \times J$ is a contiguous run and the restriction of the order to $\mathcal{V} \times \{j\}$ *is* the order of $\mathcal{V}$; its classes are $\mathcal{V}_t \times J$ and its weights q_t. On it, the model map $M_{(v,j)}[A'] \coloneqq \mathcal{I}_j^\dagger\big(M_v^{(j)}[A'_j]\big)$; data sectors (j, s), $s \in \mathbb{S}_{j,\mathrm{dat}}$, of dimension $d_{j,s}$ and support $\mathcal{V}^{(j,s)} \times \{j\}$, with $A'^{(j,s)}_{(v,j')} \coloneqq \delta_{jj'} A'^{(s)}_{j,v}$; scalar sectors (j, s), $s \in \mathbb{S}_{j,\mathrm{sc}}$, with $\phi_{(j,s)}(v, j') \coloneqq \delta_{jj'} \phi_{j,s}(v)$; and the target $F_{(v,j)}[A'] = F_v^{(j)}[A'_j]\big|_j$, the branch target in block j and zero elsewhere;

(ii) *(the coarse-grained form)* over $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ itself: the model map $M_v[A'] \coloneqq \sum_j \mathcal{I}_j^\dagger\big(M_v^{(j)}[A'_j]\big)$; the same data and scalar sectors with supports $\mathcal{V}^{(j,s)}$, $A'^{(j,s)}_v \coloneqq A'^{(s)}_{j,v}$ and $\phi_{(j,s)}(v) \coloneqq \phi_{j,s}(v)$; and the target $F_v[A'] = \bigoplus_j F_v^{(j)}[A'_j]$.

In both forms the class-marginal identity is discharged when every branch discharges its own, and the flag is set in exactly that case; and the normalizer

of Definition 11.1 acts on the composite datum *branch by branch*: its output at the data sectors of branch j is the output of the normalizer of $\mathcal{J}_j$ at A_j, so that the declared datum is $A' = (A'_j)_j$ with A'_j the branches' declared data and the declared model is $M[A']$ at those.

Corollary 10.5 (The register instance: register-conditioned squashing of instances)**.** Let $B = B_0 D$, let $\{N_j\}_{j\in J}$ be a POVM on the register D with J finite and ordered, and for each j let $\mathcal{J}_j$ be a reduction instance on B_0 over one common joint alphabet with classes and weights, with handed model $\{M_{v|j}\}_{v\in\mathcal{V}}$, channel Λ_j, transported set $\hat{O}^{(j)}$ and entry list $\mathfrak{L}_j$. Let $\mathcal{I}_j$ be the instrument of Theorem 5.13, $\mathcal{I}_j(\rho) = \mathrm{tr}_D\left[(\mathbb{1}_{B_0}\otimes N_j^{1/2})\rho(\mathbb{1}_{B_0}\otimes N_j^{1/2})\right]$. Then the branching instance of Theorem 10.4 for $\{\mathcal{I}_j\}$ and $\{\mathcal{J}_j\}$, the *register instance*, is a reduction instance on B in both forms, with the handed model

$$M_v = \sum_{j\in J} M_{v|j}\otimes N_j \quad \text{(coarse-grained)}, \qquad M_{(v,j)} = M_{v|j}\otimes N_j \quad \text{(joint-outcome)}, \tag{10.9}$$

the channel (5.7) pinched, $\Lambda = \mathcal{P}_{\mathbb{S}}\circ\bigoplus_j \Lambda_j\circ\mathcal{I}_j$, the target $\bigoplus_j F_v^{(j)}$ resp. $F_v^{(j)}|_j$, the hypotheses $\bigcup_j(\mathrm{H}_j)$ together with "$\{N_j\}$ is a POVM on D" and, for every j in a declared set $J_{\mathrm{w}}\subseteq J$ with declared lattice constants q_j, "$N_j = q_j\mathbb{1}_D$", and:

(i) *(transport)* for every branch j and sector s of $\mathcal{J}_j$, and for every $\hat{X}\in\hat{O}^{(j)}$,

$$\Lambda^\dagger\big(\Pi_{(j,s)}\big) = \Lambda_j^\dagger(\Pi_s)\otimes N_j, \qquad \Lambda^\dagger\big(\Pi_j\big) = \mathbb{1}_{B_0}\otimes N_j, \qquad \Lambda^\dagger\big(\hat{X}|_j\big) = \Lambda_j^\dagger(\hat{X})\otimes N_j; \tag{10.10}$$

(ii) *(the entry clause)* every entry $(\hat{O},\bowtie,c)$ of a branch j is carried as $\big(\hat{O}|_j - c\,\Pi_j,\bowtie,0\big)$, with pull-back

$$\Lambda^\dagger\big(\hat{O}|_j - c\,\Pi_j\big) = \big(\Lambda_j^\dagger(\hat{O}) - c\,\mathbb{1}_{B_0}\big)\otimes N_j, \tag{10.11}$$

a tensor product of the branch's certified operator with $N_j\succeq 0$, so that the certificate holds at 0 with the branch's relation; the branch-weight entry $(\Pi_j,=,q_j)$ is listed for every $j\in J_{\mathrm{w}}$;

(iii) *(the class-marginal identity)* it is discharged when every branch discharges it, and the flag is set in exactly that case;

(iv) *(the declared model)* the normalizer acts branch by branch, and the declared model is

$$M'^{\mathrm{ph}}_{v} = \sum_{j\in J} M'^{\mathrm{ph}(j)}_{v}\otimes N_j \quad \text{resp.} \quad M'^{\mathrm{ph}}_{(v,j)} = M'^{\mathrm{ph}(j)}_{v}\otimes N_j, \tag{10.12}$$

with $M'^{\mathrm{ph}(j)}$ the declared model of branch j.

Lemma 10.6 (Precomposition of a reduction instance with a channel)**.** Let B and B' be systems with separable Hilbert spaces, and let $\Phi\colon \mathcal{T}(B) \to \mathcal{T}(B')$ be a channel with adjoint $\Phi^\dagger\colon \mathcal{L}(B') \to \mathcal{L}(B)$ (Lemma 1.2).

(i) *(relations)* If $(M';O') \preceq_\Lambda (F;\hat{O})$ with $\Lambda\colon \mathcal{T}(B') \to \mathcal{T}(Q)$, then $\Phi^\dagger(M') := \{\Phi^\dagger(M'_v)\}_{v\in\mathcal{V}}$ is a POVM on B and

$$\big(\Phi^\dagger(M');\Phi^\dagger(O')\big) \preceq_{\Lambda\circ\Phi} \big(F;\hat{O}\big), \qquad (\Lambda\circ\Phi)^\dagger = \Phi^\dagger\circ\Lambda^\dagger. \quad (10.13)$$

If $\Lambda^\dagger(\hat{O}_k) \bowtie c\,\mathbb{1}_{B'}$ for a relation $\bowtie \in \{\preceq, =, \succeq\}$ and a real c, then $(\Lambda\circ\Phi)^\dagger(\hat{O}_k) \bowtie c\,\mathbb{1}_B$, with the same relation and the same constant.

(ii) *(instances)* Let $\mathcal{J}$ be a reduction instance on B' with model map $M[\cdot]$, channel $\Lambda[\cdot] = \mathcal{P}_\mathbb{S}\circ\Lambda_0[\cdot]$ and pull-backs $O[\cdot]$. The *precomposed instance* $\mathcal{J}\circ\Phi$ has the system B; the joint alphabet, data sectors, supports, handed datum, hypotheses (H), scalar values, sector set, target family, transported set, entry list and class-marginal flag of $\mathcal{J}$; the model map $A' \mapsto \Phi^\dagger(M[A'])$; the channel $A' \mapsto \Lambda[A']\circ\Phi = \mathcal{P}_\mathbb{S}\circ(\Lambda_0[A']\circ\Phi)$; and the pull-backs $A' \mapsto \Phi^\dagger(O[A'])$. It is a reduction instance on B.

(iii) *(at a datum, and the emission)* For every admissible datum A', $\mathcal{J}\circ\Phi$ at A' is $\mathcal{J}$ at A' precomposed with Φ. The emission normalizer produces the same declared datum from the handed datum of either, so the emitted target, scalar values, entries and constants of $\mathcal{J}\circ\Phi$ are those of $\mathcal{J}$; its emitted channel is the emitted channel of $\mathcal{J}$ precomposed with Φ, and its declared model is $\Phi^\dagger$ of the declared model of $\mathcal{J}$.

11 Exact emission

Definition 11.1 (The emission normalizer, the declared datum and the declared model)**.** Let $\mathcal{J}$ be a reduction instance (Definition 10.1) with handed datum $A \in \mathcal{D}_{\mathrm{adm}}$, formed exactly. Those numbers *are* the data sectors of the emitted target (Definition 10.1 (I2)), so they are this module's own declaration to its consumer. Where the exact arithmetic that formed them has left the declared lattice (9.1), they are re-established on it, per data sector s and per class t, by the following steps, the *emission normalizer*. The steps act on the elements $A_v^{(s)}$ with $v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}$, the elements of the class t in the sector's support; the elements outside the sector support are exactly zero, lie on the lattice, and are touched by no step. Every positive semidefiniteness below is decided exactly by (9.3), on matrices with entries in $\mathbb{F}_i$. The structural support of a matrix is the set of indices i whose row i or column i holds a non-zero entry (Definition "Snapping declared data onto a lattice" (foundations/def:snap)). The *designation order* ranks the elements of a class

in the sector's support by largest exact trace and, among equal traces, by the declared order of the class. *A tie is exact and ordinary here*: the traces compared are exact rationals, and a sector symmetric under a relabelling of the class ties every element of it, so "largest trace" alone names no element; the declared order is the order in which this contract reads $\mathcal{V}$, the outcome varying fastest, and it is emitted with the artefact, so the order is a function of the declaration and of nothing else. *Rounding to nearest* a matrix at κ rounds every real and imaginary part of every entry to the nearest integer multiple of $2^{-\kappa}$, *ties to even*: a part lying exactly halfway between two multiples goes to the one whose integer numerator is even. *The tie is exact*: the rounding is carried out in the substrate of Lemma 9.3 on exact rationals, so a halfway part is an exact equality rather than a floating-point near-miss. *Ties to even is odd-symmetric*: negating a part negates both of the multiples nearest it and their numerators, which preserves evenness, so the rounding commutes with $X \mapsto X^\dagger$ — mirror entries have equal real parts and opposite imaginary parts before it and therefore after it — and a matrix rounded from a Hermitian one is Hermitian; a zero entry stays zero.

(1) *Keep.* A class t at a sector s every element of which satisfies (9.1) entry by entry is *kept*: $A_v'^{(s)} := A_v^{(s)}$ for every $v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}$, and no further step acts on it.

(2) *Snap for fidelity.* At every other class and sector, for $w = 0$ and then $w = 2^{-\kappa}, 2^{-\kappa+1}, \ldots, 2^{-1}$ in this order, let P_v^w be $A_v^{(s)}$ with every off-diagonal entry multiplied by $1 - w$, that is $(1-w)A_v^{(s)} + w\,T_v$ with T_v the diagonal part of $A_v^{(s)}$. Every element of the class in the sector's support receives P_v^w rounded to nearest at κ, and the *class residue* $E^w := q_t \mathbb{1}_{d_s} - \sum_{v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}} A_v'^{(s)}$, a Hermitian matrix of lattice values, is split by its *components*: the classes of the equivalence relation on the indices i whose row i of E^w is non-zero generated by $i \sim l$ whenever $E^w_{il} \neq 0$. For a component C write E_C for the matrix that agrees with E^w on the entries whose row and column both lie in C and is zero elsewhere. Each E_C is added to the first element $A_u'^{(s)}$ in the designation order whose structural support in the handed datum, that of $A_u^{(s)}$, contains C, its *host*. The first w at which every component has a host and every element so formed is decided positive semidefinite gives the family, and w is recorded. Where no w does, the class is snapped by step (3).

(3) *The last resort.* The support-safe steps of (2) produced no snap of the class, and the class is snapped by a uniform lift, applied to the class with constant q_t and starting from the handed datum. Let n be the number of elements of the class in the sector's support, v_1 the first of

them in the designation order, and, for $w \in (0,1)$,

$$B_v^w \;\coloneqq\; (1-w)\,A_v^{(s)} + w\,\frac{q_t}{n}\,\mathbb{1}_{d_s}, \qquad v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}, \tag{11.1}$$

a family that is Hermitian, complete at q_t, with entries in $\mathbb{F}_i$, and positive definite in every element, since $q_t > 0$ at a class that is not kept (at $q_t = 0$ completeness and positive semidefiniteness make every element zero, which is on the lattice). Its designation order is that of the class, $\operatorname{tr} B_v^w$ being an increasing affine function of $\operatorname{tr} A_v^{(s)}$. For $w = 2^{-\kappa}, 2^{-\kappa+1}, \dots, 2^{-1}$ in this order, every element other than v_1 receives B_v^w rounded to nearest at κ, and v_1 receives $q_t\mathbb{1}_{d_s}$ minus their sum; the first w at which every element so formed is decided positive semidefinite gives the family. Where no w does, every element $v \neq v_1$ receives the first matrix $A_v'^{(s)}$ of the following list that is Hermitian, satisfies (9.1) entry by entry, and satisfies

$$0 \;\preceq\; A_v'^{(s)} \;\preceq\; B_v^{1/2}\,: \tag{11.2}$$

$B_v^{1/2}$ itself; then, for $r = \kappa, \kappa-1, \dots, 1$ in this order, the shift $B_v^{1/2} - 2^{-r}\mathbb{1}_{d_s}$ and the scaling $(1-2^{-r})B_v^{1/2}$, each rounded to nearest at κ; finally the zero matrix, which always qualifies, $B_v^{1/2}$ being positive semidefinite; and v_1 receives $q_t\mathbb{1}_{d_s}$ minus their sum. The weight w, or the fall-back, is recorded, and the class is flagged as one whose support is not preserved: an element may then hold a non-zero entry in a row or column in which the handed element holds none, that is, outside its structural support (Definition "Snapping declared data onto a lattice" (foundations/def:snap)).

(4) *All six conditions are decided last*, after (2) or (3), each of which changes partial sums from the touched element onward: (a)–(e) of Definition 9.1 together with the lattice invariant (9.1) of every entry, read on the target family (10.2) the datum enters. They are the six Theorem 11.2 (ii) decides; a family for which one of them fails is refused.

A class at a sector that step (1) does not keep is *moved*, whichever of (2) and (3) snapped it. Write A' for the resulting datum — the *declared datum*, an element of $\mathcal{D}_\kappa$ by (1) or the step of (2) and (3) that snapped the class, each of which makes it Hermitian, on the lattice and class complete exactly, positive semidefinite by an exact decision in (2) and in (3) at a weight of its ladder, and by construction in the fall-back of (3), and by (4), which decides all six — and

$$M'^{\mathrm{ph}} \;\coloneqq\; M[A'], \qquad \text{that is} \qquad M'^{\mathrm{ph}}_v \;=\; M_v[A'] \quad (v \in \mathcal{V}), \tag{11.3}$$

for the *declared model*: the handed model with the datum replaced by A', as the instance's model map defines it. Per data sector s, those being the sectors the normalizer touches and the only ones both data have,

$$\Delta_s \;\coloneqq\; \max_{v\in\mathcal{V}} \left\| A'^{(s)}_v - A^{(s)}_v \right\|_{\max} \tag{11.4}$$

is the *emission displacement*; it is recorded per data sector, together with the kept and the moved classes, for a moved class the step that snapped it, the weight w of (2) or (3) or the fall-back of (3), the hosts of the components of (2), and the flag of a class snapped by (3).

The normalizer has five properties, each of which Theorem 11.2, or a claim that instantiates it, uses. *First, $A' = A$ and every $\Delta_s = 0$ whenever the exact construction already satisfies* (9.1): every class is then kept, the normalizer is the identity and nothing below has an antecedent. That is the ordinary case at a nominal declaration and is not the general one: every exact step of a construction multiplies, a product of dyadics adds significand widths, and a lattice established at the instance is closed under none of them; that is the case in which a producer re-establishes its lattice at an emission. *Second, the normalizer reads data sectors and nothing else.* No object the hypotheses (H) of the instance are about — an operator without a finite presentation, a family on a part of $\mathcal{H}(B)$ the datum does not describe, a register POVM — is read, and none is repaired; so (H) holds of M'^{ph} by the stability clause of Definition 10.1 (I1), M'^{ph} is a POVM on B by (I1) at A', and every statement the instance makes about its model under (H) holds of M'^{ph}. For an instance whose datum is a family of preserved blocks below a cutoff, this is the statement that the declared model agrees with the handed one above the cutoff and satisfies every above-cutoff fact the handed one does. *Third, what the normalizer produces is a datum, and the family the emitted identity is about is the model the instance builds from it.* Two objects, two symbols, and every statement below names which it is on: A' is the declared datum, a finite list of sector blocks carried entry by entry in the emitted target; M'^{ph} is the declared model on B, the object an adjoint landing on B can equal (Theorem 11.2 (iii)). Where the model map is the identity on a finite B the two coincide; where the model has parts the datum does not describe — an above-cutoff family, further branches — they do not, and the instance's model map says how A' sits inside M'^{ph}. For a branching instance (Theorem 10.4) the normalizer acts branch by branch, so the declared model is the branches' declared models composed. *Fourth, this is a snap of what this module declares and never of what it was handed*, which is the condition under which a producer may snap at all. *Fifth, no step revives a zero row or column, save the last resort (3)*: for every data sector s and every $v \in \mathcal{V}$ whose class at s is not snapped by (3) the structural support of $A'^{(s)}_v$ is contained in that of $A^{(s)}_v$. In (2) the family $\{T_v\}$ of diagonal parts is complete at q_t, the diagonal of $\sum_v A^{(s)}_v = q_t \mathbb{1}_{d_s}$ being q_t in every place,

and T_v is diagonal, with non-negative entries, non-zero only at indices of the structural support of $A_v^{(s)}$, so P^w is a mixture with a complete diagonal family that adds only on coordinates of each element's own structural support; P_v^w has the structural support of $A_v^{(s)}$, and a rounding to nearest keeps a zero entry zero. Each E_C has its non-zero rows and columns in C, which the host's structural support contains, and the entries of E^w between two components are zero by their definition, so $E^w = \sum_C E_C$ and every non-zero entry of the residue goes to a host. In (2) the residue need not be positive semidefinite, and the family is taken only where every element is decided positive semidefinite. No element is mixed with a multiple of the identity outside (3). In (3) the elements of the class in the sector's support may gain rows and columns, which is what its flag discloses; the elements outside the support stay zero there too. And (3) always yields a family whose elements are positive semidefinite: at a weight of its ladder by decision, and in the fall-back because every element of $B^{1/2}$ is positive definite, the rounded elements satisfy (11.2), and $A_{v_1}'^{(s)} = B_{v_1}^{1/2} + \sum_{v \neq v_1} \left(B_v^{1/2} - A_v'^{(s)} \right) \succeq B_{v_1}^{1/2} \succ 0$ by the completeness of $B^{1/2}$.

Theorem 11.2 (Exact emission of a reduction instance). Let the declared data pass every refusal rule of this module's interface, so that every declared block and scalar it reads is exactly admissible on the declared lattice (Definition 9.1). Let $\mathcal{J}$ be a reduction instance (Definition 10.1) with handed datum A, and let A' be the declared datum the normalizer of Definition 11.1 produces from it (where the normalizer refuses, nothing is emitted). Let $\{F_v\} \coloneqq F[A']$, $\Lambda \coloneqq \Lambda[A']$ and $\mathfrak{L} \coloneqq \mathfrak{L}[A']$ be the target, the channel and the entry list of the instance at A', and $M'^{\mathrm{ph}} = M[A']$ the declared model (11.3). Then:

(i) *(everything on the list below is computed exactly, and the displacement Δ_s is the one emitted number that is not on it)* every number on the following list is obtained from the numbers the instance was handed by finitely many operations in $\mathbb{F}_i$, floors and comparisons: the data-sector values $A_v'^{(s)}$; the scalar values $\phi_s(v)$; the sector projectors; and every emitted constraint operator and constant of Corollary 11.3. *That list is everything this module emits with exactly one exception*: the emission displacement Δ_s of (11.4), an entrywise maximum *modulus* and hence the square root of an exact rational, in general irrational, which is emitted only as a record of what the emission normalizer did and which no statement of this contract reads. *No floating-point comparison, no eigenvalue routine, no enclosure and no interval occurs anywhere in that list*; the intermediate numbers an instance forms on the way to its datum and its scalar values are listed by the corollary that instantiates this theorem for it;

(ii) *(the emitted family is exactly admissible, and is decided so before it is*

emitted) the family $\{F_v\}$, presented as the sector-block list (10.2), satisfies conditions (a)–(e) of Definition 9.1 together with (9.1) of every entry — six conditions — and each of the six is *decided*, exactly, on the numbers about to be emitted: (a) by comparison of declared labels against the instance's sector map; (b) and (d) by entrywise equality and (c) by Lemma 9.3, per sector block; (e) by Lemma 9.5, whose hypothesis (9.5) is a declared input condition; and (9.1) entry by entry. *A family for which any of the six cannot be established is not emitted. And the same numbers discharge what the consuming declaration asks of a party's side*: the tuple $\big(Q, \mathbb{S}, \{F_v\}_{v\in\mathcal{V}}\big)$ *together with the party partition* of Definition 10.1 (I2) — the default party partition, every factor Bob's, unless the instance declares another party's label, in which case every factor is that party's — satisfies the well-formedness conditions the consuming declaration places on *that party's* side (Definition "Admissible declaration" (meat-core/def:admissible-declaration) (i) and (iii), with the party labels per tensor factor demanded through its (viii)) —

$$F_v = F_v^\dagger, \qquad F_v \succeq 0, \qquad \sum_{v\in\mathcal{V}} F_v = \mathbb{1}_Q, \tag{11.5}$$

with every F_v block diagonal for $\mathbb{S}$ and $\mathcal{H}(Q) = \bigoplus_{s\in\mathbb{S}} \mathcal{H}(Q_s)$ finite-dimensional, every factor of every $\mathcal{H}(Q_s)$ the instance's declared party's and Bob's by default. *Those are conditions on a party partition and not on a defect*, and they are a *different* condition from the conditions of Definition "Exact admissibility of a declared operator family" (foundations/def:exact-admissibility) that (a)–(e) discharge, which say nothing about who owns which factor;

(iii) *(the defining identity is about the numbers emitted, and it is an identity on B)* for every $v \in \mathcal{V}$,

$$\Lambda^\dagger(F_v) \;=\; M'^{\mathrm{ph}}_v \qquad \text{exactly,} \tag{11.6}$$

both sides operators on B, which is where $\Lambda^\dagger$ lands, Λ being a channel $\mathcal{T}(B) \to \mathcal{T}(Q)$. *Only the data sectors carry anything the normalizer produced*: M'^{ph} is the handed model with the datum replaced by A' as the instance's model map defines it, and everything the hypotheses (H) are about is the declared one, unchanged. Moreover every claim of the instance holds verbatim of the declared model — the relation (10.3) with its transported set and pull-backs, every certificate of (10.4), and the class-marginal identity (10.5) where the instance discharges it — and Theorem 3.11 and Theorem 7.4 hold of the one-link receiver segment (Definition 7.1) whose source family is (B, M'^{ph}), whose target is (Q, F) and whose channel witness is Λ, with $\mathsf{status} = \mathsf{exact}$;

(iv) *(the emitted bounds, and the direction each is rounded in)* every scalar value that was rounded to reach the lattice was rounded *down*, as the lower bound of the quantity it certifies, and every complementary value was *recomputed* from the rounded ones and never rounded on its own (Definition 10.1 (I2)); and a caller-declared constant of modulus at most one off the lattice is rounded in the direction that enlarges the feasible set for its declared relation (Corollary 11.3 (iii)), an equality constant off the lattice and any constant of modulus above one being refused. *Every other number this module emits is exact, the displacement record Δ_s of (i) aside*; these are all the rounding directions there are among the numbers a statement of this contract reads, and the direction of each is recorded;

(v) *(what relates the declared model to the handed one)* where some $\Delta_s > 0$ the emitted numbers are a declaration of a family near the constructed one, and nearness is not what relates them: *no bound relates a key length computed from M'^{ph} alone to one computed from the handed model $M[A]$, or from the supplier's own output*, and this contract has no theorem converting an operator perturbation of the target into a security penalty, nor any other relation between A' and A on a moved class: the consuming contract's impossibility (Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound)) is the whole of what is proved there; what a run rests on there instead is (vi);

(vi) *(what a run on the emission rests on where it reads a moved block, and only there)* let $B = B_{\mathrm{snap}} \sqcup B_{\mathrm{em}}$ be the blocks that the record of the run names, for this artefact, as resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), kept as two disjoint parts, since they are blocks of two different snaps and their labels may coincide: B_{em} holds the data sectors of the emission so named, and B_{snap}, where A is formed from measurement blocks of a snap (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) that the declaring party made of the measurement it states, the blocks of that snap so named; the declaring party is the instance that states the measurement of the device, or a reference of it, from which A is formed. Where B is empty, this part asserts nothing, and no part of this theorem rests on that assumption. Where B is not empty, let that assumption hold. Then on every block in B_{snap}, the block of the measurement the device performs is the snapped block, with the entries recorded for that snap, from which A is formed, in place of the block of the model the declaring party states; and every conclusion about the device that the claims of the run draw from the emitted target, under their hypothesis that its data sectors carry the handed

datum A, holds when they carry the declared datum A' on every data sector in B_{em}, which by (iii) is the datum whose model M'^{ph} the emitted target squashes exactly. The run carries no bound relating its key length to the declaring party's model on the blocks in B_{snap}, nor to $M[A]$ on the data sectors in B_{em}: no claim relates that model to the snapped block on a block in B_{snap}, nor A to A' on a data sector in B_{em}.

Corollary 11.3 (The emitted constraint entries)**.** Assume the hypotheses of Theorem 11.2 for a reduction instance $\mathcal{J}$ with declared datum A', target $\{F_v\}$, channel Λ and declared model M'^{ph}. A request *carries entries* when it selects the case in which the instance's own entries are emitted: in the flag-state submodule that case is the licensed route; in a core-only artefact it is a caller's request for an entry — for a default carried operator as an entry, or for the class-marginal entry. *What is emitted is decided by the request*: *where the request carries no entry the emitted constraint list is empty* — whatever entries the instance's own list $\mathfrak{L}[A']$ contains — and the artefact carries no attack-set licence; *where the request carries entries the emitted list is the instance's own entries, the class-marginal entry on request and any further requested entry that (iii) admits*, so that the list is non-empty and the licence of Theorem 6.7 is present. In *both* cases the certificate $(\Pi_s, \succeq, 0)$ of every sector projector of every instance is carried *per entry in the transported set*, with the certificate case of Corollary 1.7 that makes each usable, and not as a constraint entry. Then *(i) and the identity* (11.8) *of (ii) of every request, and what each of (ii) and (iii) emits of a request that carries entries*:

(i) *(the default carried operators)* every operator of the instance's *default carried set* — every sector projector Π_s, $s \in \mathbb{S}$, which Definition 10.1 (I3) puts in the transported set of every instance, together with the sums of sector projectors that instance transports besides (for a target split at a cutoff, the projector onto the blocks below it and the projector onto the flag blocks; for a branching instance, the branch projector Π_j) — is emitted at $c = 0$ with $\bowtie = \succeq$: its entries are 0 and 1, hence on the lattice at every $\kappa \geq 1$, and its pull-back

$$\Lambda^{\dagger}(\Pi_s) \;\succeq\; 0 \tag{11.7}$$

is exactly positive for every instance, as therefore is the pull-back of every sum of them. *It is emitted with the transported set*, per entry and whatever the request carries, as the certificate case of Corollary 1.7 (ii) that entry records, and it becomes an entry of $\mathfrak{L}$ only where a caller requests it *as an entry* — a request that makes the emitted list non-empty and so brings the licence with it;

(ii) *(the class-marginal entry)* For $\hat{O} = \sum_{v \in \mathcal{V}_t} F_v$ at a declared $t \in \mathcal{T}_{\mathrm{set}}$ — the expectation form of the physical class-marginal identity (4.3)

—

$$\Lambda^{\dagger}(\hat{O}) \;=\; q_t\,\mathbb{1}_B \qquad \text{exactly} \tag{11.8}$$

holds *whenever* the instance discharges the class-marginal identity (10.5), and is established by this contract *only* there — *a fact about the instance and the channel, which this part asserts of every request. Where the request carries entries*, so that the emitted list is non-empty and the licence is present, the entry is then emitted, on request, as the single equality entry $(\hat{O}, =, q_t)$ at the declared q_t; where the instance does not discharge it the request is refused, and the entry is never emitted on the strength of a branch that does not establish it;

(iii) *(any further entry — the general rule) under the same hypothesis as (ii)'s emission sentence — the request carries entries*, let $\hat{O} = \sum_{v\in\mathcal{V}}\sum_{s\in\mathbb{S}}\alpha_{v,s}\Pi_s F_v \Pi_s + Z$ with declared real lattice coefficients $\alpha_{v,s}$ and Z a fixed operator of the declared sector decomposition (a sector projector or a combination of them whose coefficients are declared lattice values, as Definition 10.1 (I4) requires), the form (10.4), with a relation $\bowtie$ and a declared real constant c. *Where a declared constant of modulus at most one does not satisfy* (9.1) *it is first rounded onto the lattice in the direction that enlarges the feasible set for* $\bowtie$ — down for $\succeq$, up for $\preceq$ — and *an equality entry whose constant is off the lattice is refused rather than split into a pair*, an equality having no loosening direction; *a declared constant of modulus above one is refused under every relation.* Suppose the certificate

$$\Lambda^{\dagger}(\hat{O}) \;\bowtie\; c\,\mathbb{1}_B \tag{11.9}$$

is established for the emitted numbers, at the constant c so obtained, in one of two ways: for an entry of the instance's own list $\mathfrak{L}[A']$, by (I4) of the instance; for a requested entry, by a *decided certificate* — a condition on the emitted numbers and the request's declared constants, decided exactly from them, that a claim stated for the instance proves sufficient for (11.9) under the instance's hypotheses (H) — that holds there. Then the entry is emitted at that c. *No other requested entry is emitted*: a requested operator of this form for which no claim stated for the instance gives a decided certificate, or whose decided certificate fails at the emitted numbers, is refused, and no certificate is taken from the request on trust.

Every entry that (i)–(iii) emit is *untagged*, $(\hat{O}, \bowtie, c)$, and is read in the form of the declared source mode (Definition 6.5).

Each certificate above is discharged for the emitted numbers themselves, those numbers being the family every part is read on (Theorem 11.2 (iii)), so no constant is shifted to pay for a displacement between the family a certificate is asserted of and the numbers its operator is assembled from.

The rounding rule of (iii) applies only to a constant a caller declares, never to one this module or an instance computes. *Together, (i)–(iii) cover every entry this module can emit*: the default carried operators of every instance, the class-marginal entry a caller may request, and every entry of the form (11.9) — which includes every entry an instance carries in its own list, such as a flag-sector entry of a weight-preserving squasher, and the branch entries a branching instance inherits (Theorem 10.4, Corollary 10.5). *A requested operator outside that form, or of that form without a decided certificate that holds, is not emitted at some other constant; it is refused.* So the three parts are exhaustive over what is *emitted*, which is what the artefact carries. *Where the request carries no entry there is nothing for (ii)–(iii) to be exhaustive about*: $\mathfrak{L} = \varnothing$ there, whatever entries the instance's own list contains, and the certificate of every default carried operator travels with the transported set alone.

Theorem 11.5 (The emission of several measuring parties)**.** Let $\mathbb{F}$ be a finite ordered set of *factors*, each f with a party label, and for each f let $\mathcal{J}^f$ be a reduction instance (Definition 10.1) on a system B_f over $\big(\mathcal{V}^f, \{\mathcal{V}^f_t\}_{t\in\mathcal{T}^f_{\mathrm{set}}}, \{q^f_t\}\big)$ to which Theorem 11.2 applies, with emitted datum A'_f, emitted target $F^f := F^f[A'_f]$ on $Q_f = \bigoplus_{s\in\mathbb{S}^f} Q_{f,s}$, channel $\Lambda_f := \Lambda_f[A'_f]$, declared model $M'^{\mathrm{ph},f} = M^f[A'_f]$ and entry list $\mathfrak{L}^f$. Put $B := \bigotimes_f B_f$, $Q := \bigotimes_f Q_f$, $\mathcal{V} := \prod_f \mathcal{V}^f$, $\Lambda := \bigotimes_f \Lambda_f$, all products in the order of $\mathbb{F}$, and for $v = (v_f)_f \in \mathcal{V}$

$$F_v := \bigotimes_f F^f_{v_f}, \qquad M'^{\mathrm{ph}}_v := \bigotimes_f M'^{\mathrm{ph},f}_{v_f}. \tag{11.10}$$

Suppose that *what is emitted* is the ordered list of the factors' artefacts, each with its party label, and the rule (11.10), and that no entry of any F_v is emitted, declared or rounded. Then:

(i) *(a measurement, block diagonal for the product sector map)* $\{F_v\}_{v\in\mathcal{V}}$ is a POVM on Q: every F_v is Hermitian and positive semidefinite and $\sum_{v\in\mathcal{V}} F_v = \mathbb{1}_Q$, as identities of the exact rational numbers the factors emitted. With the *product sector map* — index set $\prod_f \mathbb{S}^f$ in the product order, the last factor fastest, identifiers the factors' identifiers joined in that order, dimensions the products, projectors $\Pi_s = \bigotimes_f \Pi^f_{s_f}$ — every F_v is block diagonal, its block at $s = (s_f)_f$ being the Kronecker product $\bigotimes_f F^{f,(s_f)}_{v_f}$ of the factors' blocks. The party partition assigns every tensor factor of $\mathcal{H}(Q_{f,s_f})$ to the party of f.

(ii) *(the squashing relation and the declared model)* $\Lambda^\dagger(F_v) = M'^{\mathrm{ph}}_v$ for every v, exactly, so $M'^{\mathrm{ph}} \preceq_\Lambda F$; the declared joint model is the *product* of the factors' declared models. Λ is of the form the artefact

records for one factor: with $\mathcal{P}_{\mathbb{S}}$ the pinching for the product projectors, $\mathcal{P}_{\mathbb{S}} \circ \Lambda = \Lambda$, and the squashed analysis may be restricted to states block diagonal for the product sector map without excluding a reachable one. The transported set contains every product projector Π_s, every lifted projector $\mathbb{1} \otimes \Pi^f_{s_f} \otimes \mathbb{1}$ and every lifted transported operator $\mathbb{1} \otimes \hat{O} \otimes \mathbb{1}$, $\hat{O}$ transported by $\mathcal{J}^f$, with pull-back the lift of $\hat{O}$'s pull-back.

(iii) *(entries keep their constants)* for every factor f and every entry $(\hat{O}_k, \bowtie_k, c_k) \in \mathfrak{L}^f$, the certificate $\Lambda_f^\dagger(\hat{O}_k) \bowtie_k c_k \mathbb{1}_{B_f}$ which $\mathcal{J}^f$ discharges at A'_f is a certificate on that factor's channel in the sense of Theorem 6.7 (v). So the joint list $\mathfrak{L} \coloneqq \bigsqcup_f \{(\hat{O}_k, \bowtie_k, c_k; f)\}$, each entry scoped to its factor and carrying the same operator, the same relation and the same constant, is licensed with its entries read party-locally (Definition 6.5), or as scalars on the lifted operators $\mathbb{1} \otimes \hat{O}_k \otimes \mathbb{1}$, or any mixture; every squashed attack sequence against M'^{ph} is admissible for it and reproduces the protocol's output state; and the lifted operators are block diagonal for the product sector map with entries that are the factor's emitted entries and zeros and ones. No constant is re-derived and none is rounded.

(iv) *(class-marginal identities)* with classes $\mathcal{V}_t = \prod_f \mathcal{V}^f_{t_f}$ and weights $q_t = \prod_f q^f_{t_f}$, $\sum_{v \in \mathcal{V}_t} M'^{\mathrm{ph}}_v = q_t \mathbb{1}_B$ holds for every t if every factor's class-marginal identity is discharged. Where it is discharged for the factors in a subset $\mathbb{G} \subseteq \mathbb{F}$ only, the identity holds for the classes that are full in every factor outside $\mathbb{G}$: $\sum_{v:\, v_f \in \mathcal{V}^f_{t_f}\ (f \in \mathbb{G})} M'^{\mathrm{ph}}_v = \prod_{f \in \mathbb{G}} q^f_{t_f} \mathbb{1}_B$. What is emitted for it is the expectation-form entry on the pulled-back class sum and never an operator identity on Q. The weight q_t is a product of declared numbers and is formed exactly by whoever reads it; it is not emitted as a number of this module, and it could not be: each $q^f_{t_f}$ lies on its factor's declared lattice (Definition 9.1 (b), Definition 4.4 (a)), and a product of two lattice numbers lies in general on a finer lattice only and need not be a binary64 number, exactly as for the blocks of (11.10).

(v) *(what was rounded)* the numbers behind (11.10) are the factors' emitted numbers and nothing else. Every displacement, kept or moved class and rounding direction is a factor's and is recorded with that factor's emission, and Theorem 11.2 (v) applies to each factor as it stands: no bound relates a key length computed from M'^{ph} alone to one computed from the product of the handed models. The product introduces no further displacement.

(vi) *(one factor)* for $|\mathbb{F}| = 1$ parts (i), (ii), (iv) and (v) are the corre-

sponding statements of Theorem 11.2 for that instance, and what is emitted is that instance's artefact, every entry untagged. Part (iii) at one factor is Theorem 6.7 (v) at one factor, which says *more* than (i)–(iv) of that theorem in untrusted-source mode with $d_A > 1$ and the same otherwise (Definition 6.5); the untagged entry that is emitted is implied by it (Lemma 6.6 (iii)), so the one-factor emission asserts nothing beyond Theorem 6.7 (i)–(iv).

These statements are about the factors' instance links and their emitted families.

Corollary 11.6 (The emitted family of the identity instance)**.** Let the declared data pass every refusal rule of this module's interface, let $\mathcal{J}_{\mathrm{id}}$ be the identity instance of Corollary 10.2 with handed datum A, and let A' be the datum the normalizer of Definition 11.1 declares. Then Theorem 11.2 holds of $\mathcal{J}_{\mathrm{id}}$; the normalizer moves nothing, $A' = A$ and $\Delta_s = 0$ for every data sector s; and, for every $v \in \mathcal{V}$,

$$\Lambda^{\dagger}(F_v) \;=\; \mathcal{P}_{\mathbb{S}}(F_v) \;=\; F_v \;=\; M'^{\mathrm{ph}}_{v} \;=\; \bigoplus_{s\in\mathbb{S}_0} A'^{(s)}_{v} \qquad \text{exactly, on } B = Q, \tag{11.11}$$

so the emitted family *is* the declared datum, which is the handed datum, exactly admissible and decided so before emission; the emitted constraint list is *empty where the request carries no entry* — this instance's own list being empty in any case (Corollary 10.2 (I4)) — and where the request carries entries it carries, on request, the class-marginal entries $\big(\sum_{v\in\mathcal{V}_t} F_v, =, q_t\big)$, which the instance discharges, and a further requested entry $(\hat{O}, \bowtie, c)$ of the form (10.4), which Corollary 11.3 (iii) emits exactly where $\hat{O} \bowtie c\,\mathbb{1}_B$ holds at the constant c it emits, this instance carrying none of its own: $\Lambda^{\dagger}(\hat{O}) = \hat{O}$, so the certificate (11.9) of such an entry is that operator inequality on the emitted numbers, decided exactly by finitely many field operations and sign comparisons, a decided certificate in the sense of Corollary 11.3 (iii) for every operator of that form; its default carried operators — here the sector projectors and nothing else, the transported set being $\{\Pi_s\}_{s\in\mathbb{S}_0}$ — are emitted with the transported set at $c = 0$ with $\succeq$ in either case (Corollary 11.3 (i)); no scalar value is emitted and no bound of this module's own is rounded; the chain is the identity link with $\mathsf{status} = \mathsf{exact}$. These statements are about the instance's link and its emitted family. The instance is selectable only once a declared input carries an already-finite POVM.

12 The reference companion

Corollary 12.1 (The reference companion of a reduction instance). Let $\mathcal{J}$ be a reduction instance (Definition 10.1) whose handed datum A is formed, by the rule of its part (I1), from a family $K = \{K_j^{(b)}\}$ of measurement blocks, over blocks b of a sector map and outcomes j of an alphabet (one such family per setting where the blocks are given per setting), that a declaring instance states as a snap (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) of the measurement of a device. Let $G = \{G_j^{(b)}\}$ be a family over the same blocks, alphabet and dimensions, every block of which is Hermitian and positive semidefinite and satisfies exactly every completeness identity that the declaration of K states of K on that block. In the use this corollary is for, $G^{(b)}$ is the block of the handed reference (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)) where the declaring instance hands one, and $K^{(b)}$ elsewhere, so that $G = K$ where it hands none. Suppose

(H-REF) $\mathcal{J}$ is a reduction instance over the substrate $\mathbb{F} = \mathbb{R}$ (Definition 9.2), and the rule of its part (I1) forms, from every family with the properties just listed for G, a datum in $\mathcal{D}_{\mathrm{adm}}$.

Let A^{ref} be the datum that rule forms from G in place of K, and call the instance at A^{ref} — the same B, (H), sectors, supports, scalar values, constructor and entry rule, with A^{ref} in place of the handed datum — the *reference companion* of $\mathcal{J}$ at G. Then:

(i) *(an admissible target over the emitted shape)* $A^{\mathrm{ref}} \in \mathcal{D}_{\mathrm{adm}}$, the reference companion is a reduction instance, and its target $\{F_v[A^{\mathrm{ref}}]\}_{v\in\mathcal{V}}$ is a POVM on Q, sector-block-diagonal by construction. Its alphabet with classes and weights, its sector map (the sector list, its order, each sector's dimension and kind), its scalar values and its party partition are those of the target the instance has at every admissible datum, the declared datum the normalizer produces included, and it differs from such a target on the data sectors alone, where it carries A^{ref}. Where $G = K$, $A^{\mathrm{ref}} = A$: the reference companion is $\mathcal{J}$ at its handed datum;

(ii) *(an exact squash of the companion's model)* $\big(M[A^{\mathrm{ref}}]; O[A^{\mathrm{ref}}]\big) \preceq_{\Lambda[A^{\mathrm{ref}}]} \big(F[A^{\mathrm{ref}}]; \hat{O}[A^{\mathrm{ref}}]\big)$, so that $\Lambda[A^{\mathrm{ref}}]^\dagger(F_v[A^{\mathrm{ref}}]) = M_v[A^{\mathrm{ref}}]$ exactly for every v; every certificate of the entry list $\mathfrak{L}[A^{\mathrm{ref}}]$ is discharged, and the class-marginal identity (10.5) holds at A^{ref} where the instance's flag says it is discharged; and Theorem 3.11 and Theorem 7.4 hold of the one-link receiver segment (Definition 7.1) whose source family is $(B, M[A^{\mathrm{ref}}])$, whose target is $(Q, F[A^{\mathrm{ref}}])$ and whose channel witness is $\Lambda[A^{\mathrm{ref}}]$, with $\mathsf{status} = \mathsf{exact}$;

(iii) *(the companion is known through enclosures)* let every block of G be read with an entrywise enclosure (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)), with radius zero on every block where $G^{(b)} = K^{(b)}$. Where the rule of (I1) forms each entry of the datum as a finite expression in entries of the blocks and in numbers the instance was handed, and every reciprocal node of that expression receives an enclosure by part (iv) of Lemma "Arithmetic on complex enclosures" (receiver/lem:enclosure-arithmetic), as it does where the expression has none, the rules of that lemma applied to that expression give a complex enclosure of every element of A^{ref} whose centre is formed by the same expression from the centres and whose radius is zero at every entry whose inputs are all enclosed with radius zero; part (v) of that lemma, applied element by element, makes it an entrywise enclosure of A^{ref}, whose centre is the Hermitian part of that centre and whose radius is zero at an entry (i, l) where the radii at (i, l) and (l, i) both are. Every entry operator $\hat{O}_k[A^{\mathrm{ref}}]$ of (10.4), a combination of the blocks $\Pi_s F_v[A^{\mathrm{ref}}]\Pi_s$ with real coefficients $\alpha_{k,v,s}$ plus a combination Z_k of sector projectors, is then enclosed entrywise by the same combination of the centres, with radius $\sum_v |\alpha_{k,v,s}|\, R^{(s)}_{v,il}$ at the entry (i, l) of each data sector s, where (C, R) is any entrywise enclosure of the datum and $R^{(s)}_{v,il}$ is the radius of its element v at that entry, and radius zero on every scalar sector;

(iv) *(where (H-REF) holds)* (H-REF) holds for the identity instance of Corollary 10.2; and it holds for the branching instance of Theorem 10.4, and for the register instance of Corollary 10.5, whenever it holds for every branch instance whose datum is formed from blocks of K, the data of the other branches being formed from no block of K;

(v) *(what the companion rests on, and only there)* no part of (i)–(iv) rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement). Let B be the blocks on which $G^{(b)} = K^{(b)}$ that the record of the run names as resting on that assumption. Where B is empty this part asserts nothing. Where it is not, let the assumption hold: then on every block in B the block of the measurement the device performs is the snapped block $K^{(b)}$, with the entries recorded for the snap, in place of the block of the declaring instance's model, so that the companion is formed there from the block the device then measures; and no claim relates a key length computed from the companion to that model on the blocks in B;

(vi) *(entries requested beside the instance's own)* let $(\hat{O}, \bowtie, c)$ be an entry emitted at a request that carries entries (Corollary 11.3), written in

the form (10.4) with coefficients $\alpha_{v,s}$ and a combination Z of sector projectors, $\hat{O} = \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{v,s} \Pi_s F_v[A'] \Pi_s + Z$ at the declared datum A', and call $(\hat{O}^{\mathrm{ref}}, \bowtie, c)$, with

$$\hat{O}^{\mathrm{ref}} \;:=\; \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{v,s}\, \Pi_s F_v[A^{\mathrm{ref}}] \Pi_s \;+\; Z, \tag{12.1}$$

its *companion form* at that writing. Part (iii) encloses $\hat{O}^{\mathrm{ref}}$. Where $\hat{O}$ is a sum of sector projectors (Corollary 11.3 (i)), written with every $\alpha_{v,s} = 0$, $\hat{O}^{\mathrm{ref}} = \hat{O}$ and $\Lambda[A^{\mathrm{ref}}]^\dagger(\hat{O}) \succeq 0$. Where it is the class-marginal entry $\big(\sum_{v\in\mathcal{V}_t} F_v, =, q_t\big)$ of Corollary 11.3 (ii), written with $\alpha_{v,s} = 1$ for $v \in \mathcal{V}_t$, 0 otherwise, and $Z = 0$, $\hat{O}^{\mathrm{ref}} = \hat{O}$, and where the instance discharges (10.5), $\Lambda[A^{\mathrm{ref}}]^\dagger(\hat{O}^{\mathrm{ref}}) = q_t\, \mathbb{1}_B$ exactly. For any other entry this part asserts no certificate at A^{ref}. In every case, where $\Lambda[A^{\mathrm{ref}}]^\dagger(\hat{O}^{\mathrm{ref}}) \bowtie c\, \mathbb{1}_B$ holds, $\big(M[A^{\mathrm{ref}}]; O[A^{\mathrm{ref}}], \Lambda[A^{\mathrm{ref}}]^\dagger(\hat{O}^{\mathrm{ref}})\big) \preceq_{\Lambda[A^{\mathrm{ref}}]} \big(F[A^{\mathrm{ref}}]; \hat{O}[A^{\mathrm{ref}}], \hat{O}^{\mathrm{ref}}\big)$, and the companion form is an entry carrying the spectral certificate (6.9) at its constant.

Corollary 12.2 (Factor enclosures carried to the reference companion)**.** Let $\mathcal{J}$, K, G and A^{ref} be as in Corollary 12.1, the blocks $K_j^{(b)}$ indexed by a block b and an outcome j, together with the setting where the blocks are given per setting. Call the rule of part (I1) of $\mathcal{J}$ (Definition 10.1) a *non-negative combination* when for every data sector s and every $v \in \mathcal{V}$ there are non-negative rationals $\lambda_{b,j}^{(s,v)}$ such that, for every family K' with the properties that Corollary 12.1 lists for G, the datum A' the rule forms from K' satisfies

$$A_v'^{(s)} \;=\; \sum_{b,j} \lambda_{b,j}^{(s,v)}\, K_j'^{(b)}, \tag{12.2}$$

every block $K_j'^{(b)}$ with $\lambda_{b,j}^{(s,v)} > 0$ being of dimension d_s. Let the rule be a non-negative combination, and let some blocks $G_j^{(b)}$ carry factor enclosures $\big(L_j^{(b)}, \varrho_j^{(b)}, W_j^{(b)}\big)$ (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)).

(i) Let s be a data sector and $v \in \mathcal{V}$ such that every block $G_j^{(b)}$ with $\lambda_{b,j}^{(s,v)} > 0$ carries one, and list the pairs (b, j) with $\lambda_{b,j}^{(s,v)} > 0$ in a fixed order. The triple whose matrix box has as its centre and its radii those of the boxes $\big(L_j^{(b)}, \varrho_j^{(b)}\big)$ side by side in that order, and whose core is the diagonal matrix of the $\lambda_{b,j}^{(s,v)} W_j^{(b)}$ in that order, is a factor enclosure of $A_v^{\mathrm{ref},(s)}$, its number of columns the sum of theirs; where every $\lambda_{b,j}^{(s,v)}$ is zero, it has no column.

(ii) The block of the companion's target $F_v[A^{\mathrm{ref}}]$ on the data sector s is $A_v^{\mathrm{ref},(s)}$, so the triple of (i) is a factor enclosure of that block.

(iii) The rule of (I1) of the identity instance of Corollary 10.2 is a non-negative combination: each element of its datum is the block of the family it is read from at the same sector and outcome, with the coefficient 1, every other coefficient being 0.

Lemma 12.3 (A rounded constraint entry with its constant relaxed)**.** Let $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel with $(M;O) \preceq_\Lambda (F;\hat{O})$ (Definition 1.3), and let $(\hat{O}_k, \bowtie_k, c_k)$ be an entry with $\hat{O}_k$ a carried operator, $\bowtie_k \in \{\preceq, =, \succeq\}$, $c_k \in \mathbb{R}$ and the certificate $\Lambda^\dagger(\hat{O}_k) \bowtie_k c_k \mathbb{1}_B$ of (6.9). Let $\hat{O}'_k$ be a Hermitian operator on Q, let $E_k := \hat{O}'_k - \hat{O}_k$, and let $\rho_{il} \geq 0$ be numbers with $|(E_k)_{il}| \leq \rho_{il}$ for every entry in a fixed orthonormal basis of $\mathcal{H}(Q)$, and $\delta_k \geq \max_i \sum_l \rho_{il}$. Define the *relaxed entries* of $(\hat{O}_k, \bowtie_k, c_k)$ as $(\hat{O}'_k, \succeq, c'_k)$ with $c'_k \leq c_k - \delta_k$ where $\bowtie_k$ is $\succeq$; $(\hat{O}'_k, \preceq, c''_k)$ with $c''_k \geq c_k + \delta_k$ where $\bowtie_k$ is $\preceq$; and both of these where $\bowtie_k$ is $=$. Then:

(i) $-\delta_k \mathbb{1}_Q \preceq E_k \preceq \delta_k \mathbb{1}_Q$;

(ii) *(certificate)* $(M;O,\Lambda^\dagger(\hat{O}'_k)) \preceq_\Lambda (F;\hat{O},\hat{O}'_k)$, and every relaxed entry carries the spectral certificate (6.9) at its own constant: $\Lambda^\dagger(\hat{O}'_k) \succeq c'_k \mathbb{1}_B$, respectively $\Lambda^\dagger(\hat{O}'_k) \preceq c''_k \mathbb{1}_B$;

(iii) *(attack sets)* for all finite-dimensional E', E'', every $\mathcal{A}$ in the set (6.2), (6.3) or, for an entry tagged party-local with the factor f, (6.4), of the list holding the exact entry lies in the set of the list in which the exact entry is replaced by its relaxed entries, the other entries unchanged;

(iv) *(states)* for every finite-dimensional system X and every positive semidefinite ω on $\mathcal{H}(X)\otimes\mathcal{H}(Q)$, $\mathrm{tr}_Q[(\mathbb{1}_X \otimes \hat{O}_k)\,\omega] \succeq c_k \,\mathrm{tr}_Q\,\omega$ implies $\mathrm{tr}_Q[(\mathbb{1}_X \otimes \hat{O}'_k)\,\omega] \succeq c'_k \,\mathrm{tr}_Q\,\omega$, and the same with $\preceq$, c''_k in place of $\succeq$, c'_k; with X trivial these are the scalar readings $\mathrm{tr}\left[\hat{O}_k\omega\right] \geq c_k \,\mathrm{tr}\,\omega$ and the like.

13 Assumptions

Assumption 13.1 [**A-MEAS**] (Each measuring party's device as one POVM per round)**.** In every round, each measuring party's interaction with the quantum system it receives satisfies: (i) it is described by a single POVM on that system; (ii) it emits the classical outcome and nothing else, and no post-measurement quantum system is retained or reused; (iii) all downstream processing (announcements, sifting, acceptance, key map, error correction, privacy amplification) acts on classical data; (iv) the round's measurement channel is the product of one measurement channel per factor, each acting on its own factor alone, for each of the following factorizations of the measured system B_j: where the declaration declares several measured factors

(Definition "Protocol specification" (protocol-compiler/def:spec)), the factorization $B_j = \bigotimes_f B_{f,j}$ with one factor per measured factor, whichever parties carry their labels and two factors of one party included; and where several parties measure, the factorization into the parties' parts of Definition 2.1.

Assumption 13.2 [**A-ROUND-SQ**] (Round homogeneity of the physical measurement)**.** Let M be the physical POVM: the POVM on the measured system of a round that the analysis takes as the measurement the devices of the measuring parties perform on it. The family $\{M\}$ is round-homogeneous (Definition "Round homogeneity of a per-round family" (foundations/def:round-homogeneity)) in both clauses of that definition: (i) the same M describes the measurement of every round; (ii) a round's measurement acts as M describes, whatever the inputs and outputs of the other rounds, with no detector memory across rounds.

Assumption 13.3 [**A-RECV**] (Supplier artefact whose claims hold of the device)**.** Every declared block, weight, matrix and scalar that a reduction instance of this module reads from a supplier's artefact satisfies: (i) the artefact is the output of the supplying module, at the module version it declares, which is an *accepted* version of the supplying module, and on the inputs it declares, and the validation results it carries are that module's own; (ii) the facts the supplier's claims state hold of those numbers as declared, under that module's own assumption ledger.

Assumption 13.5 [**A-PROBE**] (The station's private setting drawn from its own declared randomness)**.** The private per-round setting $t \in \mathcal{T}_{\text{set}}$ of a measuring party's device satisfies, in every round: (i) it is drawn from that party's own randomness with the declared class weights q_t; (ii) it is drawn independently of the adversary's actions and of the round's quantum input; (iii) it is not inferred from any outcome.

Assumption 13.6 [**A-TSM**] (Output-state test statistics)**.** Every per-round public choice of known bias in the physical protocol that is applied at the measurement the emitted family $\{F_v\}$ describes, a private per-round setting of that measurement and any round-type designation the consumer adds by relabelling its outcomes included, (i) is a component of the emitted outcome v, and (ii) enters the analysis only through the statistics of $\{F_v\}$.

Assumption 13.7 [**A-CONST**] (Declared constants of the constraint list)**.** Every constant c_k in an emitted constraint entry (i) is fixed before the

protocol run by the device fact that certifies it, and (ii) is not a function of any statistic observed during the run.

Engine requirements

14 Requirements: channel witnesses, post-processing and relations

Requirement 14.1 (A channel witness for every emitted link). The engine emits every link with an explicit channel witness and refuses a request that names a reduction coming with none.

Requirement 14.2 (The pinched squashing channel as the emitted witness). The engine emits $\mathcal{P}_{\mathbb{S}} \circ \Lambda$, the squashing channel followed by the pinching onto the sectors of Corollary 4.7, as the artefact's channel witness (output `squashing_channel`), so that the block-diagonality restriction of that corollary's part (iii) and every other emitted constraint are sound for *one* channel.

Requirement 14.3 (Emission of the carried tuple, and constraints built from it). Beside the channel witness of each link it emits, the engine emits the carried tuple of the link's squashing relation (Definition 1.3; `transported_operators`), and it builds every constraint it emits from an operator in that tuple.

Requirement 14.4 (Refusal of approximate squashing relations). The engine refuses a declared squashing relation that carries an error term, with the reason `approximate_squashing_unsupported`, and emits no link with the status `certified_outer`, which is reserved for approximate squashing.

Requirement 14.5 (Emission of the round-homogeneous case only). The engine emits one declared model and one squashing channel for every round, both built from the one physical POVM that Assumption 13.2 holds fixed (the interface convention `round_homogeneity`). It emits no round-dependent squasher.

Requirement 14.6 (Noise and coarse-graining before the first squashing link). The engine applies a noise or coarse-graining map on the physical POVM before the first squashing link, and refuses a request to apply one after a squashing link (input `post_processing_order_declaration`, refusal `post_processing_order_violated`). The refusal reaches exactly

those two kinds of map: a stochastic relabelling applied to both sides after a link, depending on the private component of the outcome only, is not refused by it (Remark 4.5, (4.4)).

Requirement 14.7 (Exact check of a declared relation). The engine tests every declared relation (input `declared_relations`) against both necessary conditions of Lemma 4.9 and refuses one that fails either, with `declared_relation_impossible`. With X and X' as in that lemma, it decides (i) in operator form on the rational enclosures $a \leq b$ the declaration exhibits: the pair $a\mathbb{1}_Q \preceq X \preceq b\mathbb{1}_Q$ exactly by (9.3) and, where it holds, the same pair for X'; and (ii) as the exact rank over $\mathbb{F}_i$ of the matrix whose columns are the vectorized declared elements, by elimination in the field, together with its linear relations: an exact basis of the solutions (a, c) of $\sum_v a_v F_v = c\,\mathbb{1}_Q$, obtained by elimination in the field, each basis solution tested on the declared physical elements as $\sum_v a_v M_v = c\,\mathbb{1}_B$. No eigenvalue is computed and no threshold is used.

Requirement 14.8 (Refusal of the reserved receiver-stage link kind). The engine emits no receiver-stage link of the reserved kind `label-space`, and refuses a request that declares a link of that kind with `link_kind_deferred`.

Requirement 14.9 (Branch blocks emitted as supplied). The engine does not pad the branch blocks of a composite target: a reduction instance emits the blocks its branches supply, and a padded variant a submodule proves from Corollary 5.11 is not emitted unless that submodule's interface declares it.

15 Requirements: chains, constraint slots and the licence

Requirement 15.1 (Constraint and licence slots on every artefact). The engine emits every artefact with a constraint list `constraint_list` and a licence slot `attack_set_license`, even when both are empty: a record without them excludes a class of device knowledge that no classical target can carry (Proposition 6.1, Remark 6.2 (a)), and the licence slot is where the licence of Theorem 6.7 is carried.

Requirement 15.2 (Content and presence of the licence record). The engine emits the licence record `attack_set_license` naming the theorem that licenses the constraint list (Theorem 6.7); the hypothesis it was discharged on, which is the certificate (6.9) of each entry; and, per entry, what discharges that hypothesis: the claim that establishes the certificate under

Corollary 11.3 — the instance's own claim for an entry of its list, that corollary for a default carried operator and for the class-marginal entry, the claim giving the decided certificate for a requested entry — and, where that claim reads a fact a supplier's claims state, the supplier's artefact under Assumption 13.3, named by its version. It emits the record only when every entry has its certificate and what discharges it. The artefact names the proof technique the licence was issued for, since under another technique the licence must be re-established (Remark 6.8 (e)).

Requirement 15.3 (Refusal of data-dependent constants). The engine builds no constraint entry whose constant c_k is a function of a statistic observed during the run; a request for one is refused with `data_dependent_constant`. Such an entry would be unsound regardless of provenance (Remark 6.8 (c)), the constants being declared before the run (Assumption 13.7).

Requirement 15.4 (Refusal of a chain with a link that is not exact). The engine assigns a chain no composite status other than the one of Theorem 7.4 (iv): a chain containing a link that is not exact is refused, with `approximate_squashing_unsupported`, rather than given a composite status, because no theorem of this contract converts an error term into a security penalty. The refusal is a rule of the engine and not a converse this contract proves.

Requirement 15.5 (Identity link recorded for an already-finite receiver). The engine records the reduction of an already-finite receiver as the identity link of Example 7.5, never as an empty chain: an empty chain would leave the artefact unable to say whether a reduction was performed and found trivial or never considered, and the composition order the chain carries as data (Definition 7.2) has to be carried even when there is nothing to order.

16 Requirements: constraint entries

Requirement 16.1 (Emission of certified constraint entries only). Every constraint entry the engine emits carries exactly one certificate, named by its case of Corollary 1.7 (identity, positivity or spectral bounds), and is of the certified kind, case (i) of Corollary 3.13; the engine emits no entry of the assumed kind, case (ii). Each emitted entry's certificate case is recorded with the transported set, as Requirement 16.4 states.

Requirement 16.2 (Refusal of the operator form of the class-marginal identity). The engine refuses, with the reason `unsound_constraint_requested`, a request to impose on the squashed states the operator form $\sum_{v \in \mathcal{V}_t} F_v = q_t \, \mathbb{1}_Q$ of the class-marginal identity (Remark 3.14).

Requirement 16.3 (The constraint list and the licence decided by the request). Where the request carries no entry (Corollary 11.3), the engine emits an empty `constraint_list`, whatever entries the instance's own list contains, and no `attack_set_license`. Where the request carries entries, it emits in `constraint_list` the instance's own entries, the class-marginal entry on request and any further requested entry that Corollary 11.3 (iii) admits (Requirement 16.7), and it emits the licence of Theorem 6.7 in `attack_set_license`. The licence is present if and only if the list is non-empty (`licence_presence`): a non-empty list without a licence is refused with `licence_missing`, and a licence without a list with `licence_without_constraints`.

Requirement 16.4 (Certificates carried with the transported set). Whatever the request carries, the engine emits in `transported_operators`, per operator of the transported set, the certificate case of Corollary 1.7 that makes it usable. In particular the certificate $(\Pi_s, \succeq, 0)$ of every sector projector of every instance travels there and not as a constraint entry; such an operator becomes an entry of `constraint_list` only where a caller requests it as an entry (Corollary 11.3 (i)).

Requirement 16.5 (The class-marginal entry only where the instance discharges it). The engine emits the class-marginal entry $\left(\sum_{v \in \mathcal{V}_t} F_v, =, q_t\right)$ at a declared t, in the expectation form of Corollary 11.3 (ii), only on request and only where the instance discharges the class-marginal identity (10.5). A request for it of an instance that does not discharge the identity is refused with `constraint_without_certificate`, and the entry is never emitted on the strength of a branch that does not establish it.

Requirement 16.6 (Outward rounding of a caller's constant off the lattice). Where the declared constant of a requested entry does not satisfy (9.1), the engine first rounds it onto the lattice in the direction that enlarges the feasible set for the entry's relation, down for $\succeq$ and up for $\preceq$, and records the direction taken per entry in `emission_record` (Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction)). An equality entry whose constant is off the lattice is refused with `constraint_without_certificate` rather than split into a pair, and a declared constant of modulus above one is refused with `constraint_without_certificate` under every relation, before any rounding.

Requirement 16.7 (Refusal of a requested operator without a decided certificate). The engine emits an operator that a request carries as an entry only where Corollary 11.3 covers it: as a default carried operator (part (i)), as the class-marginal entry of an instance that discharges the class-marginal

identity (part (ii)), or, under part (iii), where a claim stated for the instance gives a decided certificate for the operator's form and the engine decides that condition exactly on the emitted numbers and the request's declared constants, finding that it holds. It refuses, with `constraint_without_certificate`, every other requested operator: one outside the form of Corollary 11.3 (i)–(iii), one of that form for which no claim stated for the instance gives a decided certificate, and one whose decided certificate fails at the emitted numbers. It takes no certificate (6.9) from the request on trust, and it never emits a requested operator at some other constant.

Requirement 16.8 (The party-local tag and its scope). The engine emits every constraint entry with the reading it is to be read in. An entry is emitted untagged, as $(\hat{O}, \bowtie, c)$, and is read in the form of the declared source mode (Definition 6.5); or, only where the measured system is a declared product $Q = \bigotimes_f Q_f$ of party-scoped factors and the entry's certificate is discharged on that factor's own channel Λ_f (Theorem 6.7 (v)), it is emitted with the tag `party_local` and a scope, as $(\hat{O}, \bowtie, c; f)$, with $\hat{O}$ an operator on Q_f alone carried as a sector-block list against that factor's sector map. In this module the tag is emitted only with the entries of several measuring parties (Theorem 11.5 (iii)). A request for the party-local form on a one-factor emission is answered with the untagged entry, at the same operator, relation and constant, and is not refused.

Requirement 16.9 (Lattice condition on the entries of emitted constraint operators). Where the licence is present, the engine decides, on the emitted numbers and before the licence is issued, that every entry of every emitted constraint operator satisfies (9.1), and refuses the request otherwise, with the reason `constraint_without_certificate` (the lattice half of the rule `licensed_hermiticity`; Corollary 11.3 (iii)).

17 Requirements: exact admission and emission

Requirement 17.1 (Refusal of a declaration outside the substrate). The engine refuses a declaration whose entries are not exactly representable in the field $\mathbb{F}$ chosen for the arithmetic (Definition 9.2), with the reason `declaration_not_exactly_admissible`, rather than rounding the entries into that field. No declaration whose entries satisfy (9.1) is refused on this ground.

Requirement 17.2 (Refusal of an emitted family that is not exactly admissible). The engine emits a family only if the six conditions of Theorem 11.2 (ii) — (a)–(e) of Definition 9.1 together with (9.1) of every entry

— hold of the numbers about to be emitted, decided exactly after the emission normalizer where that acted; otherwise it refuses with the reason `emitted_family_not_admissible`, naming the predicate that failed, and repairs nothing it was handed. Under the same reason, where the run declares the strict option `strict_snaps` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)) and no reference companion of Requirement 17.4 is formed beside the artefact, it refuses every moved class of Definition 11.1, one snapped by the last resort (3) included, naming the data sector, the class, the cause and the fix; where a companion is formed, whether the class rests is decided by the run after its search, and the run refuses there under the same option. Where the run does not declare the option, no class is refused for being moved. The cause of a moved class that would rest is that no reference companion bypasses the normalizer, which, by Requirement 17.4, is where no claim of this module establishes (H-REF) for the instance; for a class snapped by (3) it also names that a component of its residue has no host. The fix named is declared numbers on which the exact construction stays on the lattice, so that the class is kept; for a class snapped by (3) also a model that includes the device's loss and dark counts, so that the elements of the class are positive definite on the sector. A moved class is mixed only with the family of its diagonal parts in (2), or with a multiple of the identity in the last resort (3).

Requirement 17.3 (The record of an emission). The engine emits with every artefact, in `emission_record`: the declared lattice exponent κ; whether the run declared the strict option of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv); the reference kind of every snap the normalizer makes, *exact*, every radius zero, the reference being the handed datum itself (Definition 11.1), so that the enclosure setting, recorded as Requirement "Snapping by the declarer only" (foundations/req:who-snaps) asks of the record of every snap, does not bear on it; per data sector, the emission displacement Δ_s of (11.4); per data sector and class, whether the class was kept (Definition 11.1 (1)) or moved, and for a moved class the step (2) or (3) of Definition 11.1 that snapped it, the weight w taken or the fall-back of (3), the hosts of the components of the residue under (2), and for (3) the flag `support_not_preserved`; and the direction of every bound it rounds, each scalar value rounded down and each complementary value recomputed (Theorem 11.2 (iv)). A data sector *rests on* Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) where some class of it is moved, save where it is evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)): in a run whose key length is formed from a number on the reference companion of Requirement 17.4, its first-order bound or a number certified on it, which carries no data of this emission's normalizer, every data sector

that holds a moved class is, a kept sector being named as kept and not counted among them, and none rests on the assumption. Where the key length is formed instead from a number certified on the snapped declaration in the fallback of Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i), none is, and every data sector that holds a moved class rests on the assumption, with the cause `companion_not_certified` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)). The record names every moved sector, with the flag of any class of it snapped by (3), and carries Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) among the assumptions the artefact may rest on exactly where there is one. This record is emitted with the artefact, before the run knows whether the number its key length is formed from is formed on the reference companion; what it names is therefore provisional, and the record of the run decides which sectors rest on the assumption, each with its cause and group (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii); Theorem 11.2 (vi)).

Requirement 17.4 (Emitting the reference companion beside the artefact). Where a claim of this module establishes (H-REF) of Corollary 12.1 for the reduction instance the artefact is emitted from, and either the declaring instance hands the reference beside its snap of the measurement it states (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) (i)) on at least one block or the emission normalizer moves at least one class (Definition 11.1), the engine emits beside the artefact, in `reference_companion`, the reference companion of that instance at the family G that carries the handed reference on every handed block and the snapped block the instance declares on every other block, so that, where no block is handed, G is the instance's snapped family and the companion's datum is the handed datum (Corollary 12.1 (i)):

(i) *The datum.* Every enclosure of a handed block is first recentred by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure), so that every entry whose enclosure contains 0 has centre 0. The engine then forms, in exact arithmetic on the exact island (Definition "The exact island" (foundations/def:exact-island)), the enclosure of the companion's datum that Corollary 12.1 (iii) gives, from these enclosures and from the snapped blocks with radius zero, by the construction it applies to the snapped family: the same rule of (I1), the same weight route, scalar values, sectors and supports. It then recentres that enclosure again, element by element and block by block, by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure), after making its centre Hermitian and its radii symmetric by part (iv) of that lemma where they

are not, every block of the datum being Hermitian (Corollary 12.1 (i)); where the rule of (I1) combines entries of several blocks, a sum that is exactly zero may so have a centre that is not, and this second recentring sets it to zero wherever its enclosure contains zero. The entries of (iii) are formed from the datum's enclosure so recentred. It applies no normalizer (Definition 11.1), decides no lattice condition, rounds nothing, and decides nothing of the companion's admissibility, which is Corollary 12.1 (i) on the claims the handed reference names (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i)). It emits the centres, entries of $\mathbb{Q}(i)$, and the radii, rationals, entry by entry, so that an entry the reference knows exactly is emitted exactly, with radius zero.

(ii) *The shape.* The companion is emitted with the sector map, the alphabet with its classes and weights, the scalar values and the party partition of the artefact's target, which are equal by Corollary 12.1 (i), and with the artefact's channel witness and transported set read at the companion's datum.

(iii) *The entries.* On a route whose request carries entries (Corollary 11.3), the companion's entry list holds the instance's own entries at the companion's datum and, for every entry requested beside them that the artefact carries, its companion form (Corollary 12.1 (vi)) at the writing by which the engine recognised it, where its certificate at the companion's datum is established: for a sum of sector projectors and for the class-marginal entry, by Corollary 12.1 (vi), the operator being the artefact's, enclosed with radius zero; for another entry, where a claim of this module states for the instance a condition on an enclosure of the companion's datum under which its certificate holds, by deciding that condition exactly on the datum's enclosure recentred as (i) states (for the operator (19.13) on the flat instance's licensed route, (21.10) of Corollary 21.5). A requested entry whose certificate at the companion's datum is not so established is not formed on the companion and has no entry there in its place. For each entry $(\hat{O}_k, \bowtie_k, c_k)$ of that list, with the enclosure (C_k, R_k) of its operator that Corollary 12.1 (iii) gives, or radius zero where the operator is the artefact's: the operator $\hat{O}'_k$ whose every real and imaginary part is the binary64 number nearest to that of C_k, ties to even, which is Hermitian and has the blocks of $\hat{O}_k$; the rationals $\rho_{il} \coloneqq |\operatorname{Re}(\hat{O}'_k - C_k)_{il}| + |\operatorname{Im}(\hat{O}'_k - C_k)_{il}| + (R_k)_{il}$ and $\delta_k \coloneqq \max_i \sum_l \rho_{il}$, computed exactly; and the relaxed entries of Lemma 12.3 at δ_k, with c'_k the largest binary64 number at most $c_k - \delta_k$ and c''_k the smallest at least $c_k + \delta_k$. Where $\delta_k = 0$ the entry is $\hat{O}_k$ itself and is emitted as it is, an equality included. Each emitted entry carries the certificate case of the entry it relaxes, its δ_k, and the mark

that it is relaxed, its certificate being Lemma 12.3 (ii); its operator is added to the companion's transported set with its pull-back. The lattice condition of Requirement 16.9 binds the artefact's own entries and not these.

(iv) *No normalizer.* No step of the emission normalizer (Definition 11.1) acts on the companion's datum: the companion carries the reference, not a snap of it.

(v) *The record.* The companion carries, per block of G, whether it is the handed reference or the snapped block, the kind of every handed block (*exact*, every radius zero, or *enclosed*, with the largest radius of the block after recentring), the entries recentred in the handed blocks and in the datum, the largest radius of the datum's enclosure after its recentring, each δ_k, every requested entry formed on the companion with what established its certificate there, and every requested entry not formed with the reason; and it is marked evaluation-only: it has no run of its own, no consumer admitting it as the declaration of a run, and what a consumer decides or computes of it beside the declaration it accompanies is the consumer's (Requirement "Admission of the reference companion beside its declaration" (meat-core/req:companion-admission)). The artefact itself, its target, normalizer, emission record (Requirement 17.3) and entries, is emitted as it is without a companion.

(vi) *No companion.* Where no block is handed and the normalizer moves no class, or no claim of this module establishes (H-REF) for the instance, or a reciprocal node of the rule of (I1) receives no enclosure, so that Corollary 12.1 (iii) gives none, no companion is emitted, and the record says which of the three holds.

(vii) *Factor enclosures.* Where the handed reference carries factor enclosures (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) on some of its blocks $G_j^{(b)}$, b a handed block, as the receiver's does (Requirement "The emitted blocks are dark-composed and carry their reference" (receiver/detector-imperfections/req:dark-composed-reference)), and a claim of this module establishes that the rule of (I1) of the instance is a non-negative combination (Corollary 12.2, part (iii) for the identity instance and Corollary 21.5 for the flat instance), the engine forms, for every element of every data sector of the companion's datum every combining block of which carries one, the factor enclosure of Corollary 12.2 (i): the boxes of the combining blocks side by side in a fixed order, their centres and radii unaltered, and their cores multiplied by the coefficients in exact rational arithmetic. It emits each beside the entrywise enclosure of the datum of (i), which it does not alter, as the factor enclosure of the target's block on that data sec-

tor (part (ii) of that corollary). It forms none on any other element: none where a combining block carries none, and none for an instance no claim of this module shows to have such a rule, in particular none through a rule that maps the blocks of a data sector by a conjugation before it combines them, whose factor enclosure this module does not form. The record of (v) carries, per element of the datum, whether a factor enclosure is carried, with its number of columns, or why none is, and it names the claim that shows the rule of (I1) to be a non-negative combination, Corollary 12.2 (iii) or Corollary 21.5.

Requirement 17.5 (Party partition emitted with the sector map). The engine emits the party partition of each reduction instance with its sector map (`target_descriptor`): the default party partition where the instance declares no party label, and otherwise the declared label (`party_label`) on every factor of every $\mathcal{H}(Q_s)$. Where several parties measure, each factor's artefact carries that factor's own label with its own sector map (Theorem 11.5 (i)).

Requirement 17.6 (Emission of several measuring parties as factors). Where several measuring parties measure, the engine emits in `factor_emission` the ordered list of the factors' artefacts, each with its party label, in the declared order of the factors, and the combination rule (11.10); it emits, declares and rounds no entry of any product element F_v (Theorem 11.5).

18 Requirements: the assumption ledger

Requirement 18.1 (Rows of the emitted assumption ledger). The engine emits with every artefact the assumption ledger `assumption_ledger`: one row, carrying its machine key, for each assumption of this contract and for each assumption the interface lists as taken from another module, each under its declared presence rule. The row of A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) is present only where the emitted chain's source segment is non-empty, and there under the presence rule the interface declares for it, the row of Assumption 13.7 only where the emitted constraint list is non-empty, the row of Assumption 13.5 only where the declared setting set $\mathcal{T}_{\text{set}}$ has at least two values, the row of Assumption 13.6 only where $\mathcal{T}_{\text{set}}$ has at least two values or a public choice is applied at the measurement the emitted family describes (and also where the emission is not told whether one is), a submodule's own assumption where that submodule's instance is emitted and any narrower presence rule the submodule declares for it holds, the row of Assumption 13.3 on every emission save where Requirement 18.7 omits it, and every other row on every emission. These rows include every key of the core's own assumptions,

and every key the interface lists as taken from another module, that the closure of the claims producing the artefact contains, save the row of Assumption 13.7 where the emitted constraint list is empty, a condition on the constants of emitted entries holding vacuously of an empty list, the row of Assumption 13.5 where $\mathcal{T}_{\text{set}} = \{*\}$, whose clauses hold of the one setting drawn with weight one, and the row of Assumption 13.6 where in addition no public choice is applied at that measurement, whose clauses then concern no choice, and the row of Assumption 13.3 where Requirement 18.7 omits it, whose clauses that requirement shows to hold there (the supplier's keys ride the row of Assumption 13.3 instead, under the supplier's own presence rules, Requirement 18.3); a row may be present that no such closure contains.

Requirement 18.2 (Never-verified rows and validation of declared data). The engine marks as never verifiable (`never_verified`) the row, in the assumption ledger, of every assumption that the engine never verifies, as that assumption's verification field says, whether this contract, one of its submodules or another module states it; no code path decides what any of them says beyond its clauses on the declared data. The engine decides by validation the conditions on the declared data that the assumptions of this contract are read with, such as the declared dimensions of Definition 1.1 and the declared class weights of Definition 4.4.

Requirement 18.3 (Pass-through of the supplier's assumption ledger). On the row of Assumption 13.3 in the assumption ledger `assumption_ledger`, the engine carries, as machine keys and beside the artefact's module version `module_version`, every key the supplied ledger carries, as supplied and in the supplier's order; a key beyond those the supplier's presence rules require is carried rather than refused. The engine holds no list of the supplier's keys and adds no presence rule of its own: what it reads from the supplying module, as it is installed where the check runs, is that module's machine keys and the set of keys its rules make conditional, and it evaluates the condition itself on the artefact's own data.

Requirement 18.4 (Refusal of a supplier ledger with a repeated or missing key). The engine refuses a supplied ledger that carries one machine key more than once, and it refuses an artefact whose ledger lacks a key the supplier's own presence rules make present, rather than passing it on as a silently shortened list; both refusals carry the reason `receiver_spec_not_accepted`.

Requirement 18.5 (Refusal of a supplier artefact of an unaccepted version). The engine refuses a supplier artefact whose declared module version `module_version` is not in the caller's declared set of accepted versions

`accepted_receiver_versions`, and one that does not carry the supplying module's validation results; both refusals carry the reason `receiver_spec_not_accepted`, the reason the ledger refusals of Requirement 18.4 carry as well.

Requirement 18.6 (Supplier presence rules that omit no imported premise). The supplier's presence rules that the engine reads omit a key of the closure of the facts a reduction instance imports from that supplier only where the facts imported through it hold of the declaration without it; a change of those rules, which takes effect in the engine's check without an edit of this module, must itself meet this requirement.

Requirement 18.7 (Omission of the supplier row for a receiver artefact the run built). The engine omits the row of Assumption 13.3 from the assumption ledger `assumption_ledger` of an emission only where the reduction instance read at least one supplier artefact and, for every supplier artefact it read, all of the following hold: (i) the supplying module is the receiver, and the artefact is one that the part of the engine that ran this module obtained in the same run by running the receiver module, as installed where the run runs, on the receiver declaration the run holds, and handed to this module as it was emitted; (ii) the module version `module_version` the artefact declares is the one that installed module states for the artefacts it emits, and the artefact passes Requirements 18.5 and 18.4; (iii) the record of the run carries that artefact's own assumption ledger; (iv) the artefact records that its declaration has a model admissible exactly (Requirement "Deciding the exact route and enclosing the port Grams" (receiver/req:exact-route-decision) (iv)), and it carries the evaluation label `rigorous` (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)), every per-setting weight having been formed by the rigorous evaluation of Requirement "Rigorous evaluation of the multiclick weight" (receiver/req:rigorous-weight). There clause (i) of Assumption 13.3 holds by construction, the acceptance of the version being the effective status of the receiver's claims in the run's closure, which the record of the run reports, and clause (ii) holds of the facts the instance imports, each a theorem of the receiver about the receiver instrument the declaration is, holding of the numbers emitted under the ledger (iii) carries and resting on no premise about arithmetic. That (i) and (iii) hold is established by the part of the engine that ran the receiver module and docked its artefact into the instance, and is passed to this module beside the artefact (`receiver_provenance`); the engine never reads it from the artefact. The engine decides (ii) and (iv) on the artefact. Where it omits the row, it records the omission with the artefact's version (`receiver_row_omitted_on_provenance`), and the supplier's keys, which ride that row where it is emitted (Requirements 18.3 and 22.9),

reach the record of the run from the artefact's own ledger by (iii). Where `receiver_provenance` is not passed, or a condition fails for any artefact read, the row is emitted as Requirement 18.1 says.

Requirement 18.8 (Output-state model of the test statistics). The engine emits no state-independent test fraction, and it declares the test-statistics model `test_statistics_model` to be `output_state`.

Flag-state and weight-preserving squashers

19 Flag-state targets and their chain

Definition 19.1 (Certified weights and the flag-state and weight-preserving targets)**.** Throughout this definition and the claims of this submodule that use it, $\mathcal{H}(B) = \mathcal{H}(B_{\le N}) \oplus \mathcal{H}(B_{>N})$ is an orthogonal two-block decomposition with projections $\Pi_{\le N}$ and $\Pi_{>N}$ — a photon-number cutoff in the optical instance, where $\mathcal{H}(B_{\le N}) = \bigoplus_{m \le N} \mathcal{H}(C_m)$ is finite-dimensional — and every M_v is block diagonal for it, $M_v = M_v^{(\le N)} \oplus M_v^{(>N)}$. $\mathcal{H}(\mathrm{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ carries the orthonormal basis $\{|v\rangle\}_{v \in \mathcal{V}}$, and $\mathcal{H}(Q) = \mathcal{H}(B_{\le N}) \oplus \mathcal{H}(\mathrm{Flg})$ with block projections $\Pi_{\le N}$ (the same symbol on either space) and Π_{Flg}.

A family $(p_v)_{v \in \mathcal{V}}$ of *certified weights* for M at the cutoff consists of non-negative numbers p_v — on the emission path, integer multiples of $2^{-\kappa}$, the $p_{(y,t)}$ of (20.4) — with

$$M_v^{(>N)} \succeq p_v \, \mathbb{1}_{B_{>N}} \qquad \text{for every } v \in \mathcal{V}, \tag{19.1}$$

and we write $p \coloneqq \sum_v p_v$ for their *total weight*. Define

$$F_v^{\mathrm{FSS}} \coloneqq M_v^{(\le N)} \oplus |v\rangle\langle v|_{\mathrm{Flg}}\,, \tag{19.2}$$

$$F_v^{\mathrm{WP}} \coloneqq M_v^{(\le N)} \oplus \big(p_v \, \mathbb{1}_{\mathrm{Flg}} + (1-p)\, |v\rangle\langle v|_{\mathrm{Flg}}\big). \tag{19.3}$$

F^{FSS} is the *flag-state target* and F^{WP} the *weight-preserving target*, the multi-bound weight-preserving flag-state-squasher target, with one certified weight per outcome. The *single-bound case* is the one in which only one p_v is non-zero, and F^{FSS} is the case $p_v \equiv 0$.

Proposition 19.4 (Flag-state squashers: existence, chain and encoded weights). Let (p_v) be certified weights with $p < 1$ and define the

channels

$$\Lambda^{\mathrm{WP}}(\rho) \coloneqq \Pi_{\le N}\rho\,\Pi_{\le N} \;\oplus\; \sum_v \mathrm{tr}\big[M_v^{\mathrm{WP}}\,\Pi_{>N}\rho\,\Pi_{>N}\big]\;|v\rangle\langle v|_{\mathrm{Flg}}\,, \qquad M_v^{\mathrm{WP}} \coloneqq \frac{M_v^{(>N)} - p_v\,\mathbb{1}_{B_{>N}}}{1-p}, \tag{19.4}$$

$$\Lambda^{\mathrm{FSS}}(\rho) \coloneqq \Pi_{\le N}\rho\,\Pi_{\le N} \;\oplus\; \sum_v \mathrm{tr}\big[M_v^{(>N)}\,\Pi_{>N}\rho\,\Pi_{>N}\big]\;|v\rangle\langle v|_{\mathrm{Flg}}\,, \tag{19.5}$$

$$\widetilde{\Lambda}(\sigma) \coloneqq (1-p)\,\sigma + p\,\Pi_{\le N}\sigma\,\Pi_{\le N} + \mathrm{tr}\big[\Pi_{\mathrm{Flg}}\sigma\big]\,\mu, \qquad \mu \coloneqq \sum_v p_v\,|v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{19.6}$$

Then:

(i) *(existence, with the cutoff projectors transported)*

$$\big(M;\Pi_{\le N},\Pi_{>N}\big) \preceq_{\Lambda^{\mathrm{WP}}} \big(F^{\mathrm{WP}};\Pi_{\le N},\Pi_{\mathrm{Flg}}\big), \qquad \big(M;\Pi_{\le N},\Pi_{>N}\big) \preceq_{\Lambda^{\mathrm{FSS}}} \big(F^{\mathrm{FSS}};\Pi_{\le N},\Pi_{\mathrm{Flg}}\big); \tag{19.7}$$

(ii) *(the chain)* $F^{\mathrm{WP}} \preceq_{\widetilde{\Lambda}} F^{\mathrm{FSS}}$, and the channels compose exactly, $\Lambda^{\mathrm{FSS}} = \widetilde{\Lambda}\circ\Lambda^{\mathrm{WP}}$ on all of $\mathcal{T}(B)$, so that

$$M \;\preceq_{\Lambda^{\mathrm{WP}}}\; F^{\mathrm{WP}} \;\preceq_{\widetilde{\Lambda}}\; F^{\mathrm{FSS}} \;\preceq_{\Lambda_{\mathrm{mp}}}\; P^{\mathrm{Flg}}; \tag{19.8}$$

(iii) *(the encoded constraint)* $F_v^{\mathrm{WP}} \succeq p_v\,\Pi_{\mathrm{Flg}}$ as operators on Q, so $\mathrm{tr}\big[F_v^{\mathrm{WP}}\sigma\big] \ge p_v\,\mathrm{tr}[\Pi_{\mathrm{Flg}}\sigma]$ holds on *every* squashed state, reachable or not. Under F^{FSS} the same inequality is only a pull-back constraint: it holds on every reachable state, since $\Lambda^{\mathrm{FSS}\dagger}\big(F_v^{\mathrm{FSS}} - p_v\Pi_{\mathrm{Flg}}\big) = M_v^{(\le N)} \oplus \big(M_v^{(>N)} - p_v\mathbb{1}_{B_{>N}}\big) \succeq 0$, and it *fails* at $\sigma = |v'\rangle\langle v'|_{\mathrm{Flg}}$ for any $v' \ne v$ with $p_v > 0$;

(iv) *(no reverse relation in general)* $F^{\mathrm{FSS}} \preceq F^{\mathrm{WP}}$ can fail: at $\mathcal{H}(B_{\le N}) = \mathcal{H}(B_{>N}) = \mathbb{C}$, $\mathcal{V} = \{1,2\}$, $M_1 = \frac{1}{2}\oplus 1$, $M_2 = \frac{1}{2}\oplus 0$, $p_1 = \frac{1}{2}$, $p_2 = 0$, one has $\|F_2^{\mathrm{FSS}}\| = 1 > \frac{1}{2} = \|F_2^{\mathrm{WP}}\|$, which Lemma 4.9 (i) forbids.

Definition 19.5 (The encoded and the licensed route)**.** The *encoded* route uses the weight-preserving target F^{WP} and emits $\mathfrak{L} = \varnothing$; the *licensed* route uses the flag-state target F^{FSS} and emits the entry of Theorem 6.7.

Proposition 19.7 (Monotonicity of the weight-preserving target in its weights)**.** Let $(p_v)_{v\in\mathcal{V}}$ be certified weights for M with total weight p, let $p'_v \in [0,p_v]$ for every $v \in \mathcal{V}$, and write $p' \coloneqq \sum_v p'_v$. Write $F^{\mathrm{WP}}[p]$ and $F^{\mathrm{WP}}[p']$ for the weight-preserving targets (19.3) built from (p_v) and from (p'_v).

(i) (p'_v) are certified weights for M, and $p' \le p$; in particular, where $p < 1$ the construction of Proposition 19.4 runs unchanged with (p'_v).

(ii) If $p < 1$, put

$$\Psi(\sigma) \coloneqq \lambda\,\sigma + (1-\lambda)\,\Pi_{\le N}\sigma\Pi_{\le N} + \mathrm{tr}\big[\Pi_{\mathrm{Flg}}\sigma\big]\,\nu, \qquad \lambda \coloneqq \frac{1-p}{1-p'}, \qquad \nu \coloneqq \sum_v \frac{p_v - p'_v}{1-p'}\,|v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{19.12}$$

Then Ψ is a channel and $F^{\mathrm{WP}}[p] \preceq_\Psi F^{\mathrm{WP}}[p']$, so by Corollary 4.2 the smaller-weight target is the cleaner one.

20 The weight route

Lemma 20.1 (Descent of a dark-count matrix through a coarse-graining)**.** Let R be column-stochastic on $2^{[K]}$ and $\varpi\colon 2^{[K]} \to \mathcal{Y}$ surjective. Say that R *descends through* ϖ if the quantity $P(y \mid S') \coloneqq \sum_{S\in\varpi^{-1}(y)} R_{S|S'}$ depends on S' only through $\varpi(S')$; the resulting matrix $R^{\varpi}_{y|y'} \coloneqq P(y \mid S')$ for any $S' \in \varpi^{-1}(y')$ is then well defined and column-stochastic. In that case, for every family of operators $\{X_S\}$,

$$\sum_{S\in\varpi^{-1}(y)} \sum_{S'} R_{S|S'} X_{S'} \;=\; \sum_{y'\in\mathcal{Y}} R^{\varpi}_{y|y'} \sum_{S'\in\varpi^{-1}(y')} X_{S'}, \tag{20.1}$$

so applying R and then coarse-graining agrees with coarse-graining and then applying R^{ϖ}. A sufficient condition, which the canonical threshold-detector declaration satisfies, is that R be *monotone* ($R_{S|S'} > 0 \Rightarrow S' \subseteq S$) and that, with $\mathsf{MC} \coloneqq \{S : |S| \ge 2\}$,

$$\varpi \text{ is injective on } 2^{[K]}\backslash\mathsf{MC}, \qquad \varpi \text{ is } \textit{constant} \text{ on } \mathsf{MC}, \text{ say } \varpi(\mathsf{MC}) = \{y_{\mathrm{mc}}\}, \qquad \varpi^{-1}(y_{\mathrm{mc}}) = \mathsf{MC}; \tag{20.2}$$

the case $\varpi = \mathrm{id}$ is trivial.

On declared data both the hypothesis and the conclusion are read exactly. The declared R is exactly column-stochastic, and the descent condition, that $P(y|S') = P(y|S'')$ for every y and every pair S', S'' with $\varpi(S') = \varpi(S'')$, is an identity between the dyadic rationals the declaration carries and is decided as one (Definition 9.2). The representative is declared, not chosen: $R^{\varpi}_{y|y'} \coloneqq P(y|S'_{y'})$ with $S'_{y'}$ the first element of $\varpi^{-1}(y')$ in the declared order of $2^{[K]}$, so the descended matrix is a determinate object and not one of several. Column-stochasticity of R^{ϖ} needs no second decision: the identity $\sum_y R^{\varpi}_{y|y'} = \sum_S R_{S|S'}$ of the proof regroups one finite sum, and in the exact substrate a regrouped sum *is* the sum it regroups. So (20.1) holds of the declared numbers as an exact identity.

Definition 20.3 (The weight route)**.** Let a receiver artifact supply, for each setting $t \in \mathcal{T}_{\mathrm{set}}$, preserved photon-number blocks, $m \le N$, in one of the two input forms below, for a conditional POVM $\{M_{y|t}\}_{y\in\mathcal{Y}}$ on C block diagonal at the cutoff; a dark-count matrix R on $2^{[K]}$ that descends through the

declared coarse-graining ϖ with descended matrix R^{ϖ} (Lemma 20.1); class weights $\{q_t\}_{t\in\mathcal{T}_{\mathrm{set}}}$ as in Definition 4.4; and a slot $\{(\mathcal{O}_i, (p_i^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}})\}_{i=1}^{r}$ of certified high-photon weights, one per entry and per setting of the same $\mathcal{T}_{\mathrm{set}}$; every declared entry and scalar lies on the declared lattice (9.1), and every one of the supplier's facts about them is held under Assumption 13.3 (Proposition 20.4 (iv)). The *weight route* is the following three steps, in this order, each with the claim that licenses it. Steps 1–2 are classical post-processing of the physical outcome and precede the first squashing link.

1. *Dark counts, on the emitted alphabet.* Replace the supplied conditional POVM by

$$\widetilde{M}_{y|t} \;\coloneqq\; \sum_{y'\in\mathcal{Y}} R^{\varpi}_{y|y'}\, M_{y'|t}, \qquad y\in\mathcal{Y},\ t\in\mathcal{T}_{\mathrm{set}}. \tag{20.3}$$

   This is post-processing by a column-stochastic matrix (Lemma 4.3 (i)), so the result is a conditional POVM at each t with the same photon-number block structure, and Lemma 20.1 shows it equals the coarse-graining of the raw-level composition.

2. *Joint alphabet.* Form the joint family of Definition 4.4 (b), $M_{(y,t)} \coloneqq q_t\,\widetilde{M}_{y|t}$ on the joint alphabet $\mathcal{V} = \mathcal{Y}\times\mathcal{T}_{\mathrm{set}}$; this is a POVM by (4.2) (Lemma 4.3 (ii); the weights are the measuring party's own, Assumption 13.5), block diagonal at the cutoff, and the class-marginal identity (4.3) holds of it block by block. Its preserved blocks $M^{(=m)}_{(y,t)}$, $m\le N$, are the *post-processed datum*.

3. *The instance.* The post-processed joint family, with the routed weights below, is the datum, the hypotheses and the target data of the *flat instance* (Proposition 21.2), and its emission is that instance's.

The two input forms. The supplied preserved blocks are declared in one of two forms. In the *bare form* they are the blocks $M^{(=m)}_{y|t}$, $m\le N$, of the conditional POVM above, and step 1 acts on them. In the *dark-composed form* they are a declared family $\{\widetilde{M}^{(=m)}_{y|t}\}_{y\in\mathcal{Y}}$, $m\le N$, per setting t, each block Hermitian, positive semidefinite and complete on its block, which the model takes below the cutoff after the dark counts: in this form the family $\widetilde{M}_{y|t}$ of step 1 is defined as these blocks below the cutoff and by (20.3) above it, step 1 is not applied to the supplied blocks, and the post-processed datum is formed from them by step 2 alone. Where no snap intervened between composing and supplying, this form is the bare form composed in advance (Proposition 20.5 (ii)); where the declaring party snapped the composed blocks, they are its snapped family, and this definition asserts of them no relation to a bare family. In either form R^{ϖ} is formed and its conditions are decided as below, (20.3) defines the family above the cutoff,

$\widetilde{M}^{(>N)}_{y|t} = \sum_{y'} R^{\varpi}_{y|y'} M^{(>N)}_{y'|t}$, and the routed weights below read R^{ϖ} as they stand. In either form $\{\widetilde{M}_{y|t}\}_y$ is a POVM at each t, block diagonal at the cutoff: below it by step 1 in the bare form and by the declared properties of the supplied blocks in the dark-composed form, and above it by step 1.

Exactness of steps 1 and 2. The two steps are written on the declared numbers, and their POVM conclusions hold *exactly* of them. The declared R^{ϖ} is exactly column-stochastic with non-negative entries, decided on entry, and $\{q_t\}$ is exactly a probability distribution, as Definition 4.4 requires of class weights; and the combination is formed in the exact substrate of Lemma 9.3 (Definition 9.2), so $\{\widetilde{M}_{y|t}\}_y$ and $\{M_{(y,t)}\}$ are POVMs and every claim of this submodule reads them as such. What the two steps can nevertheless do is leave the lattice: a product of two dyadic rationals adds significand widths, so the post-processed blocks $M^{(=m)}_{(y,t)}$ need not be binary64 numbers even though every factor is. That is the one place in this submodule where exact arithmetic outruns the representation; it is where the emission normaliser of Definition 11.1 acts, and it acts on numbers this module computed and is about to declare, never on numbers it was handed.

The routed weights. They follow the same route, one bounded letter at a time. Fix an entry i whose bounded set is exactly a fibre, $\mathcal{O}_i = \varpi^{-1}(y_i)$, and whose letter is absorbing, $R^{\varpi}_{y_i|y_i} = 1$; both are declared input validity conditions, and both are decided exactly. Write

$$\begin{aligned} \hat{p}_{(y_i,t)} &:= q_t\Big(\big(1 - r_{\min,i}\big)p_i^{(t)} + r_{\min,i}\Big), \qquad p_{(y_i,t)} := \big\lfloor \hat{p}_{(y_i,t)} \big\rfloor_{\kappa}, \\ r_{\min,i} &:= \min_{y' \neq y_i} R^{\varpi}_{y_i|y'} \quad \big(:= 0 \text{ when that minimum is over the empty set}\big), \end{aligned} \tag{20.4}$$

with $p_{(y,t)} := 0$ for every other (y,t); the family $(p_v)_{v\in\mathcal{V}}$ is the *routed weights*, and $p := \sum_{v\in\mathcal{V}} p_v$. Here

$$\big\lfloor x \big\rfloor_{\kappa} := 2^{-\kappa}\big\lfloor 2^{\kappa} x \big\rfloor \tag{20.5}$$

is the largest integer multiple of $2^{-\kappa}$ that is at most x: the *downward* rounding onto the declared lattice, computed exactly on the rational $\hat{p}_{(y_i,t)}$ and equal to it whenever it is already a lattice number. The routed weights are *certified* when the element $M_{(y_i,t)}$ of step 2 satisfies, on every channel block above the cutoff,

$$M^{(>N)}_{(y_i,t)} \succeq p_{(y_i,t)}\,\mathbb{1} \qquad \text{for every } i \text{ and every } t \in \mathcal{T}_{\text{set}}, \tag{20.6}$$

which Proposition 20.4 proves from the declared input validity conditions together with the absorbing hypothesis those conditions impose, and from nothing else.

The direction of the rounding. The emitted weight is $p_{(y_i,t)}$ and not $\hat{p}_{(y_i,t)}$, and the direction of that rounding is the whole of what makes it safe: it is

the one emitted number of this submodule that a downstream inequality consumes as a *bound*, the certified-weight inequality of Definition 19.1 is preserved when a certified weight is lowered and, when one is raised, is no longer proved and can be false, and $\lfloor \cdot \rfloor_\kappa$ only lowers (Proposition 20.4; Proposition 19.7 (i)). Four features of the route are relied on, and none of them may be dropped. *(a)* $r_{\min}$ is indexed by the bounded letter: it is a property of row y_i of R^{ϖ} read across the columns $y' \neq y_i$, and one entry's value is not another's. *(b)* The absorbing condition is a hypothesis, not a consequence: without it the bound (20.6) for the weights of (20.4) is no longer proved. *(c)* A non-zero $r_{\min,i}$ can occur only at $r = 1$ (Corollary 20.6), so for $r \geq 2$ *every* entry contributes the un-improved weight $\lfloor q_t p_i^{(t)} \rfloor_\kappa$, which is sound, merely not an improvement. *(d)* $p_{(y_i,t)} \geq 0$ needs no clamp: $\hat{p}_{(y_i,t)}$ is $q_t > 0$ times $(1 - r_{\min,i})p_i^{(t)} + r_{\min,i}$, a convex combination of $p_i^{(t)} \geq 0$ and 1, since $r_{\min,i} \in [0,1]$ as an entry of the column-stochastic R^{ϖ} or as 0; and the largest lattice number at most a non-negative rational is itself non-negative.

Proposition 20.4 (Certified weights routed from the receiver artifact)**.** Let $\{M_{y|t}\}_{y \in \mathcal{Y}}$ be, for each $t \in \mathcal{T}_{\text{set}}$, a POVM on C block diagonal at the cutoff, and let R descend through ϖ with descended matrix R^{ϖ} (Lemma 20.1). Let $\{q_t\}$ be a probability distribution, and let $\{(\mathcal{O}_i, (p_i^{(t)})_{t \in \mathcal{T}_{\text{set}}})\}_{i=1}^{r}$ be the declared slot, with $\mathcal{O}_i = \varpi^{-1}(y_i)$ for pairwise distinct letters $y_1, \ldots, y_r$, with $0 \leq p_i^{(t)} \leq 1$, and with the per-setting guarantee $M_{y_i|t}^{(>N)} \succeq p_i^{(t)} \mathbb{1}$ for every $t \in \mathcal{T}_{\text{set}}$ and every i. Suppose each bounded letter is *absorbing*,

$$R^{\varpi}_{y_i|y_i} = 1 \qquad \text{for every } i = 1, \ldots, r, \tag{absorbing}$$

and put $r_{\min,i} := \min_{y' \neq y_i} R^{\varpi}_{y_i|y'}$, with the convention $r_{\min,i} := 0$ when the minimum is over the empty set (that is, when $\mathcal{Y} = \{y_i\}$), the conservative value, which reduces (20.7) to the un-improved bound. Then:

(i) The joint-alphabet POVM $M_{(y,t)} := q_t \sum_{y'} R^{\varpi}_{y|y'} M_{y'|t}$ of Definitions 4.4 and 20.3 satisfies, for every i and every $t \in \mathcal{T}_{\text{set}}$,

$$M_{(y_i,t)}^{(>N)} \succeq q_t \Big(\big(1 - r_{\min,i}\big)\, p_i^{(t)} + r_{\min,i} \Big) \mathbb{1} = \hat{p}_{(y_i,t)}\, \mathbb{1}. \tag{20.7}$$

A slot whose weights do not depend on the setting, $p_i^{(t)} = p_i$ for every t (the t-uniform form, which every one-setting slot has), is the special case, and there (20.7) is the bound in p_i.

(ii) The *emitted* weight of (20.4) is certified, because the rounding runs downwards. Every number on the right of (20.7) lies on the declared lattice: q_t and $p_i^{(t)}$ by declaration, and $r_{\min,i}$ as an entry of R^{ϖ}, which is a sum of declared entries and lies in $[0,1]$. So $\hat{p}_{(y_i,t)}$ is an exact

element of $\mathbb{F}$ formed from them; and $p_{(y_i,t)} = \lfloor \hat{p}_{(y_i,t)} \rfloor_\kappa \le \hat{p}_{(y_i,t)}$ by (20.5), whence

$$M^{(>N)}_{(y_i,t)} \succeq p_{(y_i,t)}\,\mathbb{1},$$

which is (20.6); so the emitted (p_v) are certified weights in the sense of Definition 19.1 and they lie on the declared lattice.

(iii) Under the sufficient condition (20.2) of Lemma 20.1 and with the single entry $\mathcal{O}_1 = \mathsf{MC}$, (absorbing) holds at $y_1 = y_{\mathrm{mc}}$, and its per-setting hypothesis is the supplier's slot guarantee $\sum_{S\in\mathsf{MC}} M^{(>N)}_{S|t} \succeq p^{(t)}_1\,\mathbb{1}$ at each t, read through $\varpi^{-1}(y_{\mathrm{mc}}) = \mathsf{MC}$.

(iv) On the emission path the hypotheses about the supplier's objects hold under Assumption 13.3: the conditional families, and their block diagonality in the photon number, block by block and so in particular at the cutoff, are at every setting those of the coarse-grained elements of the setting family on which the supplier states its weights, by item (a) of Definition "Setting family" (receiver/def:setting-family) and Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining); for a receiver without probe inputs that family is the one-setting family of Corollary "The unconditioned receiver is a setting family" (receiver/cor:unconditioned-family), and the conditional families and their block diagonality are given by the receiver's Theorem "Click-pattern POVM" (receiver/thm:click-povm) (iii) and Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining); the dark-count matrix R, and the block diagonality in the photon number of the elements with the dark counts composed, by Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm) and Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts); so every element of the joint family of (i), and of the post-processed joint family that Definition 20.3 forms in either of its forms, is block diagonal in the photon number in every block, below the cutoff as above it; and the slot, its per-setting guarantee (the first conclusion of Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound)) and $0 \le p^{(t)}_i \le 1$, $\sum_i p^{(t)}_i \le 1$ at every t by Definition "The strengthening slot" (receiver/def:slot), Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound) and Proposition "Range of the strengthening-slot weights" (receiver/prop:slotvalid). The other hypotheses are not supplier facts: that R is column-stochastic and descends through ϖ, that $\mathcal{O}_i = \varpi^{-1}(y_i)$ and (absorbing) are decided exactly on the declared data, a declaration failing one being refused, and $\{q_t\}$ are the declared class weights (Definition 4.4).

Proposition 20.5 (Preserved blocks supplied with the dark counts already composed)**.** Let the data of Definition 20.3 be given in the bare form, with supplied preserved blocks $B^{(m)}_{y|t}$, $m \le N$, and put

$$\widetilde{B}^{(m)}_{y|t} \;\coloneqq\; \sum_{y' \in \mathcal{Y}} R^{\varpi}_{y|y'}\, B^{(m)}_{y'|t}. \tag{20.8}$$

(i) For every t and m, $\{\widetilde{B}^{(m)}_{y|t}\}_{y\in\mathcal{Y}}$ is a family of Hermitian positive semidefinite matrices complete on its block.

(ii) The flat instance of Proposition 21.2 formed from the blocks $\widetilde{B}$ in the dark-composed form, and the one formed from the blocks B in the bare form, have the same handed datum, $A^{(m)}_{(y,t)} = q_t \widetilde{B}^{(m)}_{y|t}$, the same hypotheses (H), model map, routed weights, scalar values, channel, transported set and entries; so they are one reduction instance, and the emission normalizer gives them the same declared datum, declared model and record.

(iii) Let $\{G^{(m)}_{y|t}\}_{y\in\mathcal{Y}}$ be a reference of the bare blocks of setting t with constant 1, read through the entrywise enclosures $\big(C^{(m)}_{y|t}, r^{(m)}_{y|t}\big)$ (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)), and put $\widetilde{G}^{(m)}_{y|t} \coloneqq \sum_{y'} R^{\varpi}_{y|y'} G^{(m)}_{y'|t}$. Then $\widetilde{G}$ is a reference with constant 1, read through the enclosures $\big(\sum_{y'} R^{\varpi}_{y|y'} C^{(m)}_{y'|t}, \sum_{y'} R^{\varpi}_{y|y'} r^{(m)}_{y'|t}\big)$, and it is an exact reference where G is. If B is the snapped family of a snap of G and B' that of a snap of $\widetilde{G}$ (Definition "Snapping declared data onto a lattice" (foundations/def:snap)), then the handed data $A^{\mathrm{b}}_{(y,t)} = q_t \sum_{y'} R^{\varpi}_{y|y'} B_{y'|t}$ of the bare form and $A^{\mathrm{d}}_{(y,t)} = q_t B'_{y|t}$ of the dark-composed form differ in general. The two flat instances agree in everything but their data, so their emissions differ only in their data sectors.

Corollary 20.6 (Dark-count sharpening at a single bounded letter only)**.** Under (absorbing): if $r \ge 2$ then $r_{\min,i} = 0$ for *every* i, so (20.7) reads $M^{(>N)}_{(y_i,t)} \succeq q_t p^{(t)}_i \,\mathbb{1}$ throughout and the emitted weight is $\lfloor q_t p^{(t)}_i \rfloor_\kappa$. Consequently a positive $r_{\min,i}$, the only case in which the dark counts sharpen a weight, can occur for at most one bounded letter, and only when $r = 1$.

21 The flat instance and its emission

Corollary 21.1 (Licensed flag-sector entry of a certified weight)**.** In the situation of Proposition 19.4 with the flag-state target F^{FSS}, fix an outcome $o \in \mathcal{V}$ with certified weight p_o and put

$$\hat{O} \coloneqq F^{\mathrm{FSS}}_o - p_o\, \Pi_{\mathrm{Flg}}, \qquad \bowtie \;=\; \succeq, \qquad c = 0. \tag{21.1}$$

Then (6.9) holds, so Theorem 6.7 applies and the single-entry list $\mathfrak{L} = \{(\hat{O}, \succeq, 0)\}$ is licensed with the admissible set

$$\mathfrak{A}_{\mathfrak{L}}(E', QE'') = \Big\{\mathcal{A} : \ \mathcal{A}^{\dagger}\big(F_o^{\mathrm{FSS}} \otimes \mathbb{1}_{E''}\big) \ \succeq \ p_o\, \mathcal{A}^{\dagger}\big(\Pi_{\mathrm{Flg}} \otimes \mathbb{1}_{E''}\big)\Big\}. \tag{21.2}$$

Proposition 21.2 (The flat instance of the post-processed receiver)**.** Let the data of Definition 20.3 pass every refusal rule of this module's interface; let $\{M_v\}_{v\in\mathcal{V}}$ be the post-processed joint family of its step 2 on C, with $\mathcal{H}(C) = \bigoplus_{m\geq 0}\mathcal{H}(C_m)$, $\mathcal{H}(C_{\leq N})$ finite-dimensional, every M_v block diagonal at the cutoff N; let (p_v) be its routed weights with $p = \sum_v p_v < 1$; and let a route, encoded or licensed, be declared (Definition 19.5). Then the *flat instance* $\mathcal{J}_{\mathrm{flat}}$, given by

(I1) $B = C$; the data sectors (m), $0 \leq m \leq N$, in that order, with $d_m = \dim \mathcal{H}(C_m)$ and the default supports $\mathcal{V}^{(m)} = \mathcal{V}$; the handed datum $A_v^{(m)} \coloneqq M_v^{(=m)}$, formed from the declared blocks, R^{ϖ} and $\{q_t\}$ by steps 1–2 of Definition 20.3 in the bare form, and from the declared blocks and $\{q_t\}$ by step 2 alone in the dark-composed form; the hypotheses (H): *(H1)* for each $t \in \mathcal{T}_{\mathrm{set}}$ the above-cutoff blocks $\{M_{y|t}^{(>N)}\}_{y\in\mathcal{Y}}$ of the supplied conditional POVM form a POVM on $C_{>N}$ (the supplier's fact, Assumption 13.3, Proposition 20.4 (iv)); *(H2)* the routed weights are certified, (20.6) (Proposition 20.4); *(H3)* $p < 1$, decided on entry; and the model map

$$M_v[A'] \ \coloneqq \ \bigoplus_{m\leq N} A_v'^{(m)} \ \oplus \ M_v^{(>N)}, \qquad M_{(y,t)}^{(>N)} = q_t \sum_{y'\in\mathcal{Y}} R^{\varpi}_{y|y'} M_{y'|t}^{(>N)}; \tag{21.3}$$

(I2) the scalar sectors (f), $f \in \mathcal{V}$, in the order of $\mathcal{V}$, each of dimension 1, with

$$\phi_f(v) \ \coloneqq \ p_v + (1-p)\,\delta_{vf} \quad \text{(encoded route)}, \qquad \phi_f(v) \ \coloneqq \ \delta_{vf} \quad \text{(licensed route)}, \tag{21.4}$$

so that $\mathcal{H}(Q) = \mathcal{H}(C_{\leq N}) \oplus \mathcal{H}(\mathrm{Flg})$ and the target (10.2) is $F_v[A'] = F_v^{\mathrm{WP}}$ resp. F_v^{FSS} of Definition 19.1 for the model $M[A']$;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{WP}}$ resp. $\mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{FSS}}$ of Proposition 19.4, independent of the datum, with the transported set $\{\Pi_{\leq N}, \Pi_{\mathrm{Flg}}\} \cup \{\Pi_{(m)}\}_{m\leq N} \cup \{\Pi_{(f)}\}_{f\in\mathcal{V}}$ and pull-backs $\Pi_{\leq N}$, $\Pi_{>N}$, the photon-number projectors P_m^C, and M_f^{WP} resp. $M_f^{(>N)}$ read as operators on C supported in $C_{>N}$;

(I4) an empty entry list on the encoded route, and on the licensed route the flag-sector entries $\big(F_o^{\mathrm{FSS}} - p_o\Pi_{\mathrm{Flg}}, \succeq, 0\big)$ of Corollary 21.1 for every bounded letter $o = (y_i, t)$; the class-marginal identity discharged,

is a reduction instance over the joint alphabet of Definition 20.3. Its handed model $M[A]$ is the post-processed joint family $\{M_v\}$ itself where every M_v

is also block diagonal for the decomposition $\mathcal{H}(C_{\le N}) = \bigoplus_{m\le N} \mathcal{H}(C_m)$, the photon-number blocks the datum reads, and otherwise the family's pinching onto those blocks below the cutoff; on the emission path the first case holds, the receiver's elements being block diagonal in every photon number (Proposition 20.4 (iv)), so that there the handed model is the post-processed joint family itself. Its declared model, at the declared datum A' the normalizer of Definition 11.1 produces from A, is

$$M'_v \;:=\; M_v[A'] \;=\; \bigoplus_{m\le N} A'^{(m)}_v \;\oplus\; M_v^{(>N)}, \tag{21.5}$$

a POVM on C block diagonal at the cutoff, agreeing with the handed one above it and satisfying every certified-weight inequality (20.6) the handed one does.

Corollary 21.3 (Exact emission of the flat instance). Let the declared data pass every refusal rule of this module's interface, let $\mathcal{J}_{\mathrm{flat}}$ be the flat instance of Proposition 21.2 at the declared route with handed datum A, and let A' be the datum the normaliser of Definition 11.1 declares. Then Theorem 11.2 holds of $\mathcal{J}_{\mathrm{flat}}$, and in particular:

(i) every number on the list of Theorem 11.2 (i) and, in addition, the descended matrix R^{ϖ}, the constants $r_{\min,i}$, the exact weights $\hat{p}_v$ of (20.4), the total p and its complement $1-p$ are formed from the declared entries and scalars by finitely many operations in $\mathbb{F}_i$, floors and comparisons; the emission displacement Δ_m per preserved block is the one emitted number that is not;

(ii) the emitted family, F^{WP} resp. F^{FSS} of the declared model (21.5), satisfies conditions (a)–(e) of Definition 9.1 together with (9.1) of every entry and is decided so before it is emitted;

(iii) for every $v \in \mathcal{V}$,

$$\Lambda^{\dagger}(F_v) \;=\; M'_v \;=\; \bigoplus_{m\le N} A'^{(m)}_v \;\oplus\; M_v^{(>N)} \qquad \text{exactly, on } C, \tag{21.6}$$

both sides operators on C itself: the physical system of this instance is the channel system and the identity is on it, not on an ampliation. M' is a POVM on C block diagonal at the cutoff for which (p_v) are certified weights in the sense of Definition 19.1, and Proposition 19.4, Proposition 20.4 and Corollary 21.1 hold verbatim of it;

(iv) the per-outcome weights are $p_v = \lfloor \hat{p}_v \rfloor_\kappa$, rounded *down*, and (20.6) holds for M' at those values (Proposition 20.4); the complement $1-p$ is *recomputed* from the rounded weights and is never rounded on its own; a caller's constant off the lattice is rounded toward the enlarging side (Corollary 11.3 (iii)); and there are no other rounding directions;

(v) where some $\Delta_m > 0$ no bound relates a key length computed from M' alone to one computed from the pre-snap post-processed family, or from the receiver's own output, and nothing else relates them (Theorem 11.2 (v)).

The emitted constraint list is *empty on the encoded route*; on the licensed route it carries the flag-sector entries $(F_o^{\mathrm{FSS}} - p_o \Pi_{\mathrm{Flg}}, \succeq, 0)$ of Corollary 21.1 at $c = 0$, unshifted, and, on request, the class-marginal entries $\big(\sum_{v \in \mathcal{V}_t} F_v, =, q_t\big)$, which the flat instance discharges, a default carried operator as an entry (Corollary 11.3 (i)), and any further requested entry that Corollary 11.3 (iii) admits, which for this instance is the two-sector entry $(F_e^{\mathrm{FSS}} - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}, \succeq, 0)$ where its decided certificate of Corollary 21.4 holds, all under the licence of that route (Corollary 11.3). The instance's default carried operators, the sector projectors and the two projectors $\Pi_{\le N}$ and Π_{Flg} it transports besides, *are emitted with the transported set at $c = 0$ with $\succeq$ on either route* (Corollary 11.3 (i), Proposition 21.2 (I3)). These statements are about the instance's link and its emitted family.

Corollary 21.4 (The requested two-sector entry, decided on the emitted numbers)**.** Let $\mathcal{J}_{\mathrm{flat}}$ be the flat instance of Proposition 21.2 on the licensed route, with channel Λ and routed weights (p_v), and let A'' be an admissible datum of it over any substrate of Definition 9.2 over which it is a reduction instance, $\mathbb{F} = \mathbb{R}$ included. Let $e \in \mathcal{V}$, let a and b be declared real constants, and put

$$\hat{O}[A''] \;\coloneqq\; F_e^{\mathrm{FSS}}[A''] - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}, \tag{21.7}$$

written in the form (10.4) with coefficient 1 for e on every sector, every other coefficient 0, and $Z = -a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$. Then:

(i) its pull-back is

$$\Lambda^{\dagger}\big(\hat{O}[A'']\big) \;=\; \bigoplus_{m \le N} \big(A_e''^{(m)} - a\,\mathbb{1}_{d_m}\big) \;\oplus\; \big(M_e^{(>N)} - b\,\mathbb{1}\big); \tag{21.8}$$

(ii) if

$$A_e''^{(m)} - a\,\mathbb{1}_{d_m} \;\succeq\; 0 \quad (m \le N) \qquad \text{and} \qquad b \;\le\; p_e, \tag{21.9}$$

then $\Lambda^{\dagger}(\hat{O}[A'']) \succeq 0$, which is the certificate (6.9) of the entry $(\hat{O}[A''], \succeq, 0)$;

(iii) the first condition of (21.9) is necessary for that certificate. At the declared datum A' of Theorem 11.2, whose blocks and routed weights are emitted lattice values, and for constants a and b that are lattice values, both conditions are decided by finitely many field operations and sign comparisons on these numbers, and (21.9) is a decided certificate for the entry $(\hat{O}[A'], \succeq, 0)$ in the sense of Corollary 11.3 (iii).

Corollary 21.5 (The flat instance has a reference companion)**.** Let the flat instance $\mathcal{J}_{\mathrm{flat}}$ of Proposition 21.2 be given, at either route and in either form, with K the declared low-photon blocks from which its handed datum is formed by Definition 20.3 (the conditional POVM blocks $M^{(=m)}_{y'|t}$, $m \le N$, of each setting t in the bare form, the supplied dark-composed blocks $\widetilde{M}^{(=m)}_{y|t}$ in the dark-composed form). Then $\mathcal{J}_{\mathrm{flat}}$ satisfies (H-REF) of Corollary 12.1: for every family G over the same blocks, settings, outcomes and dimensions whose blocks are Hermitian and positive semidefinite, with $\sum_{y'} G^{(m)}_{y'|t} = \mathbb{1}_{C_m}$ exactly for every t and $m \le N$, the datum the same rule forms from G is admissible, and $\mathcal{J}_{\mathrm{flat}}$ is a reduction instance over the substrate $\mathbb{F} = \mathbb{R}$. Its rule of (I1) is a non-negative combination in the sense of Corollary 12.2: the element (y, t) of the datum it forms from a family K' with the properties just listed for G is, on the data sector (m), $q_t \sum_{y'} R^{\varpi}_{y|y'} K'^{(m)}_{y'|t}$ in the bare form and $q_t K'^{(m)}_{y|t}$ in the dark-composed form. At the datum A^{ref} so formed, its target is $F^{\mathrm{WP}}[A^{\mathrm{ref}}]$ on the encoded route and $F^{\mathrm{FSS}}[A^{\mathrm{ref}}]$ on the licensed route, and on the licensed route its entries are $\big(F^{\mathrm{FSS}}_o[A^{\mathrm{ref}}] - p_o \Pi_{\mathrm{Flg}}, \succeq, 0\big)$ for every bounded letter o, with the routed weights p_o of the handed instance. On the licensed route, let the request carry besides an entry $(\hat{O}, \succeq, 0)$ with $\hat{O} = F^{\mathrm{FSS}}_e - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$, at an outcome $e \in \mathcal{V}$ and declared real constants a and b, written in the form (10.4) with coefficient 1 for e on every sector, every other coefficient 0, and $Z = -a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$, so that its companion form (Corollary 12.1 (vi)) is $\hat{O}^{\mathrm{ref}} = F^{\mathrm{FSS}}_e[A^{\mathrm{ref}}] - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$. Let (C, R) be an entrywise enclosure of A^{ref} whose centres are Hermitian, and put $\rho_m \coloneqq \max_i \sum_l R^{(m)}_{e,il}$ for $m \le N$. If

$$C^{(m)}_e - (a + \rho_m)\,\mathbb{1}_{d_m} \;\succeq\; 0 \quad (m \le N) \qquad \text{and} \qquad b \;\le\; p_e, \tag{21.10}$$

then $\Lambda^{\dagger}(\hat{O}^{\mathrm{ref}}) \succeq 0$, with Λ the instance's channel, which is the same at every datum: the companion form carries the certificate of the entry it is formed from.

Corollary 21.6 (The flat instance of a normalized station)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, with efficiencies η_k and $\eta_{\max} > 0$, let $t \in [\eta_{\max}, 1]$, and let $\mathfrak{R}'$ be a normalized station of $\mathfrak{R}$ at the factored loss t (Definition "Normalized station and factored loss" (receiver/detector-imperfections/def:normalized-station)), both with the channel system C. Let the data of Definition 20.3 be supplied for $\mathfrak{R}'$, with $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$ and $q_{\mathrm{off}} = 1$, pass every refusal rule of this module's interface, and let $\mathcal{J}'_{\mathrm{flat}}$ be the flat instance of Proposition 21.2 at the declared route. Put $\Phi \coloneqq \Lambda^{\mathrm{loss}}_t$, the pure-loss channel of transmittance t on C (Definition "Pure-loss channel on the channel modes" (receiver/detector-imperfections/def:pure-loss)), and $\mathcal{J}_{\mathrm{flat}} \coloneqq \mathcal{J}'_{\mathrm{flat}} \circ \Phi$.

(i) *(the instance)* $\mathcal{J}_{\text{flat}}$ is a reduction instance on C with the datum, sectors, scalar values, target, transported set, entries, constants and class-marginal flag of $\mathcal{J}'_{\text{flat}}$; its channel is that of $\mathcal{J}'_{\text{flat}}$ precomposed with Φ, the pull-backs of its transported set are $\Phi^{\dagger}$ of those of $\mathcal{J}'_{\text{flat}}$, and its reachable states (Definition 3.10) are those of that composite channel. Where the supplied blocks are those of $\mathfrak{R}'$'s coarse-grained click-pattern POVM, in the bare form, or of its dark-composed family with the declared dark-count matrix, in the dark-composed form, its handed model is the post-processed joint family of steps 1–2 of Definition 20.3 formed from $\mathfrak{R}$'s own measurement with the same dark-count matrix.

(ii) *(the emission)* Theorem 11.2 holds of $\mathcal{J}_{\text{flat}}$: its declared datum, emitted family, routed weights, entries and constants are those Corollary 21.3 gives for $\mathcal{J}'_{\text{flat}}$, its channel is $\Lambda \circ \Phi$ with Λ that of $\mathcal{J}'_{\text{flat}}$, and for every $v \in \mathcal{V}$

$$(\Lambda \circ \Phi)^{\dagger}(F_v) \;=\; \Phi^{\dagger}(M'_v) \qquad \text{exactly, on } C, \tag{21.11}$$

with M' the declared model (21.5) of $\mathcal{J}'_{\text{flat}}$. Where moreover the normalizer kept every class, the right-hand side is the family of (i) formed from $\mathfrak{R}$'s measurement.

(iii) *(several measuring factors)* Where each factor f of Theorem 11.5 is such an instance $\mathcal{J}^{f}_{\text{flat}} = \mathcal{J}'^{f}_{\text{flat}} \circ \Lambda^{\text{loss}}_{t_f}$, each at its own factored loss t_f, that theorem applies, with channel $\bigotimes_f \left(\Lambda_f \circ \Lambda^{\text{loss}}_{t_f}\right)$ and every entry at the constant of its factor.

(iv) *(the reference companion)* $\mathcal{J}_{\text{flat}}$ satisfies (H-REF) of Corollary 12.1, and its reference companion at a family G is that of $\mathcal{J}'_{\text{flat}}$ at G precomposed with Φ: the same target and entries, the companion's channel precomposed with Φ, and $\Phi^{\dagger}$ of the companion's pull-backs.

22 Requirements

Requirement 22.1 (One route per artefact). The engine emits every artefact on exactly one route: the request's `route` is `encoded` or `licensed`, the emitted payload is single-valued, and a request asking for both is refused with `route_not_single_valued`. A caller wanting both issues two requests and receives two artefacts.

Requirement 22.2 (Refusal of a total weight of one or more). The engine refuses certified weights whose total weight $p = \sum_v p_v$ is not below one, with `total_weight_not_below_one`. It decides this for every declaration,

whether or not $\mathcal{H}(B_{>N})$ is trivial, and it constructs no squasher at $p = 1$, where the construction of Proposition 19.4 would divide by $1 - p = 0$.

Requirement 22.3 (Use and record of the largest certified weight). The engine uses, for each outcome, the largest weight the artefact certifies and records the weights it used in `wpfss_data`, and it never silently substitutes a larger one.

Requirement 22.4 (Exact decision of monotonicity and descent). The engine decides exactly, on the declared dark-count matrix R, that R is monotone, $R_{S|S'} > 0 \Rightarrow S' \subseteq S$, and that it descends through the declared coarse-graining ϖ in the sense of Lemma 20.1, $P(y|S') = P(y|S'')$ for every y and every pair S', S'' with $\varpi(S') = \varpi(S'')$, each an identity or a comparison between the dyadic rationals the declaration carries (Definition 9.2). It tests the descent property itself, and treats the sufficient condition (20.2) only as a shortcut that must itself be checked in full. A declaration that fails either condition is refused with the reason `dark_counts_do_not_descend`.

Requirement 22.5 (Absorbing bounded letters as an input validity condition). The engine takes (absorbing) as a declared input validity condition: it decides $R^{\varpi}_{y_i|y_i} = 1$ exactly at every bounded letter y_i of the slot, and refuses a declaration in which some bounded letter fails it, with the reason `bounded_letter_not_absorbing`. The refusal is of the whole artifact: the failing entry's weight is not set to zero and the rest kept.

Requirement 22.6 (Certified sets equal to fibres of the coarse-graining). The engine requires every bounded set of the slot to be exactly a fibre of the declared coarse-graining, $\mathcal{O}_i = \varpi^{-1}(y_i)$, validates the equality at input, and refuses a slot that fails it with the reason `bound_set_not_a_fibre`.

Requirement 22.7 (Exact column-stochasticity of the declared dark-count matrix). The engine decides *exactly* that the declared dark-count matrix R is column-stochastic, $R_{S|S'} \geq 0$ for every entry and $\sum_S R_{S|S'} = 1$ for every column, and refuses a declaration that is not, with the reason `dark_counts_not_column_stochastic`. No second rule decides the property of the descended matrix R^{ϖ} of Lemma 20.1.

Requirement 22.8 (Receiver versions from which per-setting weights are read). The engine reads the per-setting weights $p_i^{(t)}$ of a receiver artifact (`high_photon_bounds`) only where the artifact's module version resolves to an accepted receiver version that states them: one whose Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound) carries

the per-setting inequality as its first conclusion and whose Definition "The strengthening slot" (receiver/def:slot) carries a weight per entry and setting. An artifact whose version does not state them, a probe-free one included, is refused with `high_photon_slot_ill_formed`: the per-setting values it reports beside the minimum are not read from it, and its minimum is not read at every setting in their place.

Requirement 22.10 (The link of a normalized station records the factored loss). Where the receiver artefact's `efficiency_record` carries a factored loss $t < 1$, the engine emits the artefact of the precomposed instance $\mathcal{J}'_{\mathrm{flat}} \circ \Lambda_t^{\mathrm{loss}}$ of Corollary 21.6, and its link records, in place of the normalized station's: (i) the channel witness as the flat instance's witness preceded by the pure-loss channel of transmittance t on the channel modes, with t recorded under `factored_loss`; (ii) the source family as the receiver's measurement; (iii) each pull-back of the carried tuple as the pull-back through that loss of the flat instance's pull-back, not as the flat instance's pull-back itself; and (iv) the reachable-set statement as the one for that composite channel (Definition 3.10). The emitted target, routed weights, transported operators, entries, constants, certificate cases and licence record are those emitted for the normalized station, unchanged. The reference companion's link is recorded the same way: its channel and pull-backs carry the same loss in front, and its target and entries are unchanged.

Requirement 22.11 (A normalized receiver is not read as a branch of a branching instance). An artefact that embeds the flat instance as a branch of a branching instance, the register instance included, reads only a receiver artefact whose `efficiency_record` carries the factored loss $t = 1$, and refuses one with $t < 1$ with `normalized_receiver_in_branch`.

Domination links

23 The domination link and its noise channel

Definition 23.1 (The domination link)**.** Let $\mathbb{S}$ be a finite ordered set of sectors with dimensions $d_s \geq 1$, $\mathcal{H}(Q) = \bigoplus_{s\in\mathbb{S}} \mathbb{C}^{d_s}$ with sector projectors Π_s, and let $G = \{G_v\}_{v\in\mathcal{V}}$ and $F = \{F_v\}_{v\in\mathcal{V}}$ be POVMs on Q over one finite alphabet $\mathcal{V}$, both block diagonal for $\{\Pi_s\}$, $G_v = \bigoplus_s G_v^{(s)}$ and $F_v = \bigoplus_s F_v^{(s)}$. Let $\vartheta = (\vartheta_s)_{s\in\mathbb{S}}$ with $\vartheta_s \in [0, 1)$ be weights at which G dominates F, every block complete at 1 (Definition "Domination of one operator family by another at block weights" (foundations/def:domination)):

$$G_v^{(s)} - (1 - \vartheta_s)\, F_v^{(s)} \;\succeq\; 0 \qquad \text{for every } v \in \mathcal{V} \text{ and } s \in \mathbb{S}; \tag{23.1}$$

they are the *domination weights* of the pair. Let $R_v \coloneqq \bigoplus_s R_v^{(s)}$ with $R^{(s)}$ the residual family of Definition "Domination of one operator family by another at block weights" (foundations/def:domination) on sector s. A family F on Q has *indicator sectors* when there are pairwise distinct sectors $(f) \in \mathbb{S}$, one for each $f \in \mathcal{V}$, with $d_{(f)} = 1$ and $F_v^{((f))} = \delta_{vf}$ for every v; write $\Pi_{(f)}$ for the projector onto the sector (f), a rank-one projector.

(a) *(added flag sectors)* Let Q^+ be the system with $\mathcal{H}(Q^+) = \mathcal{H}(Q) \oplus \mathcal{H}(\mathrm{Flg}^+)$, $\mathcal{H}(\mathrm{Flg}^+) = \mathbb{C}^{|\mathcal{V}|}$ with orthonormal basis $\{|v\rangle\}_{v\in\mathcal{V}}$, whose sector set is $\mathbb{S}$ followed by $|\mathcal{V}|$ one-dimensional sectors $(f)^+$, $f \in \mathcal{V}$, in the order of $\mathcal{V}$, the sector $(f)^+$ spanned by $|f\rangle$. Put $F_v^+ \coloneqq F_v \oplus |v\rangle\langle v|_{\mathrm{Flg}^+}$, so that the added sectors are indicator sectors of F^+, and

$$\Phi_\vartheta^+(\sigma) \;\coloneqq\; \sum_{s\in\mathbb{S}} (1-\vartheta_s)\,\Pi_s\sigma\Pi_s \;\oplus\; \sum_{v\in\mathcal{V}} \Big(\sum_{s\in\mathbb{S}} \vartheta_s \operatorname{tr}\big[R_v\,\Pi_s\sigma\Pi_s\big] \Big)\, |v\rangle\langle v|_{\mathrm{Flg}^+} \tag{23.2}$$

for $\sigma \in \mathcal{T}(Q)$.

(b) *(shared flag sectors)* Suppose F has indicator sectors (f) and $\vartheta_{(f)} = 0$ for every $f \in \mathcal{V}$. Put

$$\Phi_\vartheta(\sigma) \;\coloneqq\; \sum_{s\in\mathbb{S}} (1-\vartheta_s)\,\Pi_s\sigma\Pi_s \;+\; \sum_{f\in\mathcal{V}} \Big(\sum_{s\in\mathbb{S}} \vartheta_s \operatorname{tr}\big[R_f\,\Pi_s\sigma\Pi_s\big] \Big)\, \Pi_{(f)}. \tag{23.3}$$

A *domination link* is the tuple (7.1) of a receiver-stage typed link (Definition 7.1) with stage = receiver, kind family-dominating and source family (Q, G); with target (Q^+, F^+) and channel witness Φ_ϑ^+ in case (a), or target (Q, F) and channel witness Φ_ϑ in case (b); with carried operators the projectors $\Pi_s \oplus 0$, $s \in \mathbb{S}$, and the flag projector $\Pi_{\mathrm{Flg}}^+ \coloneqq 0 \oplus \mathbb{1}_{\mathrm{Flg}^+}$ in case (a), and the projectors Π_s of the sectors that are not indicator sectors and the flag projector $\Pi_{\mathrm{Flg}} \coloneqq \sum_f \Pi_{(f)}$ in case (b), each with its pull-back under the channel witness; with the reachable-set statement of Definition 3.10 for that channel; with the constraint entries and licence record of the chain it ends; and with status = exact.

Theorem 23.2 (The noise channel of a domination). In the setting of Definition 23.1, let G dominate F at the domination weights ϑ and let R be the residual family.

(i) $R = \{R_v\}_{v\in\mathcal{V}}$ is a POVM on Q, block diagonal for $\{\Pi_s\}$, and $G_v^{(s)} = (1-\vartheta_s)F_v^{(s)} + \vartheta_s R_v^{(s)}$ for every v and s.

(ii) *(added flag sectors)* Φ_ϑ^+ of (23.2) is a channel $\mathcal{T}(Q) \to \mathcal{T}(Q^+)$, and

$$\big(G;\ (1-\vartheta_s)\Pi_s\ (s\in\mathbb{S}),\ \textstyle\sum_s \vartheta_s\Pi_s\big) \;\preceq_{\Phi_\vartheta^+}\; \big(F^+;\ \Pi_s\oplus 0\ (s\in\mathbb{S}),\ 0\oplus\mathbb{1}_{\mathrm{Flg}^+}\big). \tag{23.4}$$

(iii) *(shared flag sectors)* If F has indicator sectors (f) with $\vartheta_{(f)} = 0$ for every f, then Φ_ϑ of (23.3) is a channel $\mathcal{T}(Q) \to \mathcal{T}(Q)$ with $G \preceq_{\Phi_\vartheta} F$, and its adjoint takes the sector projectors and the flag projector to

$$\Phi_\vartheta^\dagger(\Pi_s) = (1-\vartheta_s)\Pi_s \quad (s \text{ not an indicator sector}),$$
$$\Phi_\vartheta^\dagger(\Pi_{(f)}) = \Pi_{(f)} + \sum_s \vartheta_s\, \Pi_s R_f \Pi_s, \qquad \Phi_\vartheta^\dagger(\Pi_{\mathrm{Flg}}) = \Pi_{\mathrm{Flg}} + \sum_s \vartheta_s \Pi_s. \tag{23.5}$$

(iv) *(output)* The outputs of both channels are block diagonal for the sectors of their target, so either channel is unchanged when the sector pinching of its target is applied after it.

(v) *(weight zero)* If $\vartheta_s = 0$ for every s, then $F = G$, and Φ_ϑ, where defined, is the sector pinching $\sigma \mapsto \sum_s \Pi_s \sigma \Pi_s$.

In particular the domination link of Definition 23.1, in either case, is a receiver-stage link of Definition 7.1: its channel witness relates its source family and carried operators to its target and carried operators exactly, as (7.1) requires with $\mathsf{status} = \mathsf{exact}$.

24 Domination in a reduction chain

Corollary 24.1 (A squashing link followed by a domination link)**.** Let $\mathcal{J}$ be a reduction instance with sector set $\mathbb{S} = \mathbb{S}_{\mathrm{dat}} \sqcup \mathbb{S}_{\mathrm{sc}}$, target family $F[\cdot]$ and model map $M[\cdot]$ (Definition 10.1), and let A° and A' be data for its data sectors such that

(a) $A' \in \mathcal{D}_{\mathrm{adm}}$; the blocks $A_v^{\circ(s)}$ are Hermitian, positive semidefinite and class complete, so that the target family $F[A^\circ]$ of (10.2) is a POVM on Q; and there are a POVM M° on B, a channel $\Lambda^\circ \colon \mathcal{T}(B) \to \mathcal{T}(Q)$ and a finite set $\hat{O}^\circ$ of operators on Q containing every sector projector, with pull-backs O°, such that $\big(M^\circ; O^\circ\big) \preceq_{\Lambda^\circ} \big(F[A^\circ]; \hat{O}^\circ\big)$;

(b) A° dominates A' at weights $\vartheta_s \in [0,1)$, $s \in \mathbb{S}_{\mathrm{dat}}$: $A_v^{\circ(s)} - (1-\vartheta_s) A_v^{\prime(s)} \succeq 0$ for every $v \in \mathcal{V}$ (Definition "Domination of one operator family by another at block weights" (foundations/def:domination)).

Put $\vartheta_s := 0$ on every scalar sector. Then:

(i) $F[A^\circ]$ dominates $F[A']$ at the domination weights $\vartheta = (\vartheta_s)_{s \in \mathbb{S}}$, as POVMs on Q block diagonal for the sectors.

(ii) Let ℓ_{dom} be the domination link (Definition 23.1) with source family $(Q, F[A^\circ])$ and target family $F[A']$, with added flag sectors (case (a) of that definition), or with shared flag sectors (case (b)) where $F[A']$ has indicator sectors among its scalar sectors; write F^{dom} and

Φ for its target and channel witness. Let ℓ_1 be the receiver-stage link with source family (B, M°), target $(Q, F[A^\circ])$, channel witness Λ° and target-side carried operators the pull-backs under Φ of the carried operators of ℓ_{dom}, with $\mathsf{status} = \mathsf{exact}$. Then $(\ell_1, \ell_{\mathrm{dom}})$ is a composable and tuple-chained receiver segment, and with an empty source segment a reduction chain (Definition 7.2).

(iii) Its receiver composite $\Phi \circ \Lambda^\circ$ satisfies $M^\circ \preceq_{\Phi\circ\Lambda^\circ} F^{\mathrm{dom}}$ exactly, its composite status is exact, and Theorem 3.1 and Theorem 3.11 apply to it: the protocol whose measurement is M° and the squashed protocol with target F^{dom} have the same output state, and security of the squashed protocol over analysed inputs that contain the composite's image transfers to the protocol with M° with the same ε.

(iv) An entry $(\hat{O}, \bowtie, c)$ declared on F^{dom} is sound for the composite, and Theorem 6.7 applies to the list of such entries with the composite channel, when $(\Lambda^\circ)^\dagger\big(\Phi^\dagger(\hat{O})\big) \bowtie c\,\mathbb{1}_B$. Every projector onto a sector of the target has a positive semidefinite composite pull-back.

(v) Where $\sum_{v\in\mathcal{V}_t} M_v^\circ = q_t \mathbb{1}_B$ for every $t \in \mathcal{T}_{\mathrm{set}}$, the class-marginal identity (10.5) of M°, the composite pulls $\sum_{v\in\mathcal{V}_t} F_v^{\mathrm{dom}}$ back to $q_t\mathbb{1}_B$ for every $t \in \mathcal{T}_{\mathrm{set}}$, so the class-marginal entry of Definition 10.1 (I4) is certified for the composite.

References